

Economic and Political Consequences of Credit Policy for Minorities: Evidence from India

Muhammad Yasir Khan S.K. Ritadhi *

[Click here for latest draft.](#)

May 26, 2025

Abstract

Can welfare policies improve the economic and political well-being of marginalized groups and promote social harmony in a fractured society? We examine this question by studying a unique policy intervention in India that encouraged commercial banks to increase lending to minority borrowers in “minority concentration” districts based on a random threshold of population. Using a regression discontinuity design, we identify substantial increases in minorities’ access to bank credit, a higher monthly household consumption of minority households, and a reduction in the consumption inequality between minority and majority households. The increase in bank credit does not come at the cost of lower lending to non-minorities or higher rate of delinquencies among the minority borrowers. These changes in economic well-being carry on to political outcomes. Elections are more likely to have minority candidates who receive a higher share of votes cast compared to minority candidates in the non-policy districts. The policy also reduced the vote share of candidates from the main right-wing party with a prominent anti-minority platform. These electoral changes lead to an increase in violence, primarily in post-election periods suggesting a blow-back effect.

JEL Classification: D63, D72, D74, G18, G51, H3,P16

*University of Pittsburgh & Occidental College. We thank Patrick Agte, Girija Borker, Francis Annan, Ritam Chaurey, Mausumi Das, Ernesto Dal Bo, Willa Friedman, Fred Finan, Osea Giuntella, Botir Kobilov, Cynthia Kinnan, Asim Khwaja, Aprajit Mahajan, Ted Miguel, Diana Ngo, Nishith Prakash, Silvia Prina, Caroline Theoharides, Noam Yuchtman, and seminar participants at Pitt, UCSC, Cornell, Notre Dame, UCLA, UIUC, and Oxford for useful comments and suggestions. This research has benefited from funding support provided by Center for Governance and Markets.

1 Introduction

Developing states, characterized by diverse and fractured societies, face a fundamental policy dilemma. On one hand, they confront strong normative and economic imperatives to design and implement inclusive policies that promote the upliftment of historically marginalized groups, thereby fostering broad-based economic growth (Marshall, 1950). On the other hand, such redistributive efforts often encounter political resistance – especially in societies marked by deep racial, religious, or ethnic faultlines – as they may be perceived as favoring one group over others, rendering these policies politically costly or even infeasible (Alesina et al., 2001). Moreover, the implementation of such policies can exacerbate social tensions or trigger conflict, especially if the majority perceives the rise of a minority group as a threat to its social or economic dominance (Mitra and Ray, 2014). These dynamics raise a fundamental challenge: can states promote equitable development without undermining political stability or social cohesion?

This paper examines this structural tension through the lens of a large-scale welfare policy in India—the world’s most populous democracy—targeted at improving the socioeconomic status of religious minorities by expanding their access to formal bank credit.¹ By analyzing the design, implementation, and consequences of this policy, we shed light on the economic benefits and political risks associated with targeted welfare interventions in setting marked with social, ethnic and religious cleavages.

India presents an ideal setting to examine the political and sectarian implications of welfare policies targeted towards marginalized groups within a fractured society. The country’s formal banking system is extensive, but prone to discriminatory practices (Fisman et al., 2020). India’s 200 million Muslim population forms the largest religious minority in the world, but remains economically disadvantaged (Sachar, 2006; Maizland, 2020), politically under-represented (Bhalotra et al., 2014; Allie, 2024), and vulnerable to religious violence (Mitra and Ray, 2014; Iyer, 2018). Against this backdrop, the Prime Minister’s New 15 Point Programme for Minorities was unveiled in 2009 by the federal government – a set of policy initiatives to enhance the socio-economic conditions for religious minorities, namely, Muslim, Christian, Sikh, Buddhist and Parsi communities. This paper focuses specifically on the

¹ Access to credit is a well-established lever for poverty alleviation and development, enabling households to smooth consumption, invest in education and health, pursue entrepreneurial activities, and improve labor market outcomes (Kaboski and Townsend, 2012; Augsburg et al., 2015; Buera et al., 2020; Cramer, 2021; Breza and Kinnan, 2021; Aydin, 2022). Yet, credit access remains deeply unequal, with significant disparities along lines of religion, caste, and gender (Blanchflower et al., 2003; Fisman et al., 2020; Brock and de Haas, 2023).

program’s directive to commercial banks to expand credit access for religious minorities.

India’s central bank, the Reserve Bank of India (RBI), implemented the directed credit policy by classifying select districts as “minority-concentration” and encouraged commercial banks to expand lending to religious minorities in these districts.² In the absence of lending targets, the RBI incentivized compliance by expanding the regulatory definition of “weaker section” to include religious minorities, expanding in the process the pool of borrowers with which banks’ could meet their regulatory targets.³

For causal identification, we leverage the administrative criteria used by the central bank to classify districts as “minority-concentration”: namely, districts where the population share of religious minorities exceeded 25%, as per the 2001 Population Census (RBI, 2007). The use of an arbitrary threshold generates plausibly exogenous variation in policy exposure in the spirit of a regression discontinuity (RD) design, allowing us to compare outcomes across districts just above the threshold (treatment), vis-a-vis those just below (control) (Lee and Lemieux, 2010). As the classification was based on data predating the policy by six years, it is unlikely that districts could have manipulated their minority share to affect inclusion. A McCrary density test (McCrary, 2008) is also unable to reject the null of no sorting around the discontinuity threshold. We also confirm balance on a diverse array of pre-treatment outcome variables.

We begin by identifying the first-order impacts of the directed credit credit policy and compare minorities’ access to bank credit across treatment and control districts within a narrow bandwidth of the discontinuity threshold. We use nationally representative household data from the All India Debt and Investment Survey (AIDIS) conducted in 2019, allowing us to estimate the long-run effects of the directed credit policy. A prior round of the AIDIS in 2003 points to substantial credit rationing among Muslim households;⁴ an Oaxaca-Blinder decomposition indicates that unobserved factors account for approximately 40 percent of the credit access gap between Muslim and non-Muslim households. Given this disparity, and the fact that Muslims constitute over 80 percent of India’s religious minority population, the focus of this paper is primarily on Muslim households.

Comparing Muslim households in minority concentration districts to observationally equivalent Muslim households in non-minority concentration areas, our RD design identifies a statistically

² Districts form the third tier of administration in India, after the federal and state.

³ Indian commercial banks are mandated to allocate at least 12.5 percent of their lending portfolio to weaker sections. Prior to 2007, these included female, *Dalit* (Scheduled Caste or SC), and *Adivasi* (Scheduled Tribe or ST) borrowers.

⁴ Financial outcomes for non-Muslim religious minorities were broadly comparable to those of Hindu forward castes households.

significant 12–18 percentage point (ppt.) increase in access to bank credit. This translates to bank credit access for an additional 300,000 minority Muslim households. We also document a sizable increase in bank credit access along the intensive margin: Muslim households in treated areas reported a INR 20,000 increase in bank loans, equivalent to 13 percent of their annual consumption.

Our preferred sample is a fixed set of 61 districts within a 0.058 bandwidth around the policy threshold, selected using the MSERD optimal bandwidth procedure recommended by Calonico et al. (2020). We verify robustness to alternate bandwidths between 0.04 to 0.10, and also confirm the results to not be driven by any single state. Our findings remain unaltered when the sample is extended to non-Muslim religious minorities. The results are also robust to a fuzzy RD specification accounting for partial non-compliance for 18 districts.⁵

A common concern with targeted welfare policies is the potential for redistributive costs to non-beneficiary groups (Holzer and Neumark, 2000; Marion, 2009). We explore three such sources of spillovers. First, comparing credit outcomes amongst non-minority households across treated and control districts, we find no evidence of negative treatment effects on bank credit access for non-minorities. Pooling together minority and non-minority households, the treatment left unaffected bank credit access for non-minorities, but resulted in a sharp differential increase for Muslim households. This rules out that the increase in bank loans for religious minorities did not stem from a credit reallocation from non-minorities to minorities. Second, bank loan delinquency did not rise for minority households in treated areas. This assuages the concern that the directed credit policy could have been deleterious for credit market efficiency by channeling loans to non-creditworthy borrowers. Finally, we offer suggestive evidence of the absence of a differential decline in bank credit access for minority households in control districts around the discontinuity threshold. This alleviates concerns that greater access to bank credit in minority concentration areas reflected a within-minority credit reallocation from control to treated districts.

To assess broader welfare implications, we examine treatment effects on household consumption and asset ownership. Minority households in treated districts reported 13 percent higher per capita monthly consumption compared to their counterparts in control districts. While we observe no statistically significant change in consumption among non-minority households, the point estimate is modestly

⁵ As no formal rationale was provided for the inclusion of these 18 additional districts, we omit them from the main specification and rely on a sharp RD design.

negative. Notably, the treatment effect for minorities closed by half the 27 percent unconditional consumption gap between Muslim and non-Muslim households in control areas. We also find a significant increase in household asset ownership among Muslim households, with no corresponding gains observed among non-minorities.

We explore three potential mechanisms underlying the observed increase in minorities' access to bank credit. First, we offer evidence of banks leveraging existing institutional arrangements to expand credit in environments susceptible to high information asymmetries. Over the past two decades, India has witnessed a steady rise in loans issued through bank-linked self-help groups (SHGs), whereby banks lend directly to registered SHGs, which subsequently decide on the internal allocation of credit. While direct loans from commercial banks too were higher in minority concentration areas, we find that up to 60 (20) percent of the extensive (intensive) margin increase in bank loans can be attributed to lending through bank-linked SHGs. There is however no corresponding impact on non-minority households, assuaging concerns that minority concentration districts witnessed a secular expansion in lending through bank-linked SHGs. Second, as minority households possessed fewer pledgeable assets, we investigate whether banks responded by easing collateral requirements. Consistent with this channel, we find a significant increase in minority borrowers' likelihood of having an unsecured loan in treated districts, both when borrowing directly from commercial banks, and through bank-linked SHGs. The rise in unsecured bank lending too is indicative of an improvement in lenders' ability to screen minority borrowers (Fisman et al., 2017). Third, we test whether banks adjusted interest rates to facilitate credit flow to minorities. We find no evidence of a lowering of interest rates for minority borrowers, suggesting that the expansion in credit access was not driven by pricing changes.

Having established that the directed credit policy improved religious minorities' access to bank credit and household consumption, we examine whether economic empowerment translated to political empowerment: a critical question when considering the under-representation of Muslims in political and public institutions (Sachar, 2006; Fisman et al., 2017). Using data compiled by the Trivedi Center for Political Data (TCPD) on Indian state elections and extending our sharp RD framework to elections conducted between 2008 and 2019, we identify sizeable increases in both the participation of minority Muslim candidates, as well as electoral support received by such candidates. The likelihood of a Muslim candidate contesting elections increased by 17 ppt. in minority concentration areas, along with a 4 ppt. increase in the share of votes accruing to these candidates. The likelihood of minority

candidates contesting on behalf of non-majoritarian parties was also significantly higher; indeed, the rise in minority vote shares was driven by candidates contesting on behalf non-majoritarian parties. This assuages concerns that economic empowerment might have increased political participation from minority individuals contesting elections solely to enhance personal visibility. This also rules out that the increase in popular support for minority candidates is a mechanical effect from a higher number of Muslim contestants.

Rising popular support translated to a 9 ppt. increase in Muslim candidates' likelihood of winning in minority concentration areas, deepening minority representation in state legislatures. The political empowerment of minorities was also accompanied by a 5 ppt. reduction in popular support for the right-wing BJP, which has historically been inimical to minority interests. Overall, our findings point to greater political empowerment in districts eligible for the directed credit policy: an uptick in the supply of Muslim candidates who received greater electoral support, and had higher winnability.

We consider three pathways through which improved access to formal credit could have facilitated political empowerment. First, the directed credit policy could have generated a positive wealth effect for minorities, allowing them to overcome entry costs to run for office. Relatedly, a positive wealth effect could have enabled minority citizens to financially aid parties sympathetic to their interests. The quality of minority candidates too could have improved in response to the directed credit policy. Second, economic empowerment could have increased voter consciousness and boosted the turnout of minority voters. Third, the directed credit policy could have altered social interactions within and across communities by increasing minority participation in self-help groups, or in the labor market.

Empirically we find that the evidence supports a positive wealth effect for Muslim politicians. Using data on politician affidavits, Muslim contestants in treated districts had significantly higher assets, and were also more likely to have finished college, pointing to the supply of better candidates. Turnout however remained comparable across minority and non-minority concentration districts. To assess the directed credit policy's impact on social interactions, we use information from the National Rural Livelihood Mission (NRLM) on membership of the bank-registered self-help groups. Using this novel data, we first identify an overall increase in the total number of Muslim members across bank-registered SHGs in minority concentration areas. This was driven by the formation of exclusively-Muslim SHGs, as opposed to SHGs comprised of both Muslim and non-minority members. Similarly, while the directed credit policy led to the reallocation of minority workers from wage work in trade

and services to self-employment in manufacturing activities, minority workers in treated districts had a significantly higher (lower) likelihood of working in sectors with a relatively high (lower) share of minority (non-minority) workers. These two pieces of evidence suggests that the directed credit policy aided social interactions between minority citizens, as opposed to social interactions across communities. If co-ordination costs were higher for religious minorities, increased social interactions through bank-registered SHGs, and the labor market could have aided the ability of minority citizens to co-ordinate and vote for politicians furthering their interests. Overall, Atkin et al. (2021) posits that heightened interactions with co-religionists can raise the salience of one’s religious identity, which too can explain the rise in popular support for Muslim politicians.

Noting India’s long history of sectarian Hindu-Muslim violence (Varshney, 2003), we conclude our empirical analysis by analysing whether the directed credit policy affected religious conflicts. Inter-group conflict can be affected by both economic and political empowerment. Economic prosperity can increase the opportunity cost of violence, while political empowerment can improve communities’ influence over law enforcement authorities. Alternately, Mitra and Ray (2014) have documented the strategic use of violence by majority communities to reassert dominance when faced with narrowing inter-group inequality.

We study this using both administrative data from the National Crime Records Bureau (NCRB), as well as data from the Armed Conflict Location and Event Data (ACLED) which sources incidences of violence from newspaper reports (Raleigh et al., 2010). In both datasets, we identify a significant increase in religious riots in treated districts. Other crime heads such as non-religious riots or murders remained comparable across treated and control districts, negating a secular increase in criminal activity in minority concentration areas.

Did the preferential treatment of religious minorities in credit markets worsen social harmony? Or does the spike in religious violence emerge as a backlash to minority political empowerment. We assess this by ascertaining the precise timing of religious violence using the ACLED data. If resentment against minorities stemmed from their economic empowerment, we would expect the increase in violence to be unrelated to the timing of elections. However, if the increase in violence is observed only prior to the elections, it is likely to be driven by an effort to suppress minority turnout in anticipation of electoral gains. Alternately, if violence occurs only after the elections, it is likely an electoral backlash to the political empowerment of minorities. Disaggregating the ACLED data into post (12 months

after) and pre-electoral periods (12 months before), we identify a significant increase in Hindu-Muslim riots only after the elections. This suggests that the violence was strategically timed as a mechanism to punish minority voters voting against majoritarian politics.

Our paper contributes to two broad strands of literature. First, we show how a welfare policy designed to improve access to formal credit for marginalized groups can reduce inter-group economic inequality and enhance political empowerment through stronger within-group coordination. While prior research has separately established that credit access improves household welfare (Breza and Kinnan, 2021) and institutional mechanisms such as electoral quotas bolster political representation (Pande, 2003; Chattopadhyay and Duflo, 2004; Besley et al., 2017; Gulzar et al., 2021), this paper, to the best of our knowledge, is the first to demonstrate that economic empowerment can translate into political agency for minority groups in a fractured society.

Furthermore, we document that these gains may provoke backlash through rising religious violence targeting the beneficiary group. In doing so, we contribute to a nascent literature on the political and social costs of redistributive policies aimed at marginalized communities (Chyn et al., 2024; Avenancio-León et al., 2024; Choi et al., 2024). While previous work has linked violence to efforts undertaken by the majority to preserve economic hierarchies (Mitra and Ray, 2014), or to suppress minority political participation (Wilkinson, 2006; Condra et al., 2018; Iyer and Shrivastava, 2018), we offer an additional mechanism: that political backlash may emerge as a form of punishment following a weakening of majoritarian political dominance. These findings speak to a central dilemma in state capacity: how efforts to correct market failures and promote inclusive development can generate countervailing political risks in polarized and fragmented societies.

Second, this paper provides evidence of the effectiveness of an at-scale implementation of affirmative action in credit markets. While affirmative action has been widely studied in labor (Leonard, 1990; Holzer and Neumark, 2000; Miller, 2017), housing (Chetty et al., 2016), and political markets (Pande, 2003; Chattopadhyay and Duflo, 2004; Jensenius, 2015; Bhavnani, 2017; Gulzar et al., 2021), little is known about its implementation within the banking industry. We examine a nationwide directive issued by the Indian government to expand credit access for religious minorities through the extant banking infrastructure – a context in which discrimination against religious minorities groups is well-documented (Fisman et al., 2020; Blanchflower et al., 2003; Brock and de Haas, 2023).

We also engage with critiques of affirmative action that focus on unintended consequences, such as

misaligned targeting (Barrow et al., 2020) or the crowding out of non-minority beneficiaries (Arcidiacono et al., 2022). Our findings indicate that neither concern materialized in this case. We find no evidence of displacement among non-minorities. Instead, banks partnered with self-help groups (SHGs) to mitigate information asymmetries and improve outreach to creditworthy minority borrowers. Our findings underscore the potential for institutional intermediation to enhance the effectiveness and precision of financial inclusion policies.

The remainder of our paper is organized as follows: Section 2 describes the policy intervention of interest; Section 3 discusses the data used for the empirical analysis and presents some descriptive trends; Section 4 presents the empirical strategy for causal identification; Section 5 presents our key findings on access to credit; and Section 6 presents our results on political outcomes.

2 Background

2.1 Prime Minister’s 15 Point Programme for Welfare of Minorities

The Prime Minister’s (PM) 15 Point Programme for the Welfare of Minority Communities was a set of policies outlined by India’s federal government, aimed at improving the socio-economic conditions of India’s religious minorities – namely Muslims, Christians, Sikhs, Buddhists and Parsis.⁶ Collectively, citizens from these religious denominations accounted for 19 percent of India’s population in 2001, with Muslims comprising the largest group of 13 percent or 138 million individuals. This makes India’s Muslim population the largest religious minority group in the world.

The policies covered the realms of education, employment, infant health, housing, sanitation, access to credit, and protection from religious violence. The overarching policies were framed by the federal government and implemented through various public agencies, financed by the federal exchequer. The initial set of policies were revised and expanded in 2009, and renamed as the Prime Minister’s New 15 Point Programme for the Welfare of Minority Communities.

Access to credit under the PM’s minority welfare programme aimed at ensuring the “smooth flow of bank credit to minority communities” from state-owned and private commercial banks (RBI, 2007). The central bank – the Reserve Bank of India (RBI) – was tasked with framing the regulations for credit allocation to religious minorities, and also responsible for overall monitoring. The RBI subsequently

⁶ Later, Jains too were included under this policy as religious minorities.

issued a set of guidelines to commercial banks in July 2007, notifying that the federal government had classified 103 districts (out of 593 districts) as “minority concentration” districts (RBI, 2007).⁷ These districts were deemed such as the share of religious minorities (as per the 2001 population census) exceeded 25% of the district’s population.⁸ Commercial banks were instructed to monitor credit flow to minority borrowers in these 103 districts. Within a few months of the initial notification, the federal government added 18 districts to this list, bringing the number of minority concentration districts to 121. No subsequent additions or deletions were made to this set. As no rationale was provided for the inclusion of these 18 additional districts, we omit them from our analysis and focus exclusively on the preliminary set of 103 districts which perfectly complied with the policy rule.⁹

2.2 Monitoring by Central Bank

The RBI in 2007 issued a set of instructions to achieve banks’ compliance with the directed credit policy for religious minorities (RBI, 2007). Almost all of these remain in effect at the time of writing and are reiterated through annual notifications issued by the RBI (RBI, 2021).¹⁰ First, each bank was instructed to set up a special division, headed by a senior officer to provide oversight in relation to the policy. Second, the “lead bank” in each minority concentration district had to assign a senior officer whose sole responsibility was to look into challenges faced by religious minorities in accessing credit.¹¹ This officer was also tasked with generating local awareness amongst minority communities regarding the policy and other related government schemes. The officer was also expected to design credit schemes to fulfill the objectives of the directed credit policy, in collaboration with other branch officers in the district (RBI, 2007).

Lead banks were also directed to co-ordinate with other non-banking financial corporations in an

⁷ Districts from 5 states and 1 union territory – namely Jammu and Kashmir, Punjab, Meghalaya, Mizoram, Nagaland and Lakshadweep were excluded from this policy as religious minorities in these areas formed a numerical majority. The policy applied to Hindus in these areas, with minority concentration districts being those where Hindus formed 25 percent of the district’s population. We omit districts from these areas in our analysis.

⁸ To the best of our understanding, no other policy within the 15-point minority welfare programme were implemented using the 25% minority population share threshold.

⁹ In addition to these 18 districts, the government also classified 4 districts whose share of minority population was between 24.6% and 24.9% as minority concentration. It is possible that the government was rounding off the minority share during treatment assignment. However, as no rationale is provided for the same, we omit these 4 districts too from the analysis.

¹⁰ These are issued by the Financial Inclusion and Development Department (FIDD), responsible for promoting financial inclusion.

¹¹ The central bank in each district assigns a “lead bank” to facilitate rural banking. The lead bank co-ordinates with other commercial banks in the district on matters of financial inclusion and credit disbursement to farm activities, as well as farm and small enterprises. Lead banks are state-owned banks.

effort to reach creditworthy borrowers in minority communities. This included issuing advertisements across print and visual media, and engaging in information campaigns at the site of religious congregations (RBI, 2007). The RBI also recommended banks to engage with self-help groups to improve their selection of underprivileged minority borrowers, and lead banks in minority concentration districts were expected to be “proactive” in this regard (RBI, 2007). Both the lead bank, and individual banks operating in minority concentration districts were advised to impart adequate training to sensitize employees to the credit needs of minority borrowers. Finally, banks were mandated to file half-yearly reports with both the RBI and the federal Ministry of Minority Affairs, detailing the disbursement of credit to minority borrowers. This bi-annual reporting forms the sole tangible monitoring of the policy undertaken by the central bank (RBI, 2007).

2.3 Incentives for Banks

The primary incentive for banks to comply with the directed credit policy is that lending to minority borrowers would allow them to meet key regulatory targets. First, every bank operating in India is required to allocate at least 40 percent of its aggregate annual loan portfolio towards the “priority sector” – farm credit, rural borrowers, and credit to small and micro-enterprises (RBI, 2020).¹² Banks are also mandated to direct 10 (presently 12) percent of their loan portfolio towards “weaker sections” – namely small and marginal farmers, village and cottage industries, and loans issued to borrowers hailing from historically marginalized *Dalit* (Scheduled Castes or SCs) and *Adivasi* (Scheduled Tribes or STs) communities. With the advent of targeted lending to religious minorities, the RBI expanded the definition of “weaker section” to include borrowers from religious minorities. This included individual loans to minority borrowers, as well as loans made to non-registered enterprises owned by religious minorities.¹³ While no explicit targets were assigned, the RBI’s annual guidelines directed banks to ensure that minority borrowers were “adequately represented” within weaker sections RBI (2007, 2021).

Second, since 1991, loans for housing, education, renewable energy and self-help groups also qualified under the priority sector. As the RBI’s directives explicitly encouraged banks to lend to SHGs in an effort to reach out to minority borrowers, banks could utilize lending to minority-concentrated

¹² Banks unable to meet this are expected to contribute the residual amount to a rural infrastructure development fund.

¹³ In the event of the firm being a partnership, it qualified towards minority credit as long as the majority of partners hailed from minority groups. However, registered companies, by virtue of being a separate legal entity were ineligible to receiving credit under this scheme, irrespective of the type of ownership.

SHGs to achieve their priority sector targets. This is particularly salient in light of the bank-SHG linkage programme unveiled in 2008, which encouraged banks to directly lend to SHGs. Consequently, the directed credit policy for religious minorities also offered banks a larger pool of potential borrowers to meet their priority sector targets.

3 Data

We describe here the data on household balance-sheets, electoral outcomes, and religious conflicts. Additional datasets are noted as discussed in the empirical analysis.

3.1 All India Debt-Investment Survey

The nationally representative All India Debt-Investment Survey (AIDIS) is conducted decennially by the National Sample Survey Organisation (NSS) in the form of repeated cross-sections. Sampling in excess of 100,000 households, it offers extensive information on household borrowings, savings and ownership of select assets. Location identifiers in the form of districts are also provided, in addition to the household’s caste, religion, and demographic characteristics. For each loan outstanding at the time of survey, the AIDIS details the initial amount borrowed, the loan origination year, amount outstanding, loan securitization, the source of credit, and interest rate. Repayment information over the past 6 months is also provided.

The AIDIS was conducted in the years 1992, 2003, 2013 and 2019. We use the 2019 AIDIS survey to estimate long-term treatment effects of the directed credit policy on household borrowing. The 2003 AIDIS survey is used to verify balance for key outcomes of interest in the pre-treatment period. Appendix Tables C1.A-C1.D present select summary statistics from the 2019 AIDIS. In all, 42 (23) percent of households had some outstanding (bank) loan at the time of the survey. While 17 percent of households borrowed directly from commercial banks, 7 percent of households received loans from a bank-linked SHG. Almost 20 percent of households had credit from informal sources, which includes money lenders, input suppliers, friends, relatives, employers and landlords. Expenditure loans accounted for most household borrowing (17 percent), followed by farm loans (13 percent). Only 4 percent of households reported taking loans for non-farm businesses.

Conditional on having some outstanding loan, Appendix Table C1.C reports average aggregate household debt to equal INR 200,000 – approximately 1.4 times aggregate annual household con-

sumption.¹⁴ The sum of loans obtained directly from commercial banks was larger – INR 250,000. Expectedly, loans from bank-linked SHGs were smaller, amounting to less than INR 50,000. Informal loans from money lenders equaled INR 120,000. Loan volumes were largest for non-farm business loans, followed by expenditure loans. Appendix Table C1.D shows that 40 percent of households had an unsecured bank loan. Households faced an average annual interest rate of 14%, with banks charging lower (12%) than money lenders (30%).¹⁵

Absence of administrative data on repayments makes it challenging to assess loan delinquency. Moreover, as the AIDIS collects information solely on outstanding household loans, it is biased towards capturing delinquencies as such loans continue to remain on a household’s balance sheet. Estimates of borrower delinquency from the AIDIS should thereby be interpreted with caution, and are likely to provide an upper bound of loan delinquency. We use information on repayments made between June 30, 2018 and the time of survey to measure loan delinquency.¹⁶ Appendix Table C1.D reports high self-reported loan delinquency, with almost 35% of households reporting an outstanding loan on which no repayment was made over the past 6 months. In line with the risk-averseness of banks, the delinquency for bank loans (26%) was lower than loans obtained from informal sources (41%).¹⁷

3.1.1 Barriers to Credit Access for Minorities: Pre-Treatment Descriptives

Descriptive evidence in Appendix A using the 2003 AIDIS offers suggestive evidence that amongst religious minorities, Muslim households faced substantial barriers to credit access. Muslim households had a lower likelihood of having a bank loan; conditional on having a bank loan, loan amounts for Muslim households were also markedly lower. Bank interest rates however were comparable across all communities, indicating no higher credit risk for Muslim households. These households however were charged the highest interest rates by professional money lenders.

Diminished access to bank credit could be a function of lower credit demand, or lower credit

¹⁴ We use initial loan sizes to measure the volume of credit. Initial loan size offers an accurate measurement of the credit extension, unaffected by capitalized interest and debt accumulation due to non-repayment.

¹⁵ Household-specific interest rates are computed as the loan volume weighted average interest rate across all outstanding loans.

¹⁶ As households were surveyed in 2019 and 2020, non-repayment since June 30, 2018 implies the loan being delinquent for at least 6 months. We are unable to obtain delinquency measures for loans obtained in 2019 and the 2019 AIDIS only informs us of the year in which the loan was obtained and not the month.

¹⁷ The vast majority of informal loans from friends and relatives were interest free, which probably did not require any regular repayments to be undertaken. Regardless, we classify these loans as delinquent if no repayment was made in the past 6 months.

supply. An Oaxaca-Blinder decomposition in Appendix A.3 offers suggestive evidence of discrimination, particularly along the extensive margin. The top panel of Appendix Figure A6 indicates that 40 percent of the extensive margin difference in bank credit for Muslim households – relative to Hindu households – could be explained by household observables. Section 5.1.6 examines whether this reflects statistical or taste-based discrimination against Muslim borrowers in India’s formal credit markets.

Taking cognizance of the descriptive evidence documenting the exclusion of Muslim households from formal credit markets and the fact that they account for 80 percent of India’s religious minority population, our paper primarily focuses on the impact of the directed credit policy on Muslim households. For the remainder of the paper, unless explicitly stated, we use the term “religious minorities” or “minorities” to refer exclusively to Muslim households.

3.2 Election Data

To assess the electoral consequences of the directed credit policy, we use data from the Trivedi Center for Political Data (TCPD), which compiles data on elections to India’s state legislative assemblies. This includes candidate names, party affiliation, and votes received, along with the number of registered voters, and voter turnout. Our primary sample comprises of elections conducted between 2008 and 2019. We append to this information from candidate affidavits pertaining to their incomes and assets. We also use the share of minority Muslim voters in each electoral constituency, estimated by Gulzar et al. (2024), based on the names of registered voters in the voter list.

To identify whether the directed credit policy affected the political empowerment of religious minorities, we compare the participation of Muslim politicians in state legislative assembly elections, and the vote share received by Muslim candidates. We use the machine learning algorithm developed in Chaturvedi and Chaturvedi (2024), and validated by Allie (2024) to ascertain the religious identity of politicians.¹⁸ As a third measure of minority political empowerment, we also identify the impact of the directed credit policy on popular support for the right-wing Hindu nationalist BJP. The BJP has stridently opposed any positive discrimination for religious minorities, and has generally been inimical to minority interests (Jaffrelot, 2021; Das, 2024).

Between 2008 and 2019, Appendix Table G1 shows that over half the elections in our sample had no Muslim contestant. While 12 candidates on average contested elections to state legislative

¹⁸ This algorithm has been specifically developed for India.

assemblies, the number of Muslim candidates equaled 1. In 13 percent of electoral contests, there was a Muslim candidate representing a non-majoritarian party.¹⁹ Muslim candidates received 7 percent of the votes cast and emerged victorious in 6 percent of elections. Over this period, the majoritarian BJP was the most successful party, winning a third of the electoral contests and capturing a fourth of the popular vote.

3.3 Conflict Data

To ascertain whether the directed credit policy affected social harmony and religious conflict, we rely on both administrative data from the National Crime Records Bureau (NCRB), as well as data from the Armed Conflict Location and Event Dataset (ACLED) (Raleigh et al., 2010). The NCRB database offers information on district-level crimes in each year, based on first information reports (FIR) filed at police stations.²⁰ Alongside traditional offenses such as murder, property crimes, rapes and abductions, the NCRB also captures riots, and since 2014, separately reports religious riots.

A variety of factors can make the administrative NCRB data prone to reporting and recording biases. Relevant to our study, majoritarian tendencies have sharply increased in India since 2014, which might lead to an under-reporting of religious violence, especially if the victims are religious minorities and the police are susceptible to political pressures. Resultantly, we supplant the administrative NCRB data on religious conflicts with the ACLED database, which covers conflict events in India from 2016 on the basis of newspaper reports. We map events to districts using the conflict location information in the ACLED data, and also classify events as riots involving Hindu and Muslim groups, based on the description of the events and actors involved. The ACLED also allows us to infer whether the violence was instigated by actors from the majority, or the minority community.

Appendix Table H1 shows that between 2014 and 2019, the administrative NCRB reported 2 incidences of religious violence in the average district. Religious riots formed a small fraction of overall rioting: during this period, the average district witnessed 128 incidences of rioting. The incidence of Hindu-Muslim violence reported in the ACLED (Appendix Table H2) is markedly lower than the NCRB data. This is possibly due to two factors: first, the administrative data relies on first information

¹⁹ We consider 5 major parties which have been vocally opposed to recent majoritarian tendencies in India: the Indian National Congress (INC), the Samajwadi Party (SP), the Rashtriya Janata Dal (RJD), the All India Trinamool Congress (AITC), and the Nationalist Congress Party (NCP). For the year 2015, the Janata Dal (United) (JD(U)) is also considered as it was part of an electoral alliance with the RJD and the INC.

²⁰ Filing a FIR forms the basis for beginning any preliminary criminal investigation in an area.

reports filed at police station while the ACLED relies on newspaper coverage by major English media outlets. This makes it probable that the ACLED omits incidents of religious conflicts in smaller urban or rural locations, or low-intensity conflicts which are unreported in the English media. Second, we focus exclusively on Hindu-Muslim riots in the ACLED, while the NCRB considers all religious conflicts, which can include conflicts between religious minorities. Similar to the administrative data, religious riots in the ACLED forms a small proportion of overall rioting. Based on the event description in the ACLED, we classify a fourth of the religious riots as those instigated by Hindu groups. In contrast, a miniscule fraction of Hindu-Muslim riots are instigated by Muslim groups.²¹

4 Empirical Strategy

The use of an arbitrary threshold – fraction of religious minorities exceeding 25% of the district’s population – for treatment assignment lends itself to causal estimation using a regression discontinuity design (Lee and Lemieux, 2010). The running variable is defined as:

$$Runvar_{ds} = ShMinority_{ds} - 0.25 \quad (1)$$

$ShMinority_{ds}$ is the population share of religious minorities in district d , located in state s . Figure C1 shows the distribution of $ShMinority_{ds}$, with the broken vertical line representing the 0.25 threshold. Using $Runvar_{ds}$ from equation (1), the district-level treatment indicator – $Treat_{ds}$ equals 1 if $Runvar_{ds} > 0$ and 0 otherwise.

Districts were assigned to treatment in 2007, using data from the 2001 population Census. This makes it implausible for districts or states to anticipate the policy and strategically manipulate their minority population shares to lie on either side of the treatment threshold. Using the first list of minority concentration districts issued by the RBI in 2007, we confirm that all districts which satisfied the treatment assignment condition were indeed assigned to treatment (RBI, 2007). Formally, Figure 1 presents the McCrary test McCrary (2008), and we are unable to reject the null of a discontinuity in the running variable at the threshold of 0.25. This alleviates concerns of any strategic sorting of districts around the discontinuity threshold.

We exploit the sharp discontinuity in treatment assignment to estimate local linear regressions:

²¹ Event descriptions in the ACLED preclude the classification for the remaining three-fourths of religious riots.

$$Y_{hds} = \alpha_s + \beta Treat_{ds} + \gamma f(Runvar_{ds}) + \delta \mathbf{X}_{hds} + \epsilon_{hds} \quad (2)$$

The unit of observation in equation (2) is the household h , located in district d of state s . $Treat$ is a dummy equaling 1 if the district is classified as a minority concentration district, based on the assignment rule described above. As recommended by Lee and Lemieux (2010) and Calonico et al. (2020), we include a linear polynomial – $f(\cdot)$ – in the running variable and its interaction with the treatment indicator. The coefficient of interest is β , comparing household outcomes across treated (minority concentration) and control districts (non-minority concentration). All specifications include state fixed effects (α), household size and binary indicators for rural location and marginalized caste groups. We use a triangular kernel, assigning greater weight to observations located near the discontinuity threshold, as well as survey weights provided in the AIDIS data. Standard errors are clustered by district – the level at which the treatment varies.

Conditional on districts being unable to strategically manipulate assignment to treatment, non-minority concentration districts serve as a valid counterfactual to minority concentration districts within a narrow window of the discontinuity threshold. Our main results use a fixed sample of 61 districts located within a bandwidth of .058 around the discontinuity threshold.²² This bandwidth is chosen using data-driven optimal bandwidth selection procedures recommended by Calonico et al. (2020) (see Section 5.1.1 for details). Appendix E shows robustness to outcome-specific MSERD bandwidths recommended by Calonico et al. (2020).

A causal interpretation of the RD coefficients is subject to the assumption that pre-determined covariates were continuous in the running variable at the discontinuity threshold. We verify this using pre-treatment data in Appendix Tables B1-B14. Out of a total of 101 pre-treatment observables, we detect a statistically significant difference for 7 variables.²³ Importantly, Appendix Table B5-B8 shows statistically indistinguishable levels of credit access across bank and non-bank sources for minority households in treated and control districts.²⁴ Pre-treatment cost of credit and delinquency rates

²² Out of these, 23 are minority concentration, and the rest non-minority concentration.

²³ The 2003 AIDIS is used to compare pre-treatment household assets and liabilities. District-level data from the administrative BSR in 2007 is used for banking outcomes, while household survey data from the 2007-08 employment-unemployment survey is used to check balance on labor market outcomes. We also verify balance on electoral and crime characteristics using data from 2001 to 2007.

²⁴ All balance checks are conducted for households in 61 districts located within a bandwidth of .058 around the discontinuity threshold. Akin to the main specification, balance checks at the level of household are conducted after controlling for state fixed effects, household size, and a dummy for the household's rural location.

were also comparable. Appendix Tables B12, B13 and B15 shows balance for pre-treatment electoral outcomes and criminal activity.²⁵ The absence of selective sorting of districts into treatment and control status, combined with the overall balance of pre-treatment household and district characteristics allow us to assign a causal interpretation to the RD coefficients estimated using equation (2).

5 Results: Minority Credit Access and Economic Empowerment

This section examines the treatment’s impact on economic outcomes. We first compare access to bank credit for minority households across treated and control areas. Subsequently, we identify treatment effects on labour market outcomes, real estate and household consumption. We end by discussing three potential mechanisms through which the directed credit policy improved credit access for religious minorities.

5.1 Access to Credit in Minority Concentration Districts

5.1.1 Baseline Results

Figure 2 graphically compares bank credit access across minority households in treated and control regions, conditional on state fixed effects and household covariates. The sample is restricted to households in 61 districts located within our preferred bandwidth of 0.058.²⁶ The horizontal axis shows the running variable. Each point in the figures show the residualized outcome variable mean corresponding to each bin. The solid lines show a smoothened linear polynomial fit; the broken lines depict 90% confidence intervals. The left panel in particular shows a sharp jump in bank credit access for religious minorities at the discontinuity threshold. A less pronounced jump is observed for the volume of bank credit.

Table 1 estimates local linear regressions corresponding to Figure 2. The outcome of interest in columns (1)-(3) is a dummy equaling 1 if the household had a bank loan; in columns (4) and (5), the amount of bank loans. Columns (1) and (4) include no covariate except for the state fixed effects; columns (2), (3) and (5) control for household size, a rural dummy, and dummies for whether the household belonged to a socially marginalized community.²⁷ Except for column (3), all specifications

²⁵ Owing to the small sample size (61 districts), these are unconditional balance checks.

²⁶ We exclude the district of North Cachar Hills where all Muslim households report having a bank loan.

²⁷ We include a dummy for *Scheduled Caste (Dalit)* households, and a dummy for households belonging to the Other

use MSERD outcome-specific optimal bandwidths recommended by Calonico et al. (2020). Column (1) identifies a positive and statistically significant treatment effect, which increases in magnitude upon the inclusion of covariates in column (2). Compared to observationally equivalent minority households in control districts, minority households in treated areas saw a 12-20 ppt. increase in the likelihood of having an outstanding bank loan. Column (4) identifies a INR 20,000 increase in the amount of bank loans received by minority households, with the coefficient increasing slightly upon the inclusion of covariates. The penultimate row of Table 1 shows that the optimal bandwidth along the extensive margin without (with) controls is .055 (.047); along the intensive margin, the bandwidth without controls is .063, and falls to .058 upon the inclusion of controls. To maximize sample size, we opt to use .058 as our preferred bandwidth in all specifications. Using this bandwidth, column (3) shows that minority households saw 18 ppt. higher bank credit access in treated districts.

The point estimates are large when compared to the outcome mean in control districts. In non-minority concentration districts around the discontinuity threshold, 12 percent of Muslim households had some outstanding bank loan, while the average bank loan amount equaled INR 20,801. Resultantly, columns (3) and (5) show that the policy doubled bank credit access for minority households along both the extensive, and the intensive margins. As the average control district had 1.6 million Muslim households, the extensive margin coefficient equates to bank credit access for an additional 0.28 million Muslim households in treated districts. Since the AIDIS only collects information on outstanding loans on a household's balance sheet, the treatment effects possibly capture a lower bound of the directed credit policy's impact on bank credit access for minorities.

Along with bank credit, the AIDIS offers an extensive break-up of credit from other institutional and non-institutional sources. Appendix Tables C5 and C6 use this information to identify how the directed credit policy affected overall household borrowings, and borrowings from non-bank sources. Column (5) of Appendix Table C5 points to a INR 22,000 increase in aggregate household debt, albeit imprecisely estimated (p-value .221). This is driven by an increase in bank borrowings, and borrowings from community networks such as friends and relatives. Access to cheaper bank loans for minority households also led to reduced borrowing from non-bank financial institutions which charge higher interest rates.²⁸ Column (5) of Appendix Table C6 also finds minority households in treated districts

Backward Classes (OBC).

²⁸ Non-bank financial institutions include non-bank financial companies, including micro-finance institutions, as well as co-operative banks.

to have a higher likelihood of having an outstanding loan from both bank and community networks. Appendix C.1 combines this piece of evidence with the noisy increase in aggregate household borrowings to contend that Muslim households in minority concentration areas faced binding credit constraints.

5.1.2 Robustness

Appendix Table C3-C4 shows the robustness of the treatment effect to alternate specifications and sampling choices. Columns (1) and (4) of Appendix Table C3 show robustness to including a quadratic polynomial in the running variable, while columns (2) and (5) show robustness to the inclusion of district covariates.²⁹ Columns (3) and (6) show that the precision of our results are unaffected if we opt to use robust standard errors, instead of clustering by district.³⁰ Columns (1) and (4) of Appendix Table C4 show robustness to excluding new districts formed from existing minority concentration districts.³¹ Columns (2) and (5) show the results to be comparable if the sample is extended to include other religious minorities – namely Christian and Sikh communities. Columns (3) and (6) expands the sample to include the 13 districts which were later classified as minority concentration, despite having less than 25% minority population. Using a fuzzy RD specification, we show our results to be unaltered by the inclusion of these districts.³² Appendix Table ?? identifies comparable treatment effects if we expand our sample to include data from the AIDIS survey conducted in 2013. Appendix Figure C2 shows that the baseline findings are not driven by any single state.

All our main specifications use a common bandwidth of .058, which is the MSERD optimal bandwidth for our intensive margin outcome. Figure 3 depicts the stability of the baseline results to a host of alternate bandwidths between .04 and .10. Appendix E replicates our key results using outcome-specific MSERD bandwidths.

²⁹ We control for pre-treatment per capita monthly household consumption, branch per capita, gender ratio, fraction of rural population, fraction of marginalized caste groups, fraction secondary educated, rates of labor force participation and unemployment, fraction of salaried and casual workers, fraction of workers in farm, manufacturing, trading and service activities.

³⁰ We use unweighted heteroskedasticity robust errors.

³¹ The list of 121 minority concentration has remained unchanged since 2008. Over this period, three of the existing minority concentration districts were bifurcated to form new districts. As the policy documents do not explicitly exclude these new districts, we continue to include them in our sample, and show our results are robust to their exclusion.

³² The fuzzy RD includes all three major religious minority communities, and district covariates. The first stage coefficients are positive and statistically significant at the 1% level.

5.1.3 Credit Access for Non-Minorities

Did the increase in bank credit access for religious minorities come at the expense of non-minorities? We assess this by comparing bank credit access for non-minority households across treated and control districts. If banks kept overall lending volumes fixed in minority concentration areas, higher credit allocation for minorities would imply a reallocation of credit from non-minority borrowers. Columns (1) and (5) in Panel A of Appendix Table F1 identify a statistically non-significant treatment effect on bank credit for non-minorities. The point estimate is positive along the extensive margin (p-value .301), but negative along the intensive margin (p-value .829). While the coefficients are non-trivial when compared to the control group means, the lack of precision preclude us from drawing any strong conclusions. There is however evidence of credit reallocation from informal sources to NBFIs for non-minority households. Taken together with Appendix Table C5, the coefficients suggest that the directed credit policy led minority households to substitute NBFi loans with credit from commercial banks. On the other hand, non-minority households substituted informal credit with credit from NBFIs.

Section 2.3 noted that a key incentive to comply with the directed credit policy was that religious minorities were now classified as “weaker sections”, and commercial banks had to allocate at least 10 percent of their aggregate portfolio to borrowers from “weaker sections”. Prior to 2007, the “weaker section” comprised of female borrowers, along with borrowers hailing from historically marginalized *Dalit* and *Adivasi* communities. While the household-level nature of the data precludes us from distinguishing between male and female borrowers, Panel B of Appendix Table F1 alleviates concerns that bank credit expansion for minorities emanated from a crowding out of *Dalit* and *Adivasi* borrowers. Similar to other non-minority borrowers, columns (2) and (6) also point to non-minority *Dalit* and *Adivasi* borrowers exiting informal credit markets in minority concentration areas. Overall, there is limited evidence that banks complied with the directed credit policy by reallocating credit across communities previously qualifying as “weaker sections.” This is not entirely unsurprising: Appendix A.2 provides a back of the envelope estimate that *Dalits* and *Adivasis* in 2003 accounted for less than 3 percent of overall bank credit. This indicates that banks enjoyed some slack within the weaker section quota to accommodate religious minorities, without having to cutback credit to other eligible groups.

5.1.4 Within District Credit Reallocation – Non-Minority to Minority

The previous section showed limited evidence that the directed credit policy crowded out non-minority borrowers from bank credit. We examine this further by pooling our sample of Hindu and minority Muslim borrowers and estimate the following sharp RD specification:

$$Y_{hds} = \alpha_s + \beta_1 Treat_{ds} + \beta_2 Muslim_{hds} + \beta_3 Treat_{ds} \times Muslim_{hds} + f(Runvar_{ds}) + \delta X_{hds} + \epsilon_{hds} \quad (3)$$

β_1 in specification 3 identifies the treatment effect on non-minority Hindu households, while β_3 identifies the differential treatment effect for minority Muslim households. If the directed bank credit resulted in increased credit access to Muslim households in minority concentration areas without crowding out non-minority Hindu borrowers, we would expect $\beta_1 = 0$ and $\beta_3 > 0$. The sample is restricted to households within a bandwidth of .058 on either side of the discontinuity threshold and we include all interactions of the treatment indicator, and an indicator for Muslim households with the running variable to estimate the treatment effect at the discontinuity threshold. Column (1) of Appendix Table ?? considers access to bank credit along the extensive margin and consistent with our hypothesis, the treatment effect for Hindu households (β_1) is positive but statistically non-significant, whereas the differential treatment effect for Muslim households (β_3) is positive and statistically significant at the 5 percent level. The sum of the coefficients is statistically significant at the 1 percent level, confirming an overall expansion in credit access. Column (2) includes district fixed effects and compares access to bank credit across the discontinuity threshold for Hindu and Muslim households within the same district. The $Treat \times Muslim$ coefficient remains positive, statistically significant at the 5 percent level and very similar in magnitude to column (1). Columns (5) and (6) consider bank credit access along the intensive margin and report comparable findings, confirming no significant decline in bank credit for non-minority households in treated districts, but a significant differential increase for Muslim households.³³ This rules out a reallocation of bank credit from non-minority to minority households in treated districts. As the uninteracted β_2 coefficient estimates the (non-causal) gap in bank credit access for minority and non-minority households in control areas, a comparison with the β_3 coefficient

³³ The point estimate corresponding to β_1 in column (5) is negative but statistically non-significant. β_3 is significant at the 10 percent level.

suggests that the directed credit policy fully reversed this gap along the extensive margin, and closed it 50 percent along the intensive margin.

5.1.5 Within Minority Credit Reallocation Across Districts

An alternate concern is whether credit expansions for minority households emanated from a reallocation of bank credit across districts. If lenders were unwilling to extend credit to religious minorities, they could have cutback lending to minorities in non-minority concentration areas, and reallocated bank credit to minorities in treated districts. The sharp RD strategy by comparing minority households on either side of the discontinuity threshold would be unable to distinguish this and misattribute it as a positive treatment effect.

We offer suggestive evidence on this by modifying specification (3). Specifically, we restrict our sample to control districts and define the binary indicator $Other_d$ to equal 1 for districts falling beyond a bandwidth of 10 – i.e. districts where the share of religious minorities is between 15 and 25 percent.³⁴ We next compare bank credit access for minority Muslim households across non-minority concentration districts relatively “close” to the discontinuity threshold, vis-a-vis, non-minority concentration districts relatively far from the discontinuity threshold. If banks were reallocating credit away from minorities in control districts around the discontinuity threshold, we would expect a significant decline in bank credit access for minority borrowers in control districts near the district threshold. Columns (3)-(4) and (7)-(8) of Appendix Table ?? however do not support this explanation: the $Muslim \times Other$ interaction is negative but statistically non-significant in each instance, suggesting that access to bank credit was comparable across non-minority and minority households in control districts around the discontinuity threshold. While none of the coefficients in this regression are causal, the point estimates offer suggestive evidence that lenders did not comply with the directed credit policy by reallocating bank credit between minority households across the discontinuity threshold.

5.1.6 Credit Quality

Pre-treatment descriptives in Appendix A pointed to the rationing of bank credit for Muslim households. An economic explanation would be that Muslim borrowers had higher risk profiles, and banks rationally

³⁴ As Figure 3 shows our baseline results to be stable across all bandwidths up to .1, we use this largest bandwidth to undertake this comparison.

avoided lending to this group. If lending to minority borrowers was indeed sub-optimal, we would expect increased bank loan delinquency following the directed credit policy. Using self-reported data on borrowers' repayment behavior, column (1) of Appendix Table C7 reports no increase in bank loan delinquency in minority concentration areas. This assuages concerns that the directed credit policy lead to a worsening in loan quality by pushing banks to lend to sub-optimal borrowers. It also indicates that banks' prior reluctance to lend to minority borrowers was unlikely to be explained by these borrowers' inherent riskiness. There is also no corresponding increase in loan delinquency from non-bank sources, alleviating concerns that borrowers were repaying banks by defaulting on other non-bank loans.

5.1.7 Statistical or Taste-Based Discrimination?

The Oaxaca Blinder decomposition in Appendix A.3 hinted at discrimination towards Muslim borrowers in formal credit markets. Appendix A.4 explores this in greater detail using information on lead banks across districts and offers evidence indicative of taste-based discrimination. Specifically, if statistical discrimination drove lenders' reluctance to lend to minority borrowers, learning about the quality of minority borrowers should have led to an updation of lenders' beliefs about these borrowers' creditworthiness. Since lead banks in minority concentration districts were tasked with the implementation of the directed credit policy, we would expect them to be the most likely to update their priors. In turn, conditional on information sharing across branches within the same bank group, we would expect increased bank lending to minority borrowers in non-minority concentration areas which shared a common lead bank with minority concentration districts. Appendix Table A1 however offers no such evidence, pointing to the presence of taste-based discrimination. See Appendix A.4 for additional details.

5.1.8 Loan Characteristics

The AIDIS provides information on the duration and purpose for which loans were undertaken. Appendix Table C8 identifies a statistically significant increase in minority households' likelihood of having a long-term bank loan.³⁵ Access to long-term bank credit at a relatively low cost, combined with a flexible repayment schedule, offer households the opportunity to pursue investments yielding higher returns (Field et al., 2013). Appendix Table C9 shows that the increase in bank credit for minority

³⁵ Long-term is defined as a loan in excess of one year.

households was driven by borrowings for farm and expenditure purposes. Along the intensive margin, farm loans accounted for 30 percent of the increase in bank borrowing. Disaggregating expenditure loans, Appendix Table C10 shows that the intensive margin increase in bank borrowings was driven by housing loans. We also identify positive and statistically significant treatment effects for bank credit for the purpose of health and education. There is no evidence though of minority households obtaining bank credit for non-farm businesses. Additional details are provided in Appendix C.2.2.

5.2 Labour Market Outcomes

A large literature has studied how access to credit affect labour market outcomes (Bruhn and Love, 2014; Breza and Kinan, 2021). We investigate this using weekly information on labour market activities from the nationally representative Primary Labour Force Survey, conducted in 2017-18. Column (7) of Appendix Table D2 shows minority workers in minority concentration areas to have a 6 ppt. higher likelihood of working in the manufacturing sector. Column (6) of Appendix Table D3 shows that this translated into 3 additional hours of manufacturing work in the week – a sizeable increase relative to the control group mean of 5.5 hours of manufacturing work. The increase in manufacturing work does not emanate from higher labour force participation, greater number of hours worked during the week, or a lower rate of unemployment. Instead Appendix Table D6 shows that minority workers shifted away from wage work in trade and services, and moved to self-employment in textile manufacturing. The findings resonate with Bruhn and Love (2014), who found improvements in bank credit access in Mexico allowed small business owners to continue operating their businesses.

5.3 Effects on Minority Well-Being

We use two indicators to measure the treatment effect on household well-being: household fixed assets, and monthly per capita household consumption.³⁶ For both outcomes, columns (1)-(2) of Table 4 identify positive and statistically significant treatment effects for minority households. The 13 percent increase in per capita monthly household consumption is substantial when considering the 23 percent unconditional consumption gap between minority Muslim and non-minority households. Overall household assets are also significantly larger for Muslim households in treated districts. Corresponding

³⁶ Household fixed assets are the sum of land, buildings, livestock, transport equipment, farm and non-farm machinery owned by the household.

RD plots in Figure 5, confirm a jump in both outcomes of interest at the discontinuity threshold. Columns (3) and (4) report comparable point estimates if the sample is extended to all minority households, although the point estimate corresponding to household consumption is not precisely estimated (p-value .115). Columns (5) and (6) replicate the results for non-minority households. We identify a negative point estimate for per capita monthly consumption and a sizeable positive coefficient for household assets. Both coefficients however are statistically non-significant. Overall, Table 4 shows a marked improvement in minority well-being in treated districts, particularly for Muslim households, with limited costs for non-minority households.

5.4 Mechanisms

We examine in this section three channels through which banks could have facilitated credit access for disadvantaged religious minorities.

5.4.1 Bank-SHG Linkages

The central bank’s annual guidelines on the directed credit policy urged lenders to collaborate with local self-help groups to reach eligible minority borrowers. The specific scheme is the bank-SHG linkage, whereby banks lend directly to SHGs registered with the bank. The SHG subsequently decides on the within-group allocation of credit. If minority borrowers had limited credit histories and lenders were unwilling to extend credit to such borrowers owing to high screening and monitoring costs, it is possible that such costs could have been ameliorated by lending through SHGs (Banerjee and Duflo, 2010). Banks too had an incentive to lend to SHGs as such loans qualified as priority sector credit.

Figure 4 and Table 2 separates direct lending by commercial banks to minority households, and bank lending through SHGs. Along the top row, the first two panels of Figure 4 show minority households to have higher credit access from both sources. Loan sizes from commercial banks and bank-linked SHGs were also larger in treated districts. Comparing the RD coefficients in Table 2, the treatment effects in relative terms are significantly larger for bank-linked SHGs: while 2 percent of minority households in non-minority concentration districts had outstanding credit from a bank-linked SHGs, the corresponding impact in minority concentration areas was 10 ppt. higher. Compared to Table 1, Table 2 suggests that 60 percent of the extensive margin increase in bank credit access for minority households could have been accounted for by loans issued through bank-linked SHGs. In

contrast, columns (2) and (4) of Appendix Table F2 show no evidence of higher lending to non-minorities through bank-linked SHGs. The treatment effects along both the extensive and intensive margins are an order of magnitude smaller and statistically non-significant. This rules out that higher lending to minorities through bank-linked SHGs can be attributed to an overall expansion in bank-SHG lending in treated districts.

5.4.2 Collateral Requirements

The second channel considered is collateral requirement by lenders. Pre-treatment descriptives in Appendix Figure A5 showed minority Muslim households to have lower levels of pledgeable assets in the form of land and real estate. If lenders use collateral to compensate for limited information pertaining to borrowers, the lack of collateral can exacerbate existing information frictions and distort households' access to credit (Fisman et al., 2017). While the central bank's policy statements did not recommend a relaxation of collateral requirements, banks could have independently exercised this to reach out to minority borrowers. The top-right panel in Figure 4 shows minority households to the right of the discontinuity threshold had a higher likelihood of having an unsecured bank loan. Correspondingly, column (1) of Table 3 identifies a 12 ppt. increase in minority households receiving an unsecured bank loan in treated districts, relative to a control district mean of 4 percent. Columns (2) and (3) show that this positive treatment effect is observed for loans obtained directly from commercial banks, and also loans from bank-linked SHGs. Columns (1)-(3) of Appendix Table F3 show no such impact for non-minority households: the treatment effects, while positive, are not statistically significant, and an order of magnitude smaller than those for minority households.

Fisman et al. (2017) posited that lower collateral requirements signal improved information acquisition as lenders are less reliant on costly collateral to secure loans. While it is not unlikely for SHGs to extend collateral-free loans, the increase in collateral-free lending by commercial banks to Muslim households points to an improvement in banks' capabilities to screen minority borrowers. The results cannot be explained by an overall improvement in banks' screening capabilities in minority concentration areas as non-collateralized loans to non-minority borrowers did not rise. Improvements in banks' information gathering about minority borrowers would also be consistent with the lack of a deterioration in credit quality for loans issued to such borrowers.

Broadly, these results showcase how a relaxation of terms in loan covenants can improve credit

allocation for disadvantaged borrowers. Brock and de Haas (2023) present experimental evidence of gender discrimination in Turkish credit markets, where loan officers were significantly more likely to require guarantors for female loan applicants, leading to lower take-up rates. Similarly, Table 3 suggests that relaxing collateral requirements can improve credit access for disadvantaged borrowers. This is particularly relevant when considering the large gap in pleadable assets across minority and non-minority households.

5.4.3 Cost of Credit

The final mechanism considered is the cost of credit. Similar to collateral requirements, the central bank’s policy documents do not contain any recommendations pertaining to interest rates. Nonetheless, it is possible that lenders on their part reduced lending rates for minority borrowers to facilitate the flow of credit. This would be true if banks held private information that prevailing lending rates constrained minority borrowers from accessing bank credit. Column (4) of Table 3 shows no overall reduction in the cost of bank credit in treated districts. Column (5) shows comparable rates of interest charged by commercial banks across underbanked and non-underbanked districts. On the contrary, column (6) points to a 3 ppt. increase in the cost of bank-SHG loans. This could either be in response to higher demand, or indicate that the recipients of bank-SHG loans possibly comprised of riskier borrowers. Commercial bank interest rates for non-minority borrowers remained comparable across treatment and control districts (column (5), Appendix Table F3), assuaging concerns that lenders were cross-subsidizing minority borrowers by charging higher rates to non-minority borrowers.

Collectively, Tables 2 and 3 offer two key mechanisms through which commercial banks increased lending to disadvantaged minority borrowers in minority concentration areas: collaborating with SHGs who are likely to have superior information and monitoring capabilities, and a relaxation of collateral requirements when lending directly to minority borrowers. This is similar to the findings of Fisman et al. (2017) who showed loan officers to exploit soft-information on co-ethnic/religious borrowers to expand access to bank credit for underprivileged communities, with lower collateral requirements. Interest rates however are not lower in minority concentration areas. On the contrary, increased borrowing from bank-linked SHGs by minority households in treated districts, despite higher lending rates, suggests that the cost of credit was not the primary barrier to bank credit for minority borrowers.

6 Economic Empowerment, Political Empowerment and Religious Conflict

Does the economic empowerment of religious minorities promote political empowerment and foster social harmony? We examine in this section the the second order effects of the directed credit policy.

6.1 Electoral Consequences of the Directed Credit Policy

Muslims have historically been under-represented in India’s political institutions.³⁷ Section 5.3 showed higher consumption and asset values for Muslim households in treated districts. If political participation entails entry costs, asset creation through increased credit access can alleviate such constraints. Alternately, a positive wealth effect can allow Muslim households to financially aid politicians and parties representing their interests. Economic empowerment of minorities can also boost civic engagement and economic ties between communities, particularly if access to credit led to a diversification of minority occupations, promoting economic collaboration between religious groups (Jha, 2013). If credit from bank-linked SHGs involve the formation of self-help groups with minority participants, such groups too can facilitate information sharing and co-ordination amongst voters. Overall, increased within-group interactions can genrate in greater identification with one’s group identity, which in turn can mobilize minorities to support co-religious politicians (?).

We study Muslim political participation in state legislative assembly elections to assess whether economic empowerment for minorities also leads to political empowerment. We also compare the performance of non-majoritarian parties which have been vocal about minority rights in India, as well as the right-wing Hindu nationalist Bharatiya Janata Party (BJP), whose political rhetoric is regularly laced with animosity towards religious minorities. The empirical specification is very similar to the cross-sectional RD specification in equation (2) but the unit of observation is the electoral constituency and we use data from elections conducted between 2008 and 2019. The sample is restricted to the 61 districts located within a bandwidth of .058 of the discontinuity threshold.³⁸

Compared to elections conducted in non-minority concentration areas, column (1) of Table 5 reports

³⁷ Despite a population share in excess of 12 percent, the share of Muslim legislators in the federal legislature has not exceeded 6 percent in the past three decades.

³⁸ Due to the presence of multiple electoral cycles, we include election year fixed effects, in addition to state fixed effects. We also control whether the constituency was “reserved” for candidates from marginalized caste backgrounds.

an 18 ppt. increase in the likelihood of a minority Muslim politician contesting an election to the state legislative assembly: a sizeable increase relative to the control group mean of 58 percent. Columns (3) and (4) also report significant increases in popular support for Muslim candidates.³⁹ Politicians mostly seek party nominations to contest elections; column (3) offers noisy evidence that Muslim politicians in minority concentration areas were more likely to secure nomination from a non-majoritarian party (p-val .144). Column (4) shows that higher electoral support for minority politicians was driven by higher vote shares for Muslim candidates contesting on non-majoritarian platforms – a statistically significant 4 ppt. rise, relative to a control group mean of 3 percent. Conditional on at least one such party nominating a Muslim candidate, column (6) shows that the vote share of Muslim politicians representing non-majoritarian parties increased by a statistically significant 14 ppt.

Column (2) confirms that not only are minority citizens more likely to participate electoral politics in treated districts, but these politicians also also more successful in securing nomination from political parties sympathetic to minority interests. This indicates that economic empowerment facilitated substantive participation of minority groups in competitive politics, led by serious candidates backed by major political formations. Along similar lines, column (5) assuages concerns that the rise in popular support for Muslim politicians is not a mechanical increase arising from a higher supply of Muslim candidates. As the average electoral constituency in control districts had 189,000 registered voters and turnout equaled 64 percent, a 4 ppt. increase in vote shares for Muslim candidates representing non-majoritarian parties amounts to an additional 5,200 votes received by these politicians. As the average districts witnesses 6 elections to the state legislative assembly, the district-level increase in popular support for Muslim politicians in minority concentration areas equates to over 30,000 votes.

Columns (7) and (8) show that the rise in electoral support for minority politicians is matched by lower support for the right-wing BJP. BJP vote shares fell by a statistically significant 6 ppt. in treated areas – a 25 percent drop relative to the control group mean.⁴⁰ The rise in electoral support for minority politicians contesting for non-majoritarian parties is very similar in magnitude to the loss in electoral support for the majoritarian BJP. Appendix Table G2 shows that increased popular support translated into a higher win likelihood for minority Muslim candidates. The likelihood of a Muslim candidate

³⁹ We use two measures of popular support. Column (3) compares the vote share for each Muslim contestant, while column (4) aggregates votes received across all Muslim candidates in the constituency.

⁴⁰ The point estimate in column (7) is conditional on the BJP contesting elections, and is statistically significant at the 10 percent level (p-value .094). The coefficient remains largely unchanged in magnitude, but the standard errors are larger in column (8) when we assign a vote share of 0 to constituencies in which the BJP did not contest elections.

winning the election to the state legislative assembly was 9 ppt. higher in minority concentration areas. This was driven by a 8 ppt. increase in the likelihood of minority Muslim candidates winning when nominated by non-majoritarian parties. Over a 12-year period, this amounts to an additional 47 Muslim legislators being elected from minority concentration areas.⁴¹

In summary, Table 5 and Appendix Table G2 offers strong evidence that areas witnessing the economic empowerment of religious minorities also saw heightened political representation for these groups: minority Muslim politicians are significantly more likely to contest elections, and garner a higher share of votes. The increase in popular support also increased their chances of winning, leading to greater representation of religious minorities in state legislatures. Moreover, the rise in popular support for minority candidates is matched by a decline in electoral support for the right-wing BJP. Figure 4 shows RD plots consistent with our findings in Table 5.

6.1.1 Voting Along Communal Lines?

Was the rise in electoral support for minority candidates driven by Muslim citizens voting for Muslim candidates, or were minority candidates also able to elicit support from non-minority voters? Lacking information on individual voting decisions, we assess this by exploring treatment heterogeneity across the share of Muslim voters in a constituency. In the absence of administrative data on the religious composition of the electorate, we use data compiled by Gulzar et al. (2024) on the fraction of Muslim voters in electoral constituencies. Constituencies are classified as “high” Muslim concentration if the share of registered Muslim voters exceeded the sample median.⁴² We note that in almost no constituency did Muslims voters form a plurality, making it infeasible for any politician or political party to win solely with support from this community.⁴³

Appendix Table G4 shows that while the participation of Muslim politicians in treated districts was significantly higher in both sets of constituencies, the treatment effects for Muslim politicians

⁴¹ Between 2008 and 2019, districts within our preferred bandwidth witnessed 615 (449) electoral races to the state legislative assembly in control districts. A .076 increase in the chance of victory in an individual election in minority concentration areas translates in expectation to the election of an additional 47 Muslim legislators representing non-majoritarian parties.

⁴² Gulzar et al. (2024) estimated the shares of Muslims from the list of registered voters for every polling station. This data was shared with us after aggregating the data to the level of electoral constituencies. We express our gratitude to the authors for sharing this data. The sample median is based on electoral constituencies located in districts falling within our preferred bandwidth.

⁴³ The share of Muslims amongst registered voters for the median constituency within our bandwidth equaled 14 percent. The 25th and 75th percentile of this distribution equaled 8 and 24 percent.

contesting for non-majoritarian parties were substantially larger in areas with a relatively high share of Muslim voters. Column (5) in Panel A shows that the likelihood of a Muslim (BJP) candidate winning (losing) the election was also much larger in constituencies with a relatively higher share of Muslim voters. This is consistent with minority voters co-ordinating to support minority candidates.

6.1.2 Mechanisms of Electoral Effects

We explore three non-exclusive channels to explain the electoral impacts of the directed credit policy: changes to voter turnout, wealth effects for minority politicians and social interactions amongst citizens.

6.1.2.1 Voter Registration and Turnout

We begin by identifying whether the economic empowerment of minority citizens led to increased engagement with the electoral process. If economic empowerment generated an overall sense of empowerment and deepened civic engagement, it could have boosted the turnout of minority citizens in treated areas. If the turnout of non-minority citizens remained unaffected, this would result in greater overall turnout in minority concentration areas. Relatedly, improved access to bank credit could have induced the migration of minorities to treated districts, or an out-migration of non-minorities from such areas. Such demographic shifts could have altered overall voter registration in minority concentration areas.

Columns (4) and (5) of Appendix Table G2 shows that voter turnout and the number of registered voters remained comparable across treated and control districts, while Appendix Table C15 shows no impact of the treatment on district demographics. While the RD coefficients corresponding to voter turnout and the number of registered voters are positive, they are relatively small in magnitude and statistically non-significant. This rules out that the rise in popular support for Muslim politicians emanated from greater engagement of minority voters with the electoral process.

6.1.2.2 Wealth of Muslim Candidates

Improved access to bank credit could have generated a positive wealth effect, enabling aspiring politicians to overcome entry costs for participating in electoral politics, and using their personal wealth for campaign finance. A lowering of entry barriers could also have improved candidate quality and led to higher support for such contestants. Following a ruling by India's Supreme Court in 2003, contestants to federal and state legislatures in India are mandated to disclose their assets and liabilities through sworn

affidavits, in addition to educational qualifications, occupation and pending criminal charges. We use this data to compare the wealth of Muslim politicians across minority and non-minority concentration areas. As the increase in popular support for Muslim politicians was driven by those representing non-majoritarian parties, we separately compare the characteristics of Muslim contestants in minority and non-minority concentration areas, conditional on contesting for a non-majoritarian formation.

Column (1) of Appendix Table G5 shows that Muslim politicians in treated districts reported an additional INR 4 million assets, but the standard errors are too wide to draw any strong conclusion. Restricting the sample to contestants nominated by non-majoritarian parties, we identify a statistically significant INR 17 million increase in the asset of Muslim politicians in treated districts, relative to a control group mean of INR 20 million. Column (4) shows that Muslim candidates nominated by non-majoritarian parties also had significantly higher levels of debt, but the treatment effect on liabilities is a third of the treatment effect on politician assets.⁴⁴ Subtracting liabilities from assets, column (6) shows that Muslim politicians nominated by non-majoritarian parties in minority concentration areas reported INR 21 million higher net assets. In summary, while we find limited evidence of an overall increase in the wealth of Muslim politicians, the wealth of Muslim politicians nominated by non-majoritarian parties are significantly higher in treated districts. As seen from Section 6.1, the positive treatment effects on electoral support for minority contestants was also driven by this subset of politicians. This points to the presence of a wealth effect as a mechanism explaining the increased participation of minorities in electoral politics in minority concentration areas.

Appendix Table G6 uses information from affidavits to compare the quality of minority politicians across treated and control districts. Columns (1)-(4) show minority Muslim candidates in minority concentration districts had greater experience (proxied by age) and were also more likely to have a college degree. Akin to politician assets, these treatment effects are substantially larger for minority politicians contesting on behalf of non-majoritarian parties. Across treated and control districts, minority politicians were no more or less likely to have criminal charges or convictions against them.

Overall, we find that the directed credit policy did increase the personal wealth of Muslim politicians who were nominated to contest elections by non-majoritarian parties. If politicians use personal wealth for campaign finance, it can improve candidate visibility, which would be consistent with higher votes received by these contestants. The quality of minority politicians in these districts – as measured

⁴⁴ Lacking information on the source of liabilities, we cannot ascertain whether this was driven by higher bank loans.

by college education and experience – too was significantly higher, which also could have aided their ability to garner popular support.

6.1.2.3 Social Interactions Through Self-Help Groups

The political economy literature has studied the role of social interactions in influencing both turnout and voter behaviour. In our setting, we hypothesize two channels through which the directed credit policy could have affected electoral outcomes by altering social interactions. First, by encouraging banks to lend to religious minorities through bank-linked SHGs, the directed credit policy could have fostered SHG formation, which in turn could have generated social interactions and influenced voting decisions. Second, by altering the labor market choices of minority workers, the directed credit policy could have affected citizens' social interactions in the labor market. For both the above channels, a prime question of interest is whether the directed credit policy facilitated social interactions between minorities and non-minorities, which could have mitigated societal prejudices and improved social harmony.

Beginning with the social composition of self-help groups, the National Rural Livelihood Mission (NRLM) conveys information on the gender, caste, religion and economic status for every member of a bank-registered self-help group.⁴⁵ We use this data to compute the number of Muslim members in each SHG and whether the SHG had a mix of minority and non-minority members, or comprised exclusively of minority members. As SHGs are registered with banks, we aggregate the data to the district-bank for each year, and compare SHG characteristics on either side of the discontinuity threshold. The unit of observation is the district-bank, and we augment specification (2) with bank and SHG registration year fixed effects. The empirical analysis is disaggregated over two time periods: one, the full duration between 2008 and 2019, and a shorter duration between 2014 and 2019. If SHG formation evolves over time, the latter time period allows us to test for the intensification of treatment effects.

The results are shown in Appendix Table C14. Panel A considers bank-linked SHGs formed between 2008 and 2019; Panel B restricts the sample to SHGs formed between 2014 and 2019. Over the full post-treatment period, column (1) shows that overall registration of bank-linked SHGs were comparable across treated and control areas. This again rules out concerns that minority concentration areas witnessed a secular increase in the overall registration of bank-linked SHGs. Column (2) of

⁴⁵ Members are classified as poor or non-poor.

Appendix Table C14 shows that relative to 88 Muslim members across all new SHGs registered in a year in the average bank in a control district, treated districts had an additional 78 Muslim members (across all SHGs), significant at the 10 percent level (p-value .058).

Exploring the social composition of bank-linked SHGs, column (3) reports 6 additional SHGs with a Muslim member in the average bank in treated districts, but the confidence intervals are too wide to rule out a null effect. Column (4) shows that this increase is driven entirely by the formation of SHGs with only Muslim members (p-value .057): there is no evidence of a treatment effect for the formation of bank-linked SHGs with both Muslim and non-minority members. Appendix Table C14 replicates the analysis for all minorities and confirms that the positive treatment effect on SHG group composition emanated from increased participation of Muslims in self-help groups. Panel B offer modest evidence of treatment intensification over time: an additional 117 Muslim members were part of new bank-linked SHGs in the average bank in a treated district and 8 additional bank-linked SHGs were formed with only Muslim members. Both the coefficients are now statistically significant at the 5 percent level.

Appendix Table C14 shows that the directed credit policy led to a significant increase in the number of Muslim members in bank-linked SHGs. This however was driven by self-help groups comprised exclusively of minority Muslim members, as opposed to self-help groups comprised of both minority and non-minority members. In terms of magnitudes, within our optimal bandwidth, there are 178 district-bank combinations in treated districts. Scaling the treatment effect in column (2) of Panel B, our coefficients indicate that between 2014 and 2019, an additional 21,000 Muslim members were incorporated as members in bank-linked SHGs in each year, across all treated districts. Compared to Section 6.1, the effect size is two-thirds of the treatment effect on votes received by Muslim politicians.

6.1.2.4 Social Interactions In the Labour Market

Along with SHG formation, the directed credit policy could also have affected social interactions by altering minority workers' labor market choices. Section 5.2 discussed that minority workers in treated districts were significantly more likely to work in self-employed manufacturing activities. This was accompanied by a decline in minorities engaged as wage workers in trade and services. If the directed credit policy resulted in minority workers moving to industries with a relatively high concentration of non-minority workers, this could have encouraged social interactions between communities through the labor market. We explore this hypothesis by redefining our binary labor market outcome variables as

whether a minority worker was employed in a minority or a non-minority concentration industry.⁴⁶ Appendix Table D7 offers limited evidence of increased social interactions between minorities and non-minorities through the labor market. On the contrary, we identify a shift away of minority workers from trade and service sectors dominated by non-minority workers. This is accompanied by a corresponding increase of minorities to work in manufacturing sectors with a relatively high share of minority workers.

Summing up, we find no evidence that the directed credit policy increased social interactions across minority and non-minority communities either through the creation of mixed member SHGs, or altering the labor market choices of minority workers. Instead, the evidence suggests that the directed credit policy likely improved social interactions of religious minorities through the formation of SHGs comprised solely of Muslim members, and increasing minority workers' likelihood of working in industries with a relatively high share of minority workers. If minority citizens faced higher costs of co-ordination, increased social interactions in the labor market and through exclusively minority SHGs could have improved co-ordination amongst minority voters and allowed them to vote in favour of candidates supportive of their interests, and against parties inimical to minorities. Increased social interactions with co-religionists could also have heightened the salience of religious identity for minorities, leading them to support minority Muslim candidates during elections.

6.2 Communal Violence

The final part of empirical analysis examines whether the economic and political empowerment of religious minorities affected religious conflicts. Overall, economic prosperity can reduce conflicts by increasing the opportunity cost of violence. Economic advancement of religious minorities can also mitigate conflicts through improved access to law enforcement authorities or legal resources. Alternately, Mitra and Ray (2014) and Bonomi et al. (2021) have shown that majority groups can resort to violence to assert their dominance when faced with minority economic empowerment. Additionally, the formation of exclusively Muslim self-helped groups in minority concentration areas could have improved in-group cohesion amongst minorities, and accentuated religious violence (Mitra and Ray, 2014). In light of the competing possibilities, the impact of economic and political empowerment of

⁴⁶ We use data from the nationally representative employment-unemployment survey conducted by the NSS in 2011-12. For each 2-digit industry, we compute the share of Hindu and Muslim workers in each industry. Industries with an above median share of minority (non-minority) workers are deemed to be minority (non-minority) concentration industries.

minority groups on religious violence is ex-ante ambiguous.

We begin by comparing religious conflicts across minority and non-minority concentration districts using administrative data from the National Crime Records Bureau (NCRB), which since 2014 has reported the number of religious riots occurring in each district. Scaling religious conflicts by district population, column (1) of Table 6 shows that minority concentration areas witnessed 3 additional incidences (per million population) of religious riots in a year. Column (2) considers raw incidences of religious riots and reports 5 additional religious conflicts per year. Both coefficients are statistically significant at the 1 percent level, and reflect a doubling in the incidence of religious conflicts relative to non-minority concentration areas. Columns (3)-(6) affirm that only religious riots increased in minority concentration areas: non-religious riots and murders remained comparable across treated and control areas, alleviating concerns that the rise in religious conflicts emanated from an overall increase in local violence in treated areas. Columns (1) and (2) of Appendix Table H3 reports very similar results if we use the ACLED data as opposed to administrative records. Religious conflicts were significantly higher in minority concentration areas in both per capita and overall terms. In contrast, non-religious riots remained unaffected across treated and control areas.

The detailed description of each event in the ACLED database permit us to distinguish riots based on the religious identity of the instigating group. Columns (4) and (5) of Appendix Table H3 report a statistically significant increase in Hindu-instigated religious violence in minority concentration areas. Scaling the treatment effects on Hindu-instigated riots in column (5) with the overall increase in Hindu-Muslim violence in column (2), two-thirds of the increase in religious violence in treated districts can be attributed to Hindu groups. This alleviates concerns that the increase in religious violence was an upshot of increased assertion of Muslim groups as a result of economic and political empowerment.⁴⁷ Overall, our treatment effects are in line with the theoretical predictions of Mitra and Ray (2014).

6.2.1 Mechanisms of Conflict

We conclude our empirical analysis by teasing out the mechanisms of religious conflicts. Higher incidence of religious conflicts suggests that while the directed credit policy promoted economic well being for religious minorities, it also led to a worsening of social harmony by eliciting a violent response

⁴⁷ There are but a handful of religious conflicts in the ACLED perpetrated by Muslims. This precludes our ability to estimate a RD coefficient for this subset of conflicts.

from the majority group. As these districts witnessed both the economic and political empowerment of religious minorities, the source of conflict can emerge from relative improvements in minority well-being, as theorized by Mitra and Ray (2014), or a majoritarian response to the political assertion of minorities. In the Indian context, (Iyer and Shrivastava, 2018) showed that religious violence yields electoral dividends to right-wing majoritarian forces. Alternately, increased violence can emanate as a blowback to minority assertion following the elections.

6.2.2 Electoral Politics and Religious Conflicts

We begin by exploring whether the timing of religious conflicts is linked to the incidence of elections. We hypothesize that if economic convergence across minorities and non-minorities fueled religious conflicts, the timing of conflicts should be orthogonal to the timing of elections. Every year, a select number of states in India witness elections to the state legislature, with the precise timing of the elections varying by state.⁴⁸ As the administrative data from the NCRB is aggregated by year, we are unable to use this to precisely ascertain the timing of religious conflicts, and rely instead on the ACLED data, which provides the event date for each reported conflict.

Using the ACLED event dates, we split our sample into religious conflicts occurring prior to the elections, and religious conflicts occurring within 1 year of the elections being concluded. Columns (1)-(3) of Appendix Table H4 shows that the rise in religious conflicts in minority concentration areas was driven by conflicts occurring in the year following the elections. While the point estimate in column (1) is positive and statistically significant (p-value .059), column (2) shows conflicts in other years to be comparable across treated and control districts. Column (3) specifically focuses on religious conflicts in the year prior to elections and reports statistically indistinguishable levels of religious conflicts across minority and non-minority concentration areas.

Columns (1)-(3) allows us to make two inferences. First, it is unlikely that the economic channel posited by Mitra and Ray (2014) were driving the results: if the increase in religious violence emanated in response to a lowering of the consumption gap between Hindus and Muslims, we would have expected conflicts to be unrelated to the timing of the elections. Second, the lack of an increase in religious conflict prior to the elections suggest that political actors were not using violence as a tool of voter

⁴⁸ For a given year, multiple states can go to elections at different times. For instance, in 2014 and 2019, the states of Andhra Pradesh, Orissa and Arunachal Pradesh conducted elections in the month of May; the states of Haryana, Maharashtra and Jharkhand conducted elections in the months of October and November.

intimidation. This leaves us with our third channel; that heightened religious conflicts in minority concentration areas following elections represents a “backlash” to rising political assertion of religious minorities, which results in a larger number of Muslim politicians being elected to the state legislative assembly, and reduces electoral support and the chances of winning for the right-wing BJP.

6.2.3 Location of Violence

Section 6.1.1 noted that the treatment effects on electoral support for minority Muslim candidates was substantially larger in constituencies with a relatively high share of Muslim voters. The right-wing BJP was also less likely to win elections in such constituencies. If religious violence emanated as a right-wing backlash against minority political empowerment, we would expect this to be concentrated in areas with a relatively high share of Muslim voters. We test this by mapping event locations in the ACLED database to electoral constituencies. Using the data from Gulzar et al. (2024), columns (4) and (5) of Appendix Table H4 splits our sample into constituencies with high and low Muslim voters and compares the incidence of Hindu-Muslim violence from the ACLED across treated and control districts. Consistent with the hypothesis that the rise in religious violence was a backlash to minority political empowerment, column (4) shows the increase in religious conflicts was entirely driven by areas with a relatively high share of Muslim voters.

6.2.4

Second, we can test for strategic use of violence in relation to elections using information in the ACLED data on the timing of the violent events. For each assembly constituency falling in the sample of districts within our preferred bandwidth, we identify Hindu-Muslim riots at monthly levels. We split the data into 12-months before and 12-months after election time periods, and run separate analysis using our regression discontinuity design specification. The analysis eliminates the month of election from the data as we cannot identify if the violent event happened before or after the election with certainty in the month of the election. Columns 2 and 3 of Appendix Table ??, reports the effects of credit policy on the number of Hindu-Muslim riots separately for twelve months before and twelve months after the elections. We find no evidence for increase in violence before the elections, but we do find that violence spiked after the elections in minority concentration districts. This pattern of violence suggests that it was strategically used as a response to the electoral effects.

In order to further confirm that the violence was indeed due to electoral effects, we probe if violent events are concentrated in places where the right-wing party lost electorally. For this analysis we divide the sample of districts based on whether the main right-wing party won most of the races or not, and separately report the effect of policy on violent events using the same RD specification. Column 4 of Appendix Table ?? reports there is no effect on violent incidents in districts where the right-wing party won most of the races (more than 50%), however, Column 5 shows that Hindu-Muslim riots increase in post election months in districts where the right-wing party did not win majority of the races, further confirming that the negative effect on social harmony in the society is a blowback for the electoral empowerment of the minority group.

7 Conclusion

Many polarized countries have large minorities that are marginalized economically and politically. If governments want to improve the economic well-being of these minorities, increasing access to credit is one important policy lever available to them. This paper shows that contrary to the expectation, such a policy can be successful at expanding credit and improving well-being without a cost to the majority group. However, there are important considerations that are the key mechanisms for the success of the program. The policy has to be designed to work in the context of the institutional setup of the country. As the paper shows, Indian banks were asked to work with Self Help Groups of citizens which reduced the likely discrimination the minority citizens faced in the status quo when dealing with banks individually and also reduced the costs of screening that banks had to otherwise incur to extend credit to the minority borrowers. Further, banks have to be flexible in their requirement of collateral if they are to get more clients from the marginalized sections of society.

While such policies targeting the economic well-being of minorities are not explicitly designed to politically empower minorities, it is natural that with better economic conditions, minorities are more likely to claim a stake in politics. But this does come at the cost of violence. While it is possible that violence may be used as a tool to keep minorities out of politics, the paper sheds light on a subtle use of violence: as a punishment after the election for minorities becoming more active politically.

Credit policies are commonly used as a means to help people out of poverty. This paper opens a new avenue for research by highlighting that political empowerment may be a consequence of economic empowerment. But since this comes at the cost of violence, it is not clear if, in the long run, the

well-being effect may persist. Even though this paper uncovers the positive effects of directed credit policy on economic well-being and political empowerment ten years after the policy was enacted, it will still be important to uncover if the policy managed to reduce inequality in the long run.

References

- ALESINA, A. F., E. L. GLAESER, AND B. SACERDOTE (2001): “Why doesn’t the US have a European-style welfare system?” Tech. rep., National bureau of economic research Cambridge, Mass., USA.
- ALLIE, F. (2024): “The Representation Trap: How and Why Muslims Struggle to Maintain Power in India,” *Working Paper*.
- ARCIDIACONO, P., J. KINSLER, AND T. RANSOM (2022): “What the Students for Fair Admissions Cases Reveal About Racial Preferences,” *Journal of Political Economy Microeconomics*.
- ATKIN, D., E. COLSON-SIHRA, AND S. MOSES (2021): “How Do We Choose Our Identity? A Revealed Preference Approach Using Food Consumption,” *Journal of Political Economy*, 129, 1193–1251.
- AUGSBURB, B., R. DE HAAS, H. HARMGART, AND C. MEGHIR (2015): “The Impacts of Micro-Credit: Evidence from Bosnia and Herzegovina,” *American Economic Journal: Applied Economics*, 7, 183–203.
- AVENANCIO-LEÓN, C. F., T. HOWARD, AND W. MULLINS (2024): “The Electoral Persistence of ‘Welfare Queen’ Politics: Dissecting the Racial Politicization of Public Policies,” .
- AYDIN, D. (2022): “Consumption Responses to Credit Expansions: Evidence from Experimental Assignment of 45, 307 Credit Lines,” *American Economic Review*, 112, 1–40.
- BANERJEE, A., S. COLE, AND E. DUFLO (2004): “Banking Reform in India,” in *The India Policy Forum: Volume 1*, ed. by S. Bery, B. Bosworth, and A. Panagariya, New Delhi: Brookings Institution Press, 277–332.
- BANERJEE, ABHIJIT, V. AND E. DUFLO (2010): “Giving Credit Where it is Due,” *Journal of Economic Perspectives*, 24, 61–80.
- (2014): “Do Firms Want to Borrow More? Testing Credit Constraints Using a Directed Lending Program,” *The Review of Economic Studies*, 81, 572–607.
- BARROW, L., L. SARTAIN, AND M. DE LA TORRE (2020): “Increasing access to selective high schools through place-based affirmative action: Unintended consequences,” *American Economic Journal: Applied Economics*, 12, 135–163.
- BESLEY, T., O. FOLKE, T. PERSSON, AND J. RICKNE (2017): “Gender quotas and the crisis of the mediocre man: Theory and evidence from Sweden,” *American economic review*, 107, 2204–2242.
- BHALOTRA, S., I. CLOTS-FIGUERAS, G. CASSAN, AND L. IYER (2014): “Religion, politician identity and development outcomes: Evidence from India,” *Journal of Economic Behavior & Organization*, 104, 4–17.
- BHAVNANI, R. R. (2017): “Do the effects of temporary ethnic group quotas persist? Evidence from India,” *American Economic Journal: Applied Economics*, 9, 105–123.
- BLANCHFLOWER, D. G., B. LEVINE, PHILLIP, AND J. ZIMMERMAN, DAVID (2003): “Do Firms Want to Borrow More? Testing Credit Constraints Using a Directed Lending Program,” *The Review of Economics and Statistics*, 85, 930–943.
- BONOMI, G., N. GENNAIOLI, AND G. TABELLINI (2021): “Identity, beliefs, and political conflict,” *The Quarterly Journal of Economics*, 136, 2371–2411.

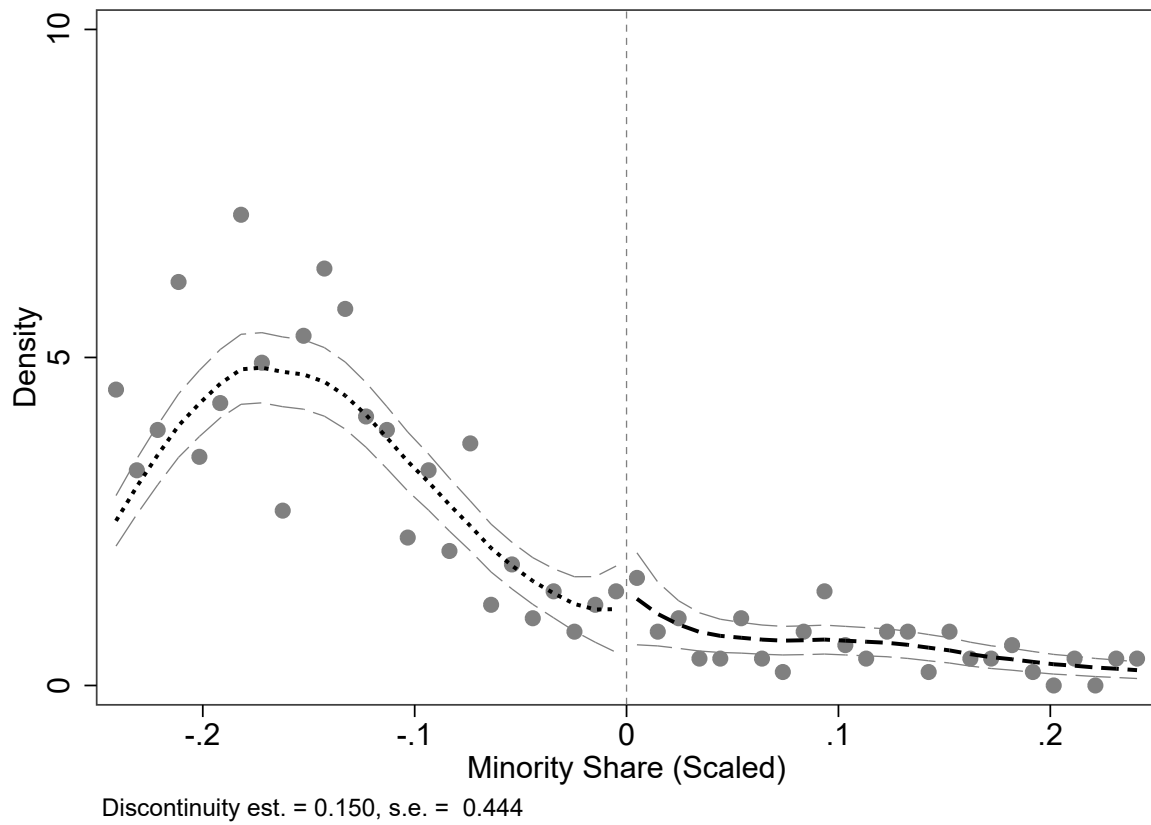
- BREZA, E. AND C. KINAN (2021): “Measuring the Equilibrium Impacts of Credit: Evidence from the Indian Microfinance Crisis,” *The Quarterly Journal of Economics*, 136, 1447–1497.
- BREZA, E. AND C. KINNAN (2021): “Measuring the Equilibrium Impacts of Credit: Evidence from the Indian Microfinance Crisis,” *The Quarterly Journal of Economics*, 136, 1447–1497.
- BROCK, MICHELLE, J. AND R. DE HAAS (2023): “Discriminatory Lending: Evidence from Bankers in the Lab,” *American Economic Journal: Applied Economics*, 15, 31–68.
- BRUHN, M. AND I. LOVE (2014): “The Real Impact of Improved Access to Finance: Evidence from Mexico,” *The Journal of Finance*, 69, 1347–1376.
- BUERA, FRANCISCO, J., P. KABOSKI, JOSEPH, AND Y. SHIN (2020): “The Macroeconomics of Microfinance,” *The Review of Economic Studies*, 88, 126–161.
- BUSTOS, P., B. CAPRETTINI, AND J. PONTICELLI (2016): “Agricultural Productivity and Structural Transformation: Evidence from Brazil,” *American Economic Review*, 106, 1320–1365.
- CALONICO, S., M. D. CATTANEO, AND M. H. FARRELL (2020): “Optimal bandwidth choice for robust bias-corrected inference in regression discontinuity designs,” *The Econometrics Journal*, 23, 192–210.
- CHATTOPADHYAY, R. AND E. DUFLO (2004): “Women As Policy Makers: Evidence From a Randomized Policy Experiment in India,” *Econometrica*, 72, 1409–1443.
- CHATURVEDI, R. AND S. CHATURVEDI (2024): “It’s all in the name: A character-based approach to infer religion,” *Political Analysis*, 32, 34–49.
- CHETTY, R., N. HENDREN, AND L. F. KATZ (2016): “The effects of exposure to better neighborhoods on children: New evidence from the moving to opportunity experiment,” *American Economic Review*, 106, 855–902.
- CHOI, J., I. KUZIEMKO, E. WASHINGTON, AND G. WRIGHT (2024): “Local economic and political effects of trade deals: Evidence from NAFTA,” *American Economic Review*, 114, 1540–1575.
- CHYN, E., K. HAGGAG, AND B. A. STUART (2024): “Inequality and Racial Backlash: Evidence from the Reconstruction Era and the Freedmen’s Bureau,” .
- CONDRA, L. N., J. D. LONG, A. C. SHAVER, AND A. L. WRIGHT (2018): “The logic of insurgent electoral violence,” *American Economic Review*, 108, 3199–3231.
- CRAMER, KIM, F. (2021): “Bank Presence and Health,” .
- DAS, S. (2024): “Democratic Backsliding in the World’s Largest Democracy,” *Unpublished Working Paper*.
- EMERICK, K. (2018): “Agricultural productivity and the sectoral reallocation of labor in rural India,” *Journal of Development Economics*, 135, 488–503.
- FIELD, E., R. PANDE, J. PAPP, AND N. RIGOL (2013): “Does the Classic Microfinance Model Discourage Entrepreneurship Among the Poor? Experimental Evidence from India,” *American Economic Review*, 103, 2196–2226.
- FISMAN, R., D. PARAVISINI, AND V. VIG (2017): “Cultural Proximity and Loan Outcomes,” *American Economic Review*, 107, 457–492.

- FISMAN, R., A. SARKAR, J. SKRASTINS, AND V. VIG (2020): “Experience of Communal Conflict and Inter-Group Lending,” *Journal of Political Economy*, 128, 3346–3375.
- GULZAR, S., F. ALLIE, V. ENSSLE, T. GOYAL, AND G. PATHAN (2024): “Increasing Polarization of Hindu-Muslim Identity in India,” *Working Paper*.
- GULZAR, S., A. LAL, AND B. PASQUALE (2021): “Representation and Forest Conservation: Evidence from India’s Scheduled Areas,” Tech. rep., Working Paper. SocArXiv.
- GUPTA, N. AND R. DEHEJIA (2021): “Financial Development and Micro-Entrepreneurship,” *Journal of Financial and Quantitative Analysis*, 57, 1834–1861.
- HOLZER, H. AND D. NEUMARK (2000): “Assessing affirmative action,” *Journal of Economic Literature*, 38, 483–568.
- IYER, S. (2018): *The economics of religion in India*, Harvard University Press.
- IYER, S. AND A. SHRIVASTAVA (2018): “Religious riots and electoral politics in India,” *Journal of Development Economics*, 131, 104–122.
- JAFFRELOT, C. (2021): *Modi’s India: Hindu Nationalism and the Rise of Ethnic Democracy*, Princeton University Press.
- JENSENIUS, F. R. (2015): “Development from representation? A study of quotas for the scheduled castes in India,” *American Economic Journal: Applied Economics*, 7, 196–220.
- JHA, S. (2013): “Trade, institutions, and ethnic tolerance: Evidence from South Asia,” *American political Science review*, 107, 806–832.
- KABOSKI, JOSEPH, P. AND M. TOWNSEND, ROBERT (2012): “The Impact of Credit on Village Economies,” *American Economic Journal: Applied Economics*, 4, 98–133.
- LEE, D. S. AND T. LEMIEUX (2010): “Regression discontinuity designs in economics,” *Journal of Economic Literature*, 48, 281–355.
- LEONARD, J. S. (1990): “The impact of affirmative action regulation and equal employment law on black employment,” *Journal of Economic Perspectives*, 4, 47–63.
- MAIZLAND, L. (2020): “India’s Muslims: An increasingly marginalized population,” *Council on Foreign Relations*, 20, 21.
- MARION, J. (2009): “How costly is affirmative action? Government contracting and California’s Proposition 209,” *The Review of Economics and Statistics*, 91, 503–522.
- MARSHALL, T. H. (1950): *Citizenship and social class*.
- MCCRARY, J. (2008): “Manipulation of the running variable in the regression discontinuity design: A density test,” *Journal of Econometrics*, 142, 698–714.
- MILLER, C. (2017): “The persistent effect of temporary affirmative action,” *American Economic Journal: Applied Economics*, 9, 152–190.
- MITRA, A. AND D. RAY (2014): “Implications of an economic theory of conflict: Hindu-Muslim violence in India,” *Journal of Political Economy*, 122, 719–765.

- PANDE, R. (2003): “Can mandated political representation increase policy influence for disadvantaged minorities? Theory and evidence from India,” *American economic review*, 93, 1132–1151.
- RALEIGH, C., R. LINKE, H. HEGRE, AND J. KARLSEN (2010): “Introducing ACLED: An armed conflict location and event dataset,” *Journal of peace research*, 47, 651–660.
- RBI (2007): *Master Circular – Priority Sector Lending – Credit Facilities to Minority Communities*, Reserve Bank of India, Mumbai, India.
- (2020): *Master Directions – Priority Sector Lending (PSL) – Targets and Classification*, Reserve Bank of India, Mumbai, India.
- (2021): *Master Circular – Credit Facilities to Minority Communities*, Reserve Bank of India, Mumbai, India.
- SACHAR, R. (2006): “Social, Economic, and Educational Status of the Muslim Community in India,” Tech. rep., Cabinet Secretariat, Government of India.
- VARSHNEY, A. (2003): *Ethnic conflict and civic life: Hindus and Muslims in India*, Yale University Press.
- WILKINSON, S. (2006): *Votes and violence: Electoral competition and ethnic riots in India*, Cambridge University Press.

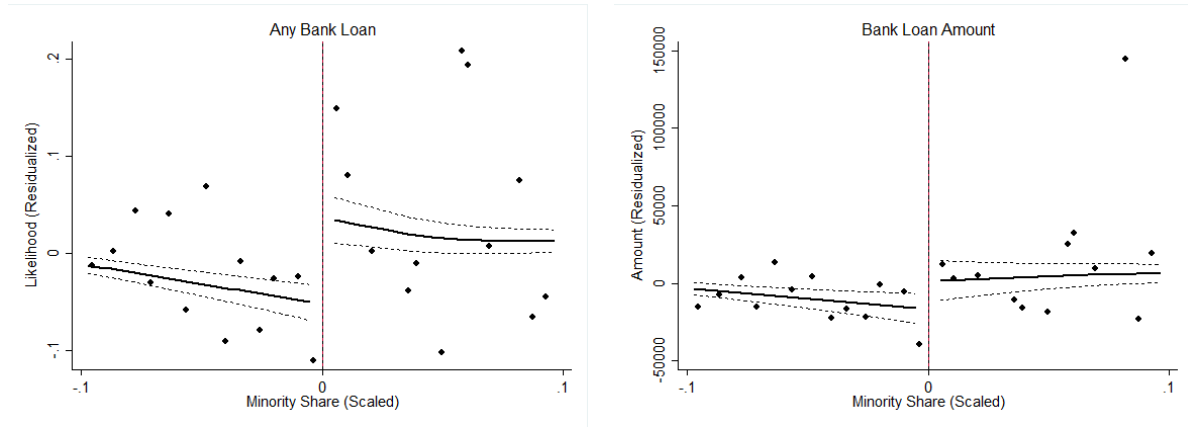
8 Figures

Figure 1: Smoothness in Treatment Assignment at Discontinuity Threshold



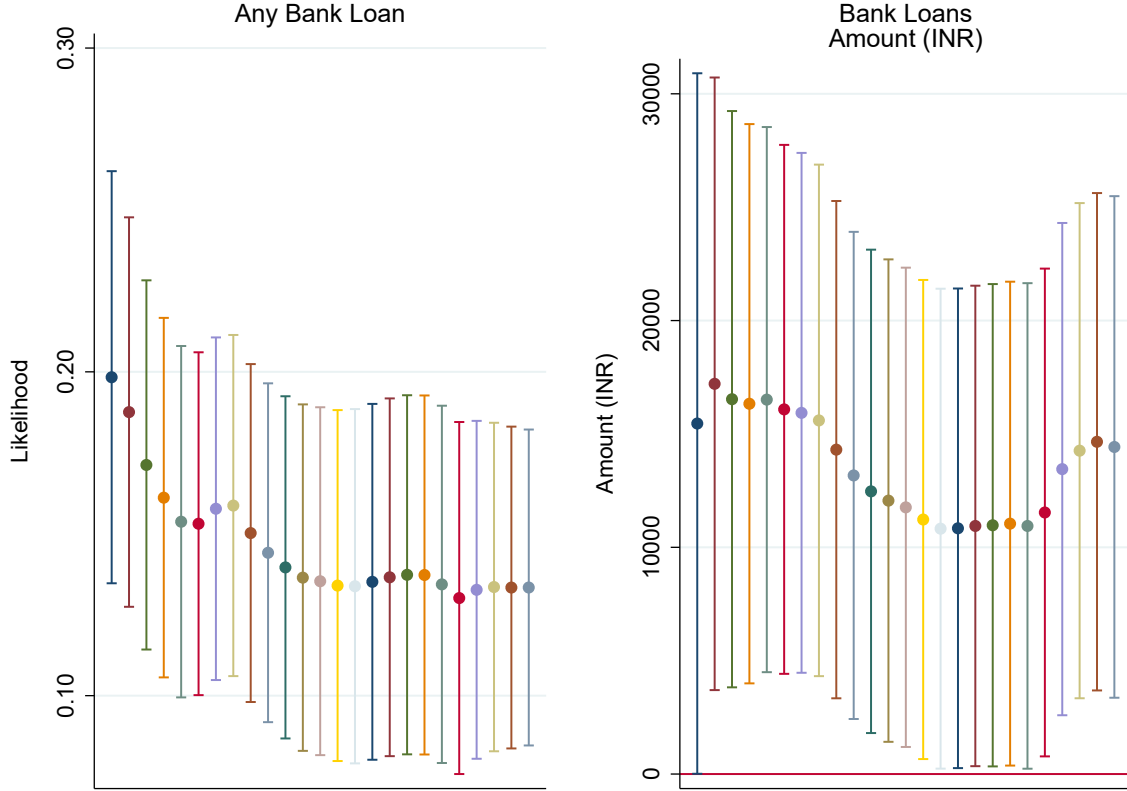
Notes: This figure presents the McCrary test (McCrary, 2008), testing for selective sorting of districts as minority concentration around the 0.25 threshold. The horizontal axis depicts the running variable – defined in equation (1). The horizontal axis shows the range of the running variable between -0.25 and 0.25. The discontinuity threshold of 0 is indicated by the broken vertical line. The discontinuity estimate and standard error is depicted below the figure.

Figure 2: Bank Credit for Minority Households



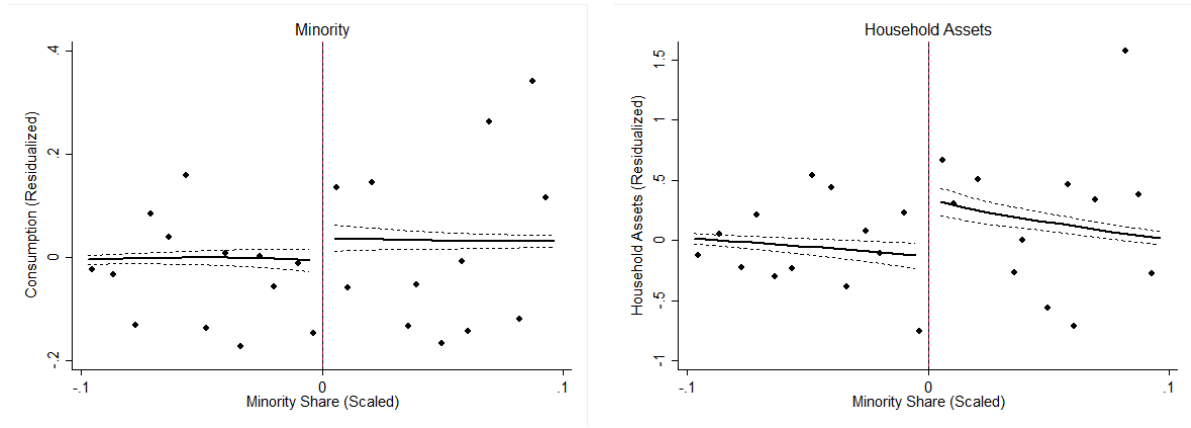
The above figures show the treatment effect for access to bank credit for minority households. The unit of observation is the household. The outcome of interest in the left-panel is a dummy equaling 1 if the household has any bank loan; in the right-panel, the volume of bank loans (initial loan value). The outcome variable is residualized, conditional on state fixed effects and household covariates. The solid dots show residualized means of the outcome variables, corresponding to each bin of the running variable. The horizontal lines show the linear fit from a local linear regression, and the dashed lines show 90% confidence intervals. Estimates are weighted using a triangular kernel and household specific weights provided by the AIDIS.

Figure 3: Bank Credit for Minority Households: Robustness to Alternate Bandwidths



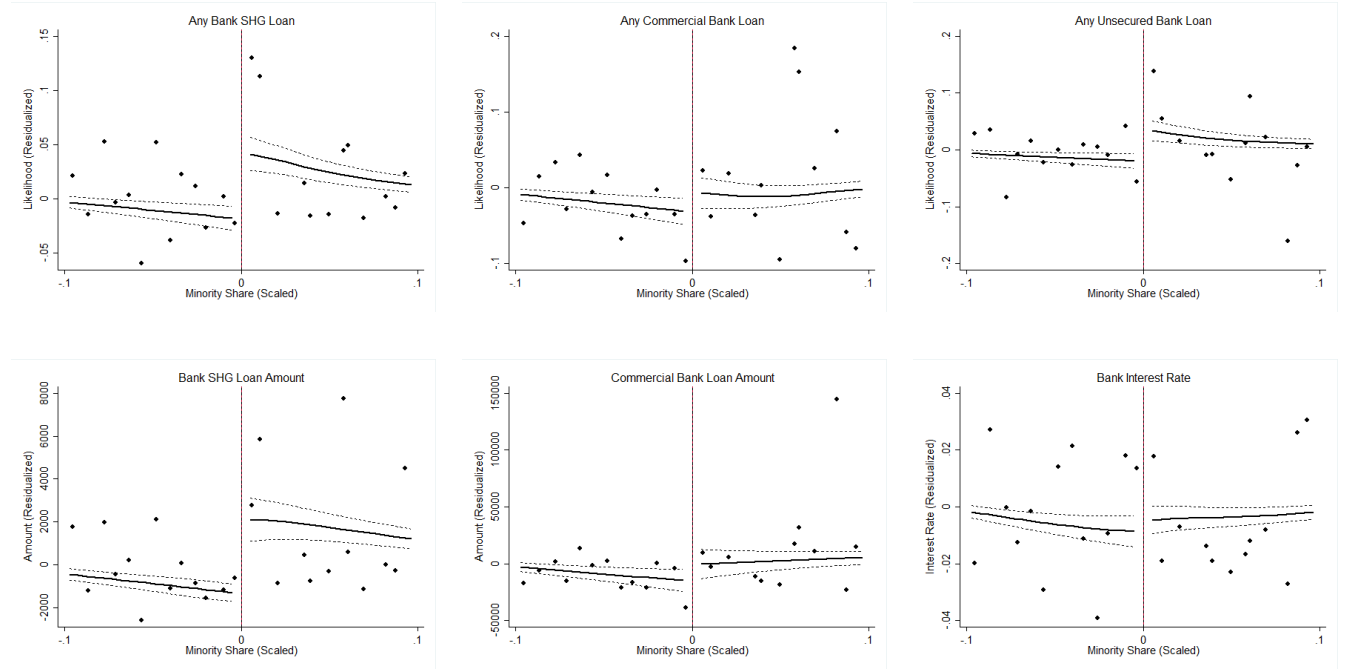
Notes: This figure shows the robustness of the baseline results to alternate bandwidths. The unit of observation is the household. *Minority Share (Scaled)* denotes the running variable, defined in equation (1). The outcome of interest in the left-panel is a dummy equaling 1 if the household has any bank loan; in the right-panel, the volume of bank loans (initial loan value). The vertical lines denote 95% confidence intervals of the coefficient estimates. The first coefficient is computed using a bandwidth of .04. Subsequent estimations sequentially increase the bandwidth by 0.003. The last point estimate is based on a bandwidth of .10. All specifications include state fixed effects, household covariates and a linear polynomial in the running variable. Standard errors are clustered by district. Estimates are weighted using a triangular kernel and household specific weights provided by the AIDIS.

Figure 5: Consumption and Assets for Minority Households



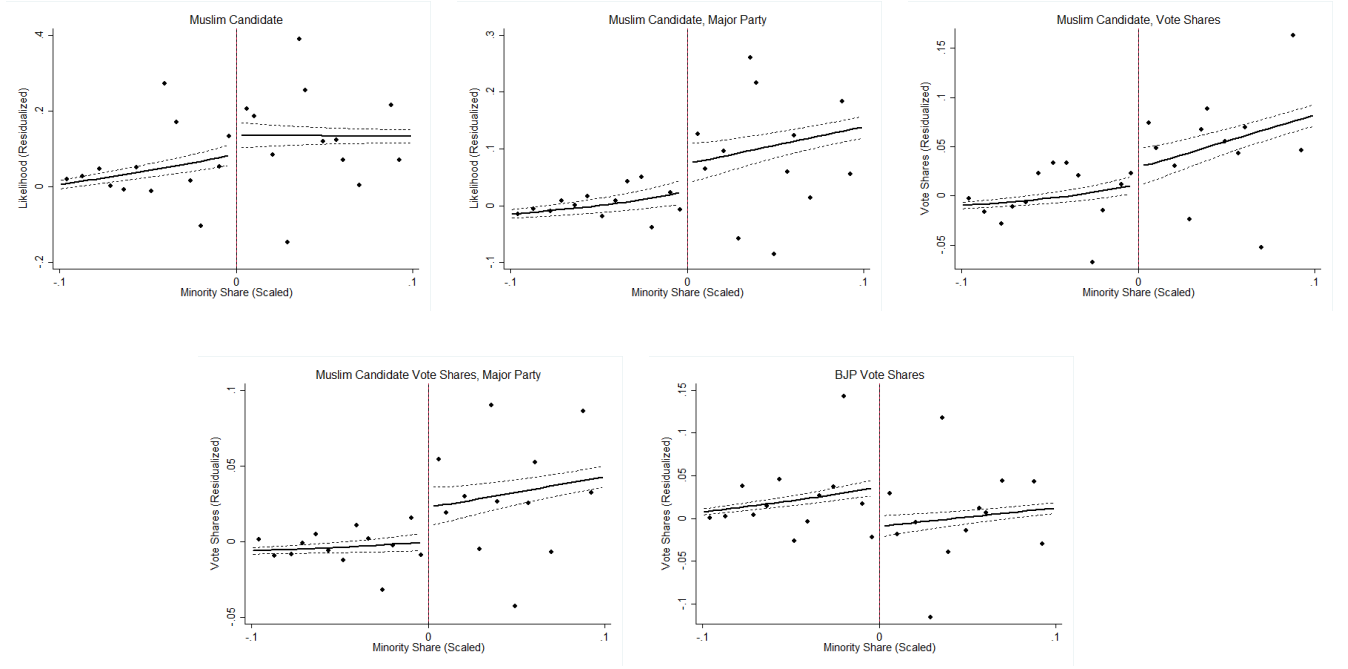
The above figures show the treatment effect for household consumption and household assets for minority households. The unit of observation is the household. The outcome of interest in the left-panel is logged per capita monthly household consumption; in the right-panel, logged household assets. The outcome variable is residualized, conditional on state fixed effects and household covariates. The solid dots show residualized means of the outcome variables, corresponding to each bin of the running variable. The horizontal lines show the linear fit from a local linear regression, and the dashed lines show 90% confidence intervals. Estimates are weighted using a triangular kernel and household specific weights provided by the AIDIS. Household assets include value of real estate, livestock, fixed assets, farm and non-farm machinery, and transport.

Figure 4: Mechanisms: Source of Bank Credit, Loan Securitization and Cost of Credit for Minority Households



Notes: The above figure shows RD plots corresponding to the three mechanisms examined in the paper. The unit of observation is the household. *Minority Share (Scaled)* denotes the running variable, defined in equation (1). The outcome of interest in the top-left panel is a dummy equaling 1 if the household has a loan from a bank-linked SHG; in the top-middle panel, a dummy equaling 1 if the household has a loan from a commercial bank; in the top-right panel, a dummy equal to 1 if the household has an unsecured bank loan; in the bottom-left panel, the amount of loans from bank-linked SHGs; in the bottom-middle panel, the amount of loans from commercial banks; in the bottom-right panel, bank interest rates. All outcome variables are residualized, conditional on state fixed effects and household covariates. The solid dots show residualized means of the outcome variables, corresponding to each bin of the running variable. The horizontal lines show the linear fit from a local linear regression, and the dashed lines show 90% confidence intervals. Estimates are weighted using a triangular kernel and household specific weights provided by the AIDIS.

Figure 4: Electoral Impacts of the Directed Credit Policy



Notes: The above figure shows RD plots showing the electoral impacts of the directed credit policy. The outcome of interest in the top-left panel is a dummy equaling 1 if there is a Muslim candidate contesting the election; in the top-middle panel, if a Muslim candidate contests the election from a major non-BJP party; in the top-right panel, the total votes received by Muslim candidates, as a share of votes cast in the constituency; in the bottom-left panel, votes received by Muslim candidates contesting on behalf of major non-BJP parties; in the bottom-right panel, vote share of the right-wing Hindu nationalist BJP. All outcome variables are residualized, conditional on state and year fixed effects. The solid dots show residualized means of the outcome variables, corresponding to each bin of the running variable. The horizontal lines show the linear fit from a local linear regression, and the dashed lines show 90% confidence intervals. Estimates are weighted using a triangular kernel and the number of registered voters in the constituency.

9 Tables

Table 1: Treatment Effect on Bank Credit: Minority Households

	(1)	(2)	(3)	(4)	(5)
	Pr(Loan = 1)			Loan Amount (INR)	
Treat	.120*** (.036)	.197*** (.033)	.178*** (.031)	20089.469*** (7630.758)	21064.418*** (6328.599)
Observations	1505	1378	1765	1930	1765
Dep Var Mean	.131	.128	.115	20344.221	20801.635
Bandwidth	.0547	.0469	.0580	.0634	.0580
Controls	N	Y	Y	N	Y

Notes: The above table shows the treatment effect on bank credit access for minority households. The unit of observation is the household. The outcome of interest in columns (1)-(3) is a dummy equaling 1 if the household had any outstanding bank loan; in columns (4)-(5), the amount of bank loans received by the household. All specifications include state fixed effects and a linear polynomial in the running variable and its interaction with the treatment indicator; columns (2), (3) and (5) also control for household size and a rural indicator. The bandwidth in columns (1), (2), (4) and (5) is the MSERD optimal bandwidth, computed as recommended by Calonico et al. (2020). In column (3), the sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table 2: Mechanism: Nature of Bank Lending in Minority Concentration Districts

	(1)	(2)	(3)	(4)
	Pr(Loan = 1)		Loan Amount (INR)	
	Commercial Bank	Bank SHG	Commercial Bank	Bank SHG
Treat	.096*** (.023)	.102*** (.018)	17107.647*** (6012.407)	3941.329*** (1310.464)
Observations	1765	1765	1765	1765
Dep Var Mean	.096	.024	20046.831	754.804

Notes: The above table shows the treatment effect by type of bank loan for minority households. The unit of observation is the household. The outcome of interest in columns (1)-(2) is a dummy equaling 1 if the household has any outstanding loan from the source mentioned; in columns (3)-(4), the amount of outstanding loan from the source mentioned. *Commercial Bank* refers to loans obtained directly from commercial banks; *Bank SHG* refers to loans received through bank-linked SHGs. All specifications include state fixed effects, a linear polynomial in the running variable, household size, and a rural indicator. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table 3: Mechanisms: Treatment Effects on Collateral Requirements and Cost of Credit

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Unsecured Loan = 1)			Interest Rates		
	All Bank	Commercial Banks	Bank SHG	All Bank	Commercial Banks	Bank SHG
Treat	.118*** (.044)	.051** (.021)	.087*** (.033)	.017 (.017)	-.001 (.011)	.027** (.012)
Observations	1765	1765	1765	473	383	98
Dep Var Mean	.044	.035	.044	.108	.107	.108

Notes: The above table shows the treatment effect on bank loan securitization and interest rates for minority households. The unit of observation is the household. The outcome of interest columns (1)-(3) is a dummy equaling 1 if the household has any unsecured bank loan; in columns (4)-(6), the interest rate faced by the household for bank loans. The outcome of interest in columns (1) and (4) include all bank loans; in columns (2) and (5), loans obtained directly from commercial banks; in columns (3) and (6), loans obtained from bank-linked SHGs. Average household interest rates are weighted by loan volume. All specifications include state fixed effects, a linear polynomial in the running variable, household size, and a rural indicator. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table 4: Minority Credit Access, Household Consumption and Household Assets

	(1)	(2)	(3)	(4)	(5)	(6)
	Muslims		Minorities		Non-Minority	
	Consumption	Assets	Consumption	Assets	Consumption	Assets
Treat	.130*** (.047)	.916*** (.269)	.110 (.069)	.692*** (.188)	-.111 (.071)	.401 (.253)
Observations	1765	1674	2480	2359	9763	9210
Dep Var Mean	2524.275	1092694	2654.092	1140989	3295.129	1637987

Notes: The above table shows the treatment effect on household consumption and assets. The unit of observation is the household. Columns (1) and (2) restrict the sample to Muslim households; columns (3) and (4) restricts the sample to all religious minorities; columns (5) and (6) restrict the sample to non-minority households. The outcome of interest in the odd-numbered columns is monthly per capita household consumption; in even-numbered columns, total household assets. Household assets are defined as the sum of land, buildings, livestock, farm and non-farm machinery, and transportation assets. All specifications include state fixed effects and household covariates. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using household-specific weights and a triangular kernel. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table 5: Directed Credit Policy and Electoral Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Minority Candidates						Right Wing Party	
	Any Muslim	Any Muslim Major Party	Vote Share	Vote Share	Vote Share Major Party	Vote Share Major Party	Vote Share	Vote Share
Treat	.178*** (.058)	.081 (.056)	.028*** (.008)	.043** (.020)	.043** (.018)	.139*** (.050)	-.059* (.032)	-.051 (.050)
Observations	1064	1064	1672	1064	1064	190	876	1064
Dep Var Mean	.581	.146	.050	.084	.028	.190	.329	.291

Notes: This table identifies the impact of the directed credit policy on electoral outcomes for politicians from minority communities and those representing the right-wing party. The unit of observation is the electoral constituency. The outcome of interest in column (1) is a dummy equaling 1 if there was a Muslim candidate contesting the election; in column (2), a dummy equaling 1 if there was a Muslim candidate contesting the election on behalf of a major party; in column (3), the vote share received by individual Muslim candidates; in column (4), the total vote share received by all Muslim candidates; in column (5), vote share received by Muslim candidates contesting on a major party ticket; in column (6), vote share received by Muslim candidates contesting on a major party ticket, conditional on a Muslim candidate contesting on a major party ticket; in column (7), the vote share received by the BJP candidate, conditional on participation; in column (8), the vote share received by the BJP candidate. Vote share is the number of votes received by a candidate, divided by the total votes cast in the constituency. All specifications include state and electoral year fixed effects, in addition to a linear polynomial in the running variable. All specifications restrict the sample to a bandwidth of .058 around the discontinuity threshold, and are weighted using a triangular kernel. Standard errors are in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table 6: Treatment Effects on Religious Conflicts

	(1)	(2)	(3)	(4)	(5)	(6)
	Religious Riots		Non-Religious Riots		Murders	
	Riots Per Capita	Total Riots	Riots Per Capita	Total Riots	Per Capita	Total
Treat	3.705*** (1.161)	6.190*** (2.093)	20.670 (15.014)	-28.612 (35.211)	2.108 (3.186)	-2.978 (17.144)
Observations	323	323	323	323	388	388
Dep Var Mean	1.010	2.170	48.977	135.911	26.390	71.163

Notes: This table compares religious and non-religious conflicts across minority and non-minority concentration districts. The unit of observations is the district. Columns (1)-(2) pertain to religious riots; columns (3)-(4) pertain to non-religious riots; columns (5)-(6) pertain to murders. The outcome of interest in columns (1), (3) and (5) scales riots by the district's population in 2001. All specifications include state and year fixed effects, in addition to a linear polynomial in the running variable. All specifications restrict the sample to a bandwidth of .058 around the discontinuity threshold, and are weighted using a triangular kernel and the district population in 2001. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

A Pre-Treatment Minority Credit Access: Descriptive Evidence

This section offers descriptive evidence from the 2003 AIDIS and highlights that in the pre-treatment period, Muslim households faced substantial barriers to credit access and reported significantly lower financial assets.

A.1 Credit Access Across Communities

We present the descriptive analysis by disaggregating households into 5 mutually exclusive groups: Hindu forward castes, Hindu marginalized castes, Muslims, and other religious minorities.¹ We also disaggregate the source of credit into bank loans, loans from non-bank financial institutions (NBFIs) such as co-operative banks and non-banking financial corporations (NBFCs), and loans from informal sources. The latter is further disaggregated into loans from professional money lenders and input suppliers, and loans from community networks such as friends, relatives, employers and landlords.

Appendix Figure A1 shows the share of outstanding credit accounted for by each of the communities, disaggregated by the source of credit. The top-left panel shows that overall, Muslims accounted for 7 percent of outstanding credit across all sources, which shrunk to 6 percent when considering only bank credit (middle-panel, top row). For reference, Muslims accounted for 13 percent of India's population according to the 2001 Census. This under-representation is seen across all credit sources, with the exception of loans from community sources. In contrast, other non-Muslim religious minorities accounted for 7 percent of bank credit, while comprising 5 percent of the population. Unsurprisingly, Hindu forward castes accounted for 40 percent of bank credit, despite accounting for less than a quarter of India's population.

The top-left panel of Appendix Figure A2 shows that between 30 and 40 percent of households across all five groups had some outstanding loan in 2003, with informal loans being the primary source of credit. While less than 10 percent of households had any outstanding bank loan, this was particularly low for Muslim households (3 percent), relative to Hindu forward castes (5.6 percent) or other religious minorities (5.6 percent). Similarly, the top-right panel of Appendix Figure A2 shows Muslim households to also have lower credit access along the intensive margin. Relative to Hindu forward castes or other religious minorities, and conditional on having an outstanding bank loan, Muslim households on average had 30 percent lower bank loan amounts. The bottom row of Appendix Figure A2 compares the cost of credit across communities and shows no variation in bank interest rates across the five groups. This indicates that Muslim households were not inherently riskier borrowers. However, these households were charged the highest rates of interest by professional money lenders. Collectively, Appendix Figure A2 points to a rationing of formal credit for Muslim households along both the extensive and intensive margins. This is unlikely to be explained by a lower demand for credit as the fraction of Muslim households with outstanding loans is very similar to Hindu forward castes. In the absence of formal credit, Muslim households were pushed towards informal credit markets, where they faced the highest cost of credit.

A.1.1 Loan Applications

As the AIDIS has no information on loan applications, we use data from the nationally representative Indian Human Development Survey (IHDS) to compare trends in loan application and denial across bank and non-bank sources, and communities. The IHDS in their 2011-12 survey inquired whether households had ever applied for a loan, and whether the loan application was approved or denied. The

¹ We refer to *Dalits*, *Adivasis*, and Other Backward Classes (OBCs) when referring to Hindu marginalized castes. The descriptive analysis also distinguishes between SC/STs and OBCs.

top panel of Appendix Figure A3 show that while 50-60 percent of households had applied for a loan over the past five years, Muslim households were the least likely to apply for a bank loan. Thus, only 11 percent of Muslim households applied for a bank loan, as opposed to 24 percent of Hindu forward caste households, or other religious minorities. The low rate of bank loan applications cannot be explained by low credit demand, as 34 percent of Muslim households over the same period applied for loans from informal sources – a rate of application comparable to other communities. The bottom left panel of Appendix Figure A3 shows that conditional on applying, bank loan applications of Muslim households were also more likely to be rejected. Thus, while the bank loan rejection rate was 10 percent for Hindu forward castes and other religious minorities, the corresponding denial rate for Muslim borrowers equaled 15 percent. While the evidence in Appendix Figures A1, A2 and A3 is purely descriptive, it is consistent with Muslim borrowers facing discrimination in formal credit markets.

A.1.2 Household Assets

Appendix Figures A4 and A5 shows that relative to Hindu forward castes and other religious minorities, Muslim households had lower levels of households savings – both bank deposits and retirement savings – and immovable assets in the form of land and real estate. As land and real estate often serve as collateral in loan covenants, lower values of pledgeable assets could also have contributed to the exclusion of these borrowers from credit markets, and a shrinking of loan size (Banerjee and Duflo, 2010).

In summary, Figures A2 - A5 show substantially lower access to bank credit for Muslim households along both the extensive and intensive margins. They were least likely to apply for a bank loan, and their bank loan applications were also more likely to be rejected. Muslim households also faced the highest rates of interest from informal money lenders, and had significantly lower levels of financial and physical assets. This suggests that the directed credit policy was unlikely to be inframarginal for Muslim households. In contrast, the financial status of other religious minorities in terms of credit access and asset ownership were comparable to Hindu forward castes.

A.2 Credit Allocation for Historically Marginalized Communities

To facilitate the flow of credit to under-represented minorities, the central bank mandates commercial banks to allocate 10 percent of their aggregate credit to “weaker sections” – namely borrowers from *Dalit* (Scheduled Caste) and *Adivasi* (Scheduled Tribe) communities, along with female borrowers. However, it is unclear as to how well the central bank monitors this, or what penalties are levied if banks fail to meet this target.² To facilitate lending to religious minorities, the RBI expanded the definition of weaker sections to include religious minorities. Banerjee et al. (2004) showed that the priority sector guidelines were binding for commercial banks, with banks being unwilling to extend more than 40 percent of their loan portfolio towards the priority. If banks viewed the 10 percent allocation for weaker sections in a similar manner, it is possible that the inclusion of religious minorities in this category would lead to banks reallocating credit from women and socially marginalized groups, to religious minorities.

In the absence of administrative data, we assess this by combining administrative data from the Basic Statistical Returns (BSR), with household lending data from the AIDIS. The RBI hosted publicly available BSR offers a sectoral break up of bank credit, which separately classifies agricultural loans directly issued to farmers (direct finance), as well as personal loans issued to individual borrowers. We assume these two categories to correspond to the household-level borrowing data in the AIDIS.³ In the

² This is contrary to the priority sector regulations, where the RBI sets explicit guidelines for banks if they fail to allocate 40 percent of their credit to the priority sector.

³ The remaining loan categories are manufacturing loans, loans to trading and service enterprises, loans to financial

pre-treatment period in 2003, the value of outstanding bank loans to households in the AIDIS data equaled INR 1,359 billion. In the administrative BSR data, outstanding personal loans and direct agricultural loans equaled INR 1,730 billion. While we cannot account for female borrowers owing to the household-level nature of the AIDIS data, *Dalit* and *Adivasi* borrowers accounted for 13 percent of total household loans in the AIDIS. As farm and personal loans accounted for 22 percent of aggregate commercial bank credit in 2003, loans to socially marginalized groups comprised of 3 percent of banks' lending portfolio. As long as banks did not issue over 7 percent of their loans to female borrowers, this suggests that banks could have extended loans to religious minorities without breaching the 10 percent cap.

A.3 Oaxaca Blinder Decomposition

We formally use an Oaxaca-Blinder decomposition to gauge whether the descriptive difference in credit access for minority Muslim households, discussed in Section A.1, can be explained by observable characteristics. We use the 2003 AIDIS and restrict the sample to Hindu and Muslim households. District fixed effects are used to partial out aggregate district characteristics which uniformly affect credit access for both communities. Consequently, our decomposition tests whether observable characteristics can explain the difference in bank credit access for Muslim households, relative to Hindu households residing in the same district. We condition for demographic features such as household size and rural location, as well as educational characteristics such as whether the household has a secondary educated member. Additional covariates includes whether the household's earnings are primarily from self-employment or wage work, and the broad industry of occupation. To factor in variations in households' demand for credit, we also control for monthly per capita household consumption. Finally, as the ability to extend collateral often determines credit access Banerjee and Duflo (2010), we control for the value of land and buildings owned by the household. The sample is restricted to households in non-minority concentration districts and standard errors are clustered by district.

Appendix Figure A6 shows that observable characteristics cannot fully explain the variation in bank credit access across Hindu and Muslim households, pointing to the presence of discrimination. After conditioning on the fixed effects and covariates, Muslim households have a statistically significant 2.3 percentage point lower likelihood of having a bank loan, relative to Hindu households. Observable characteristics can only explain 38 percent of this difference. Along the intensive margin, unobservable characteristics explain 30 percent of the difference in the amount of bank credit, although the point estimate is not statistically significant (p-value .229).

A.4 Testing for Statistical Discrimination

Section 5.1.1 showed that the directed credit policy resulted in higher bank lending to religious minorities. This did not come at the expense of non-minority borrowers. Neither was it accompanied by a deterioration in credit quality, which suggests that banks' aversion to lend to these groups stemmed from private information pertaining to the poor creditworthiness of these borrowers. On aggregate, Appendix A.1 showed that Muslim households accounted for 6 percent of aggregate bank credit, despite comprising 13 percent of India's population. The Oaxaca-Blinder decomposition in Appendix A.3 indicated that observable characteristics could explain only 30 percent of the extensive margin difference in bank credit across Hindu and Muslim households. Taken together with the finding of Fisman et al. (2020) that childhood exposure of Hindu loan officers to religious conflicts resulted in lower credit access for Muslim borrowers, the descriptive evidence points to discrimination against minority borrowers in formal credit markets.

institutions, and loans to transport operators. These are likely to be loans issued to firms, as opposed to households.

Discrimination can be statistical or taste-based. If lenders were previously rationing credit to religious minorities owing to statistical discrimination, information gained on borrower quality from the directed credit policy should have led to increased lending to religious minorities, ex-post. The ideal experiment thereby would have been to compare bank credit received by religious minorities across minority and non-minority concentration areas, following the discontinuation of the directed credit policy. As the policy remains in effect, we propose an alternate *suggestive* test for statistical discrimination.

We exploit the fact that the primary responsibility of implementing the directed credit policy was assigned to lead banks. Lead banks in minority concentration districts were also lead banks in other non-minority concentration areas. If lead banks learnt about the quality of minority borrowers through the directed credit policy and shared this information with other branches, we would expect increased credit access from these banks to minority borrowers in control areas, relative to other banks which were less likely to update their priors regarding the quality of minority borrowers.⁴ The empirical specification is the following:

$$Y_{hds} = \alpha_d + \beta_1 \text{Muslim}_{hds} + \beta_2 \text{MatchLeadBank}_{ds} \times \text{Muslim}_{hds} + \delta \mathbf{X}_{hds} + \epsilon_{hds} \quad (\text{A4})$$

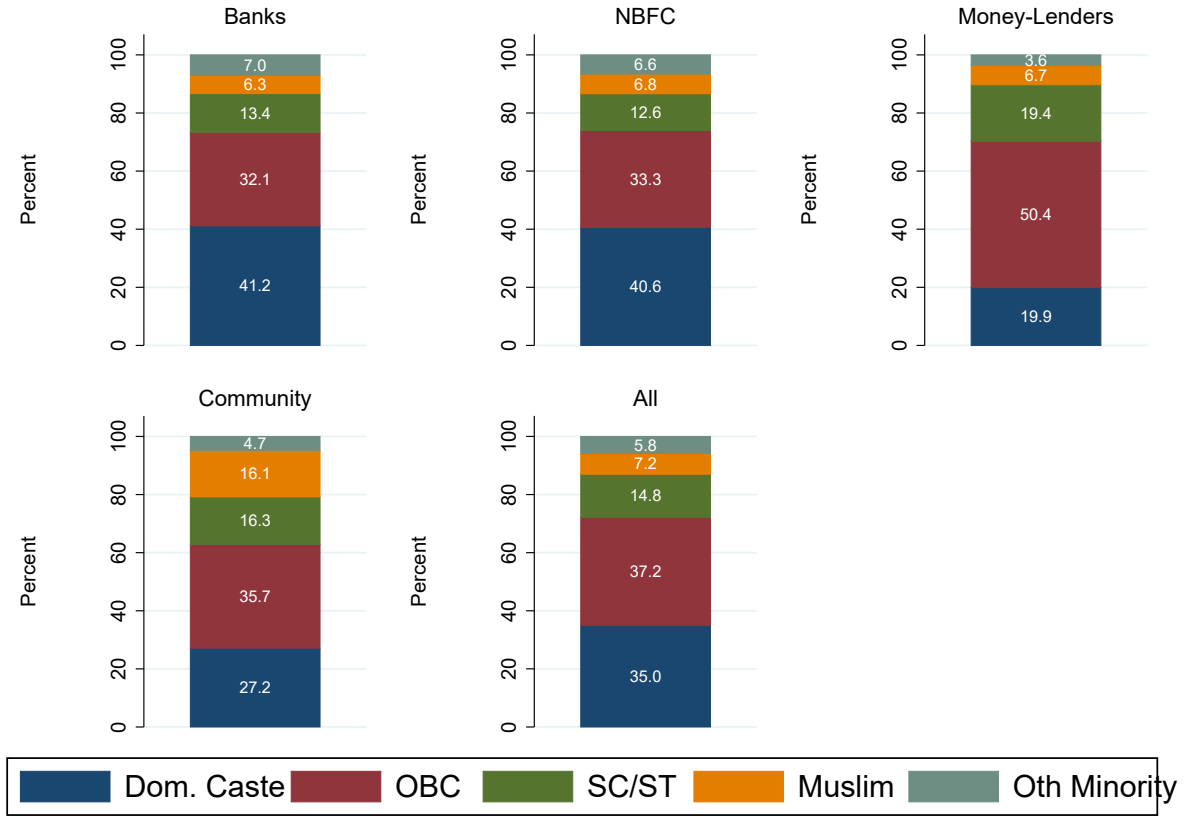
In specification (A4), *MatchLeadBank* is a dummy equaling 1 if the lead bank in district d is also a lead bank in one of our treated districts. The sample is restricted to Hindu and Muslim households in non-minority concentration areas. Our preferred specification conditions on district fixed effects (α) and household covariates ($\mathbf{X}$). This restricts the comparison of bank credit access to Hindu and Muslim households in the same district. Assuming information sharing across lead banks, a positive coefficient on β_2 would offer suggestive evidence in favour of statistical discrimination.

Appendix Table A1 disallows us from arriving towards any such conclusion. The odd-numbered columns restrict the sample to Muslim households, and test whether bank credit access for Muslim households is significantly different in districts which have a common lead bank with minority concentration areas. As the State Bank of India (SBI) was the lead bank in a fifth of minority concentration districts, we separately estimate the results by comparing credit to Muslim households in control districts where the SBI was the lead bank. When comparing access to bank credit across districts with a common lead bank for Muslim households, the coefficient of interest is negative. When comparing bank credit access for Muslim households, relative to Hindu households in the same district, conditional on districts having a common lead bank, the interaction term remains negative and statistically non-significant. While the empirical strategy of specification (A4) precludes us from making any causal claims, the fact that the negative correlation estimated in β_1 for access to bank credit is not over-turned in β_2 offers suggestive evidence that either there was no information sharing across lead bank branches, or the reluctance of banks to lend to Muslim borrowers did not emanate from statistical discrimination.

⁴ As the vast majority of lead banks are state-owned banks, information sharing can occur through the transfer of loan officers. As detailed in Fisman et al. (2017), loan officers are regularly rotated across bank branches, and such rotations can occur across districts.

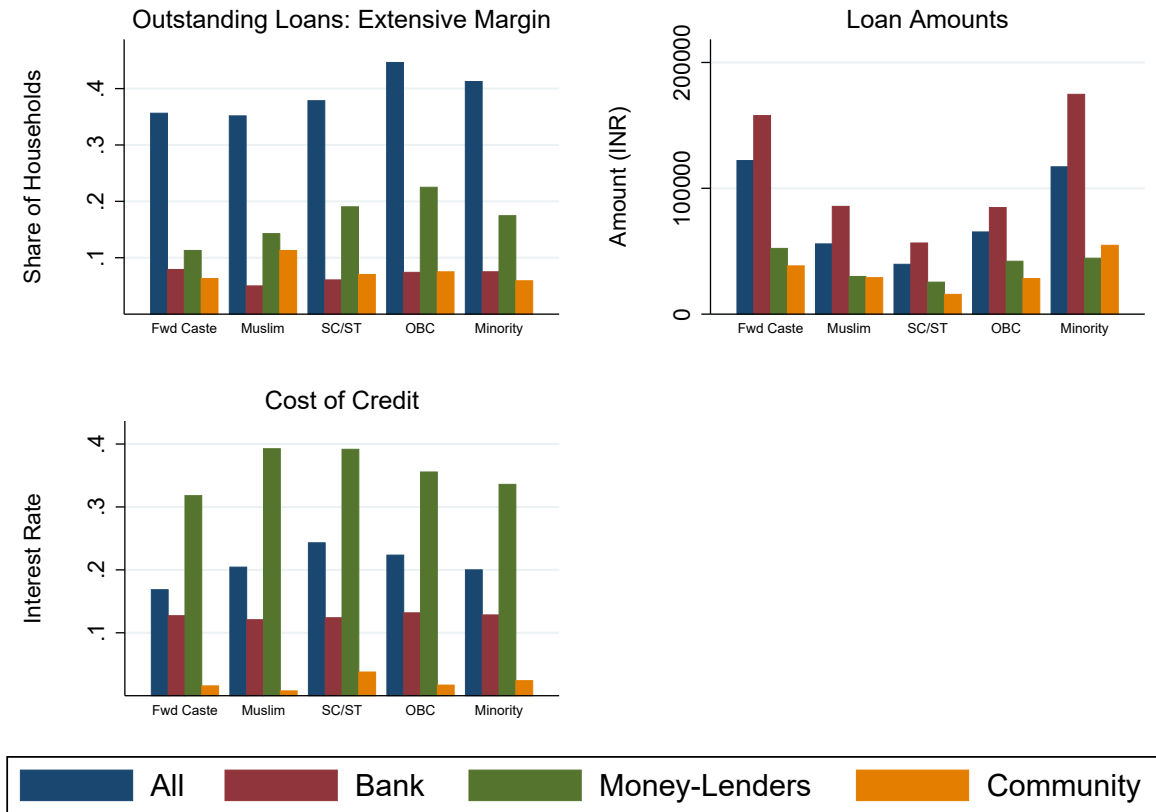
A.5 Figures

Figure A1: Distribution of Credit Across Communities and Credit Source: Pre-Treatment Shares



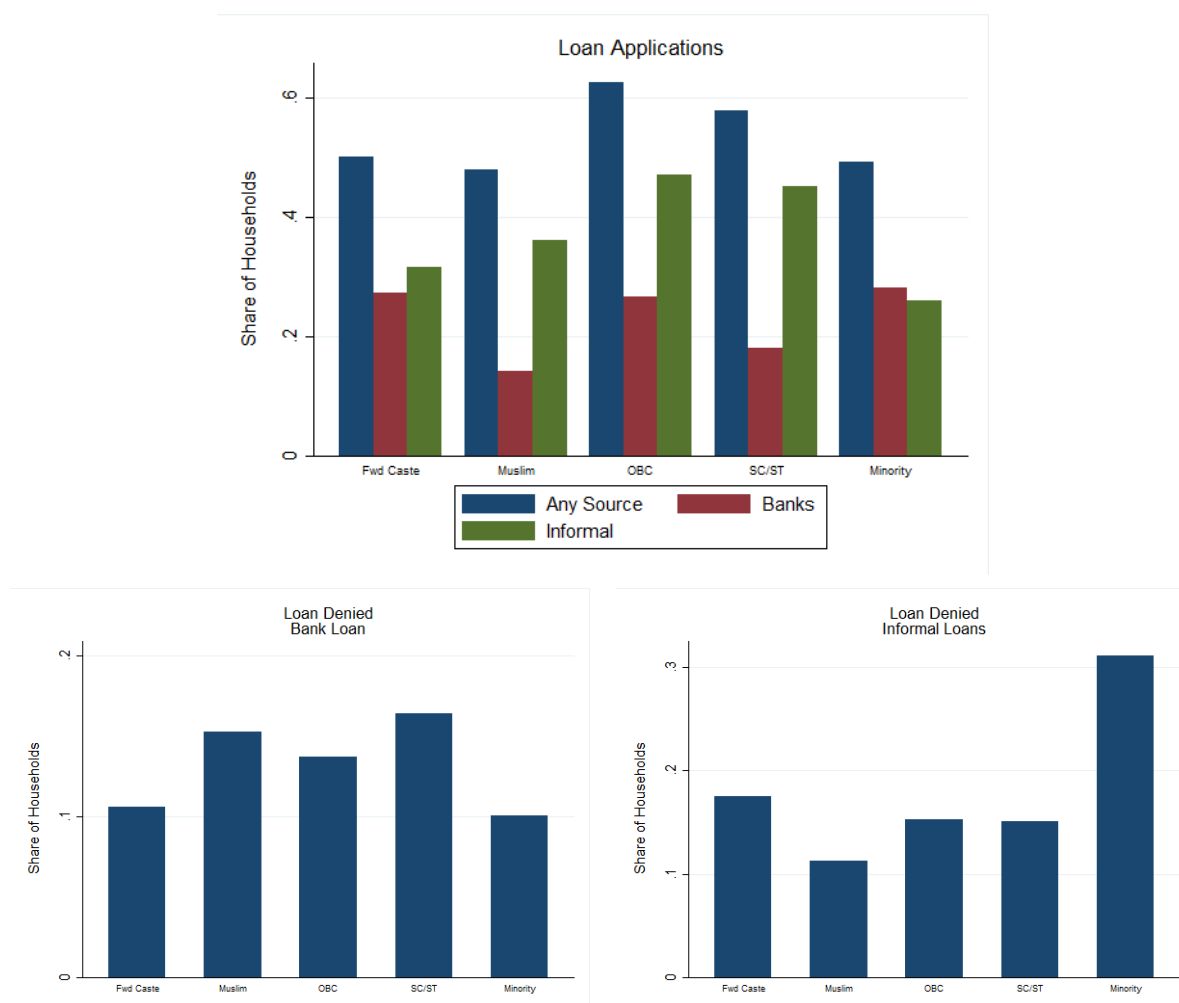
Notes: The above figure shows the pre-treatment distribution of credit across communities and credit source. The bar graph shows the share of outstanding credit for each community, based on credit source. *Oth Minority* refers to non-Muslim religious minorities; *Dom Caste* refer to non-SC/ST/OBC Hindu households. Data is from the AIDIS, 2003.

Figure A2: Pre-Treatment Household Credit Across Communities



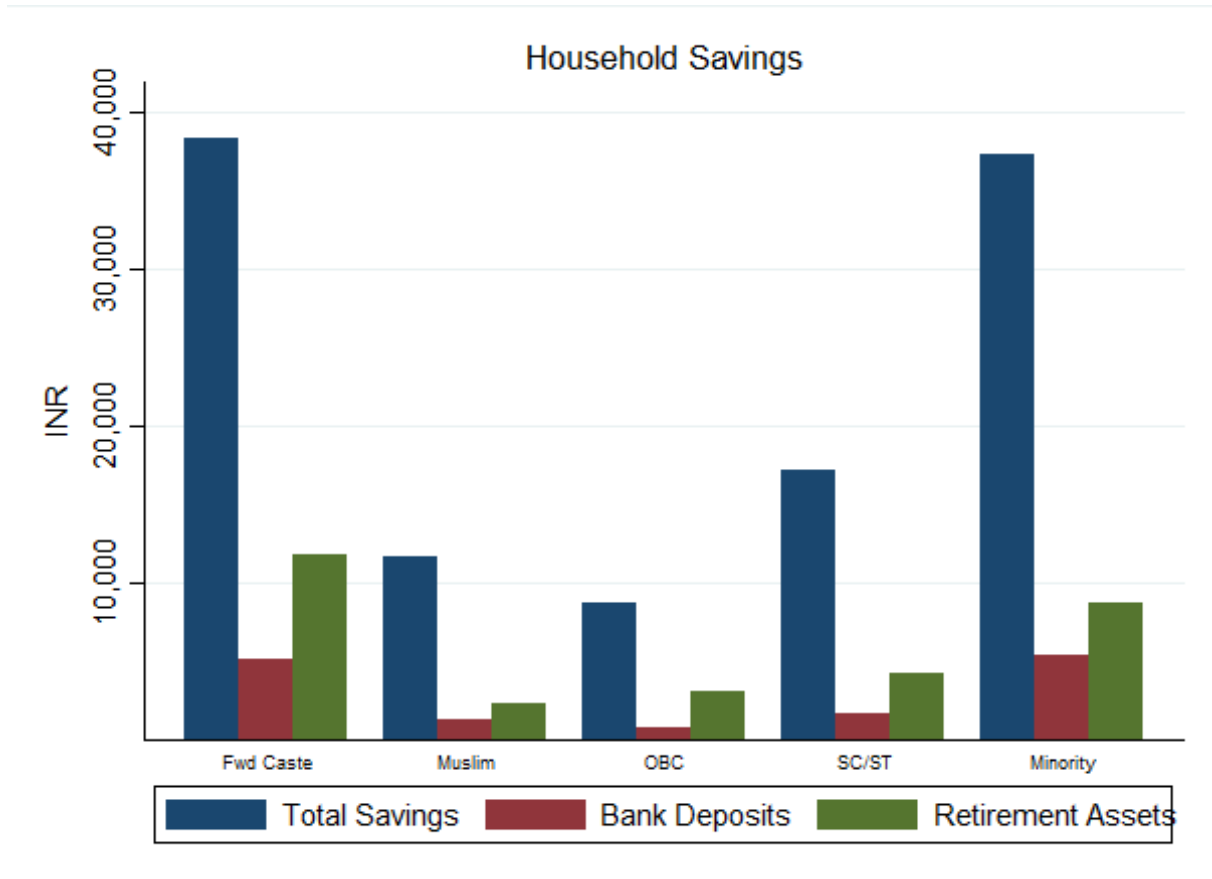
Notes: The above figure presents pre-treatment descriptive statistics for loan outcomes by community. The top-left panel shows the likelihood of households having loans from any of the above-mentioned sources; the top right-panel shows the average volume of credit received by households from each source; the bottom-right panel shows the average rate of interest from each source. *Minority* refers to non-Muslim religious minorities; *Fwd Caste* refer to non-SC/ST/OBC Hindu households. Data is from the AIDIS, 2003.

Figure A3: Loan Application and Denial Across Communities and Credit Source



The above figures compares loan application and denial rates across communities using data from the Indian Human Development Survey (IHDS), conducted in 2011-12. Loan applications cover any loan application made by the household in the past five years, from the source mentioned. Loan denial is conditional on the household applying for the loan from the source mentioned.

Figure A4: Pre-Treatment Household Financial Assets Across Communities



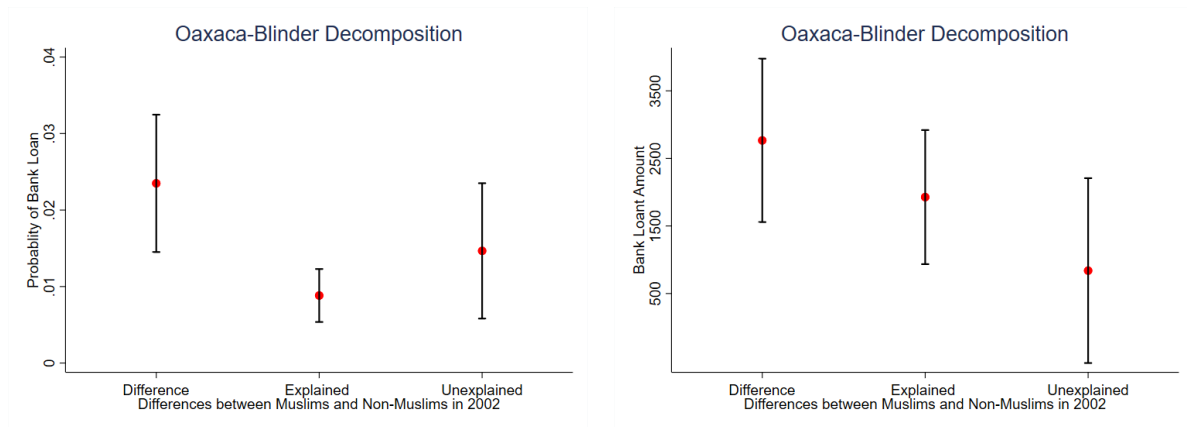
Notes: The above figure presents pre-treatment descriptive statistics for risk-free financial assets held by households, across communities. *Bank* refers to bank deposits. *Minority* refers to non-Muslim religious minorities; *Fwd Caste* refer to non-SC/ST/OBC Hindu households. Data is from the AIDIS, 2003.

Figure A5: Pre-Treatment Household Land and Real Estate Across Communities



Notes: The above figure presents pre-treatment descriptive statistics for land and real estate held by households, across communities. *Real Estate* includes both residential and non-residential real estate. *Minority* refers to non-Muslim religious minorities; *Fwd Caste* refer to non-SC/ST/OBC Hindu households. Data is from the AIDIS, 2003.

Figure A6: Pre-Treatment Credit Access Across Minority and Non-Minority Households: Oaxaca-Blinder Decomposition



Notes: The above figures uses a Oaxaca-Blinder decomposition to test the role of unobservables in explaining the difference in bank credit access across Hindu and Muslim households in the pre-treatment period. The sample is restricted to Hindu and Muslim households. All specifications condition on district fixed effects and a rich set of household covariaes. The outcome of interest in the left-figure is a dummy equaling 1 if the household has any bank loan; in the right-figure, the value of bank loans. Standard errors are clustered by district.

A.6 Tables

Table A1: Access to Bank Credit for Religious Minorities in Control Districts: Heterogeneity by Common Lead Banks

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pr(Loan = 1)				Loan Amount (INR)			
	Muslims	All	Muslims	All	Muslims	All	Muslims	All
Lead Bank Match	-.101** (.045)				-27003.067** (13433.791)			
Muslim $\times$ Lead Bank Match		-.012 (.026)				-174.951 (11163.227)		
Lead Bank SBI			-.070* (.040)				-14879.008 (16335.524)	
Muslim $\times$ Lead Bank SBI				-.003 (.033)				5002.103 (22065.990)
Observations	4443	65498	4443	65498	4443	65498	4443	65498
Dep Var Mean	.188	.188	.188	.188	37092.773	37092.773	37092.773	37092.773
Covariates	Y	Y	Y	Y	Y	Y	Y	Y
Fixed Effects	State	District	State	District	State	District	State	District

Notes: The above table shows the treatment effect on bank credit access for minority households in non-minority concentration districts. The unit of observation is the household. The outcome of interest in columns (1)-(4) is a dummy equaling 1 if the household had any outstanding bank loan; in columns (5)-(8), the amount of bank loans received by the household. *Lead Bank Match* is a dummy equaling 1 if the lead bank in the district has also been a lead bank in a minority concentration district. Columns (3)-(4), and (7)-(8) consider only matches with the State Bank of India (SBI) as the lead bank. All specifications include state fixed effects and household covariates; the even numbered columns include district fixed effects. The odd numbered columns restrict the sample to Muslim households; the even-numbered columns restrict the sample to Hindu and Muslim households. The sample is restricted to non-minority concentration districts, not located within a bandwidth of .058 around the discontinuity threshold. Specifications are weighted using household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

B Appendix: Balance Checks

Table B1: Pre-Treatment Balance on Household Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	Demographics				Education		Employment			Consumption	
	Rural	HH Size	Females	Children	Female Head	Secondary Educated	Higher Educated	Self Employed	Non-Farm Self Employed	Farm Work	Per Capita
Treat	.089 (.055)	.885** (.403)	.089 (.191)	.102 (.167)	.039 (.027)	.064 (.047)	.018 (.015)	.063 (.057)	.055 (.095)	.077 (.103)	27.319 (33.164)
Observations	1993	1993	1993	1993	1993	1993	1993	1993	1993	1993	1993
Dep Var Mean	.627	5.584	2.695	2.496	.110	.251	.049	.525	.351	.316	554.779

Notes: The above table shows the pre-treatment balance on household demographics, employment characteristics and consumption. The unit of observation is the household. All specifications include state fixed effects, a linear polynomial in the running variable. Columns (3)-(10) also include household covariates. All specifications are weighted using a triangular kernel and AIDIS assigned household weights. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B2: Pre-Treatment Balance on Household Farm Land and Livestock

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Land					Livestock	
	Any Irrigated Area	Irrigated Farm Area	Irrigated Farm Value	Farm Area	Farm Value	Any Livestock	Livestock Value
Treat	-.004 (.042)	.061 (.046)	-4143.607 (13141.852)	.137* (.078)	15595.739 (11901.311)	-.011 (.079)	119.653 (390.440)
Observations	1993	1993	1993	1993	1993	1993	1993
Dep Var Mean	.169	.082	23500.044	.202	37660.714	.339	1934.010

Notes: The above table shows the pre-treatment balance on household farm and livestock ownership for minority households. The unit of observation is the household. All specifications include state fixed effects, a linear polynomial in the running variable. Columns (2)-(10) also include household covariates. All specifications are weighted using a triangular kernel and AIDIS assigned household weights. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B3: Pre-Treatment Balance on Household Farm Machinery

	(1)	(2)	(3)	(4)	(5)	(6)
	Farm Machinery		Farm Transport			
	Any Machine	Machinery Value	Any Transport	Transport Value	Any Tractor	Tractor Value
Treat	-.116 (.110)	-1021.445 (978.701)	-.014 (.014)	-96.632 (187.739)	-.005 (.004)	-15.926 (205.007)
Observations	1993	1993	1993	1993	1993	1993
Dep Var Mean	.488	1004.220	.033	174.769	.002	98.378

Notes: The above table shows the pre-treatment balance on household farm machinery and transport for minority households. The unit of observation is the household. All specifications include state fixed effects, a linear polynomial in the running variable and household covariates. All specifications are weighted using a triangular kernel and AIDIS assigned household weights. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B4: Pre-Treatment Balance on Household Assets

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Real Estate		Financial Assets		Business Assets		Transport	
	Buildings	Land and Buildings	Bank Savings	Financial Assets	Any Machine	Machine Value	Any Transport	Transport Value
Treat	-3741.423 (15782.407)	40518.854 (28890.198)	-358.433 (823.554)	8926.687** (4200.977)	-.004 (.018)	-262.098 (337.308)	.025 (.103)	2367.358 (3268.986)
Observations	1993	1993	1993	1993	1993	1993	1993	1993
Dep Var Mean	67847.308	1.45e+05	1158.907	7429.465	.092	766.218	.560	4150.895

Notes: The above table shows the pre-treatment balance on household assets for minority households. The unit of observation is the household. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. All specifications are weighted using a triangular kernel and AIDIS assigned household weights. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B5: Pre-Treatment Balance on Household Credit: Extensive Margin

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Any Loan = 1)					
	Any Source	Bank	Non Bank	Informal	Money Lender	Community
Treat	-.000 (.033)	.004 (.016)	-.014 (.015)	.047 (.042)	.028 (.028)	.014 (.030)
Observations	1993	1993	1993	1993	1993	1993
Dep Var Mean	.285	.027	.032	.211	.124	.092

Notes: The above table shows the pre-treatment balance on household credit for minority households. The unit of observation is the household. The outcome of interest is a dummy equaling 1 if the household has a loan from the source mentioned. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. All specifications are weighted using a triangular kernel and AIDIS assigned household weights. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B6: Pre-Treatment Balance on Household Credit: Intensive Margin

	(1)	(2)	(3)	(4)	(5)	(6)
	Loan Amount (INR)					
	Any Source	Bank	Non Bank	Informal	Money Lender	Community
Treat	-2895.303 (3474.605)	106.123 (2505.643)	-856.669 (1517.745)	-1935.327 (1667.373)	-871.601 (1080.519)	-1063.726 (998.547)
Observations	1993	1993	1993	1993	1993	1993
Dep Var Mean	11312.726	1457.078	2736.800	5946.208	3236.216	2709.991

Notes: The above table shows the pre-treatment balance on household credit for minority households. The unit of observation is the household. The outcome of interest is the value of loans obtained from the source mentioned. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. All specifications are weighted using a triangular kernel and AIDIS assigned household weights. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B7: Pre-Treatment Balance on Bank Loan Purpose

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Any Bank Loan = 1)			Bank Loan Amount (INR)		
	Farm	Non Farm	Consumption	Farm	Non Farm	Consumption
Treat	-.002 (.011)	-.000 (.007)	.003 (.006)	150.173 (2246.359)	-449.626 (680.935)	573.479 (428.136)
Observations	1993	1993	1993	1993	1993	1993
Dep Var Mean	.014	.006	.007	540.775	451.256	350.700

Notes: The above table shows the pre-treatment balance on purpose of bank credit for minority households. The unit of observation is the household. The outcome of interest in column (1)-(3) is a dummy equaling 1 if the household has any bank loan for the purpose mentioned; in columns (4)-(6), the value of bank loans obtained for that purpose. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. All specifications are weighted using a triangular kernel and AIDIS assigned household weights. The sample is restricted to a bandwidth of .06 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B8: Pre-Treatment Balance on Loan Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Cost of Credit			Pr(Default = 1)			Pr(Long Term Loan = 1)			Pr(Unsecured Loan = 1)		
	Any Source	Bank	Informal	Any Source	Bank	Informal	Any Source	Bank	Informal	Any Source	Bank	Informal
Treat	-.007 (.074)	.005 (.012)	-.105 (.072)	.104 (.082)	-.007 (.282)	-.051 (.101)	.034 (.040)	-.006 (.016)	.011 (.038)	.046 (.035)	.006 (.010)	.027 (.041)
Observations	877	116	633	748	107	488	1993	1993	1993	1993	1993	1993
Dep Var Mean	.192	.131	.211	.646	.749	.656	.167	.022	.114	.229	.016	.188

Notes: The above table shows the pre-treatment balance on loan characteristics for minority households. The unit of observation is the household. The outcome of interest in column (1)-(3) is the interest rate from the loan sources mentioned; in columns (4)-(6), a dummy equaling 1 if the household has an unsecured loan from the source mention. For households with multiple outstanding loans, the average interest rate is weighted by initial loan size. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. All specifications are weighted using a triangular kernel and AIDIS assigned household weights. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B9: Balance on Weekly Labour Market Activities: Extensive Margin

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pr(Outcome = 1)							
	Labour Force Participation	Unemployed	Self Employed	Wage Work	Farm	Non Farm	Manufacturing	Trade and Services
Treat	.019 (.023)	-.034 (.029)	.078*** (.028)	-.033 (.031)	.028 (.026)	-.001 (.022)	.002 (.021)	-.039 (.025)
Observations	5050	5050	5050	5050	5050	5050	5050	5050
Dep Var Mean	.566	.067	.302	.250	.205	.347	.095	.206

Notes: The above table shows the treatment effect on weekly labour market outcomes along the extensive margin for minority working-age individuals. The unit of observation is the working-age individual. Working-age individuals are individuals aged between 18 and 60 years. The outcome of interest in column (1) is a dummy equal to 1 if the individual participated in the labour force in any of the 7 days preceding the survey; in column (2), a dummy equal to 1 if the individual was unemployed in any of the 7 days preceding the survey; in columns (3)-(6), a dummy equal to 1 if the individual was engaged in the sector referred. All specifications include state and survey month fixed effects; a linear polynomial in the running variable; controls for demographic and education covariates at the individual level; household level controls. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B10: Pre-Treatment Balance on District Banking Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Bank Branches			Deposits		Credit	
	All Banks	State-Owned Banks	Private Banks	Accounts	Amount	Accounts	Amount
Treat	-6.065 (30.870)	-1.706 (27.755)	-4.359 (4.573)	-.195 (.235)	-22395.830 (34629.274)	-.040 (.066)	-18953.575 (33846.067)
Observations	59	59	59	59	59	59	59
Dep Var Mean	67.404	62.730	4.674	.427	23212.736	.099	19430.789

Notes: The above table shows the pre-treatment balance on district-level banking outcomes. The unit of observation is the district. The outcome of interest in column (1)-(3) is the number of bank branches in the district; in columns (4)-(5), bank deposits; and columns (6)-(7), bank loans. Credit and deposit amounts are in millions of rupees. All specifications include a linear polynomial in the running variable and are weighted using a triangular kernel. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B11: Pre-Treatment Balance on District Sectoral Lending

	(1)	(2)	(3)	(4)	(5)	(6)
	Farm Account	Farm Amount	Manufacturing Account	Manufacturing Amount	Personal Loans Account	Personal Loans Amount
Treat	.002 (.009)	924.474 (1497.737)	.001 (.001)	-12897.358 (22206.290)	-.042 (.059)	-608.173 (3218.739)
Observations	59	59	59	59	59	59
Dep Var Mean	.039	3772.214	.003	10909.024	.043	3962.045

Notes: The above table shows the pre-treatment balance on district-level bank lending by sector. The unit of observation is the district. Columns (1)-(2) refer to farm bank loans; columns (3)-(4) refer to bank manufacturing loans; and columns (5)-(6) refer to personal loans made by banks. The outcome of interest in the odd-numbered columns refer to loan accounts; in the even-numbered columns, loan amounts. Loan accounts and amounts are expressed in millions. All specifications include a linear polynomial in the running variable and are weighted using a triangular kernel. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B12: Directed Credit Policy and Electoral Outcomes: Pre-Treatment Balance

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Minority Candidates						Other Parties		
	Any Muslim	Any Muslim Major Party	Vote Share	Vote Share	Vote Share Major Party	Vote Share Major Party	BJP Vote Share	BJP Vote Share	INC Vote Share
Treat	.017 (.053)	-.027 (.091)	-.017 (.033)	.033 (.043)	.076 (.053)	.013 (.028)	-.035 (.032)	-.011 (.023)	.017 (.053)
Observations	422	422	393	422	422	422	284	422	422
Dep Var Mean	.196	.466	.079	.079	.133	.029	.278	.205	.196

Notes: This table shows pre-treatment balance on electoral outcomes for politicians from minority communities and those representing the right-wing party. The unit of observation is the electoral constituency. The outcome of interest in column (1) is a dummy equaling 1 if there was a Muslim candidate contesting the election; in column (2), a dummy equaling 1 if there was a Muslim candidate contesting the election on behalf of a non-majoritarian party; in column (3), the vote share received by individual Muslim candidates; in column (4), the total vote share received by all Muslim candidates; in column (5), vote share received by Muslim candidates contesting on a non-majoritarian party ticket; in column (6), vote share received by Muslim candidates contesting on a non-majoritarian party ticket, conditional on a Muslim candidate contesting on a major party ticket; in column (7), the vote share received by the BJP candidate, conditional on participation; in column (8), the vote share received by the BJP candidate; in column (9), the vote share received by the INC candidate. Vote share is the number of votes received by a candidate, divided by the total votes cast in the constituency. The sample is restricted to elections conducted between 2003 and 2007. All specifications include state and electoral year fixed effects, in addition to a linear polynomial in the running variable. All specifications restrict the sample to a bandwidth of .058 around the discontinuity threshold, and are weighted using a triangular kernel. Standard errors are in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B13: Balance Checks for Pre-Treatment Electoral Wins and Political Participation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Pr(Electoral Victory = 1)			Political Participation		
	Muslim Candidate	Muslim Candidate Major Party	BJP Candidate	INC Candidate	Voter Turnout	No. of Parties	Registered Voters
Treat	.086 (.058)	-.006 (.038)	.021 (.039)	-.112 (.093)	.058 (.045)	.286 (.436)	-48110.474** (20619.682)
Observations	422	422	422	422	422	422	422
Dep Var Mean	.056	.022	.298	.172	.585	8.685	160053.171

Notes: This table shows balance checks for pre-treatment electoral characteristics. The data is based on elections conducted between 2003 and 2007. The unit of observation is the electoral constituency. The outcome of interest in column (1) is a binary variable equaling 1 if a Muslim candidate won the election; in column (2), a binary variable equaling 1 if a Muslim candidate contesting from a major political party won the election; in column (3), a dummy equaling 1 if the BJP won the election; in column (4), a dummy equaling 1 if the INC won the election; in column (5), overall voter turnout; in column (6), the number of candidates contesting elections; in column (7), the total number of registered voters. All specifications include state and electoral year fixed effects, in addition to a linear polynomial in the running variable. All specifications restrict the sample to a bandwidth of .058 around the discontinuity threshold, and are weighted using a triangular kernel. Standard errors are in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B14: Composition of Bank-Linked Self-Help Groups: Pre-Treatment Balance

	(1)	(2)	(3)	(4)	(5)	(6)
	SHGs Formed				SHG Size	
	All SHG	Total Muslims	Some Muslim	Only Muslim	All SHG	Only Muslim
Treat	-19.361 (21.173)	31.634 (54.948)	.927 (7.477)	2.684 (5.091)	-1.948 (2.513)	-.079 (.250)
Observations	2972	2972	2972	2972	2972	2972
Dep Var Mean	45.313	36.697	5.998	1.845	4.061	11.384

Notes: This table shows pre-treatment balance on the social composition of bank-linked self-help groups (SHG). The unit of observation is the district-bank. The sample is restricted to bank-linked self-help groups registered between 2003 and 2007. The outcome of interest in column (1) is the total number of bank-linked SHGs registered; in column (2), the total number of Muslim members in bank-linked SHGs; in column (3), the number of bank-linked SHGs with a Muslim member; in column (4), the number of bank-linked SHGs registered with only Muslim members; in column (5), the number of bank-linked SHGs registered with both Muslim and non-minority members; in column (6), average SHG size; in column (7), the average size of SHGs with only Muslim members. All specifications include state, bank and SHG formation year fixed effects, along with a linear polynomial in the running variable. All specifications are weighted using a triangular kernel. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table B15: Pre-Treatment Balance on Crimes

	(1)	(2)	(3)	(4)	(5)	(6)
	Riots	Murder	Rape	Property Crimes	Kidnapping	All Crimes
Treat	-8.279 (12.398)	-6.806 (5.026)	2.184 (3.149)	18.504 (213.196)	1.209 (4.980)	-23.994 (301.200)
Observations	297	297	297	297	297	297
Dep Var Mean	57.211	32.938	16.981	517.451	45.462	1991.724

Notes: This table shows pre-treatment balance in crime outcomes across minority and non-minority concentration districts. The unit of observations is the district. The outcomes of interest in columns (1)-(6) scales the total incidents of crime by the district's population in 2001. All specifications include state and year fixed effects, in addition to a linear polynomial in the running variable. All specifications restrict the sample to a bandwidth of .058 around the discontinuity threshold, and are weighted using a triangular kernel and the district population in 2001. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

C Appendix: Additional Results on Household Credit Access

C.1 Are Minority Households Credit-Constrained?

The AIDIS provides detailed information of household borrowings from both bank and non-bank sources. We use this data to understand whether Muslim minority households face binding credit constraints. We extend the methodology of Banerjee and Duflo (2014) to test for credit constraints. Specifically, as banks are the cheapest source of credit, we would expect households to substitute non-bank sources of credit with bank credit in the absence of credit constraints.¹ In particular, there would not be an increase in aggregate household borrowings. Alternately, if minority households faced binding credit constraints, access to cheaper sources of credit should result in an expansion in aggregate household debt (Banerjee and Duflo, 2014).

Columns (2) and (4) of Appendix Table C5, along with columns (6) and (8) point to a reallocation of household borrowings from banks to non-bank financial institutions (NBFIs) along both the extensive, and intensive margins.² Importantly, column (5) points to a INR 22,000 increase in aggregate household borrowings. While not statistically significant (p-value .221), the point estimate is large and reflects a 49 percent increase in household credit for minority households in treated areas. In addition to higher bank loans, the increase in household debt emanates from a statistically non-significant increase in borrowing from informal sources (column (7)).

Appendix Table C6 disaggregates informal loans into those from professional money lenders and community networks, such as friends and relatives.³ Columns (1) and (2) suggest that the treatment reduced minority households' likelihood of borrowing from professional money lenders and input suppliers, but increased their likelihood of having an informal loan through community networks. Column (7) suggests that the increase in informal borrowings in response to the directed credit policy came solely through borrowings from community networks. However, the lack of precision of the point estimates limits our ability to draw any definite conclusions from these results. Columns (3)-(5) offer an alternate test for credit constraints by comparing whether minority households had an outstanding loan from both bank and informal sources. Column (3) shows a 3.6 ppt. (p-value .059) increase in minority households' likelihood of having loans from both banks and informal sources. Column (5) shows this to be driven by a statistically significant (p-value .017) increase in minority households having both bank loans, and loans from community networks.

While not always precisely estimated, Appendix Tables C5 and C6 offer evidence consistent with minority households facing binding credit constraints. Access to directed bank credit allowed these households to substitute credit from relatively more expensive institutional sources such as NBFCs. The remaining borrowing needs were accounted for by community networks, with minority households in treated areas being more prone to having outstanding loans from both banks and community networks.

C.2 Loan Characteristics

This section documents characteristics of bank loans in minority concentration areas in terms of loan duration and purpose of borrowing.

¹ Commercial bank lending rates were 1 percentage point lower than co-operative bank lending rates, and 5 percentage points lower than rates charged by non-banking financial corporations.

² Non-bank financial institutions include both co-operative banks, as well as non-banking financial corporations such as formal microfinance institutions.

³ Professional money lenders also include input suppliers. Loans from community networks also include loans from landlords due to anecdotal evidence on extreme residential segregation along religious lines in India.

C.2.1 Loan Duration

The AIDIS provides qualitative indicators on loan duration, which we recode to define long-term loans as loans exceeding 1 year in duration. Column (1) of Appendix Table C8 identifies a doubling in minority households' likelihood of having a long-term bank loan. There is also a corresponding reduction in the likelihood of having a long-term loan from non-bank financial institutions [column (3)], pointing to the substitution of non-bank credit with cheaper bank loans by minorities in treated districts. Access to long-term bank credit at relatively low cost, combined with a flexible repayment schedule, offer households the opportunity to utilize these loans for long-term investments yielding higher returns.

C.2.2 Purpose of Borrowing

We use information on the self-reported purpose of household borrowing to identify the purpose for which minority households obtained bank credit in response to the directed credit policy. Appendix Table C9 identify a positive and significant impact for farm and expenditure loans. The INR 6359 (15232) increase in bank loans for farm (expenditure) purposes is substantial, when considering that the corresponding control group mean equaled INR 3910 (11635). Overall, a third of the intensive margin increase in minority bank credit can be attributed to farm loans. Appendix Table C10 disaggregates expenditure loans into its three major constituents. Columns (1)-(3) identify positive treatment effects on the likelihood of receiving a bank loan for all three of housing, consumption, and health and education purposes. The point estimates are both statistically and economically significant. Along the intensive margin though, the increase in bank expenditure loans is driven by loans obtained for the purpose of housing, which include both loans to purchase new residential properties, or augment existing properties. The RD estimate for health and education loans is also positive and statistically significant, but an order of magnitude smaller than housing loans. As farm, education, and housing loans all qualify for priority sector lending, higher loan disbursement in these categories would be consistent with banks' incentive to use the directed credit policy to meet their regulatory targets.

C.3 Directed Credit Policy and Household Assets

Section C.2.2 showed that the expansion in bank credit for minority households in minority concentration areas were primarily in the form of farm and expenditure loans. The increase in the latter category was driven by housing loans. To this effect, we use the rich data on household assets from the AIDIS to identify whether the expansion in bank credit also translated into increased holding of farm assets and real estate by minority households.

C.3.1 Farm Assets

The AIDIS provides information on household land holdings, and the value of livestock, farm machinery and transport equipment owned by the household. Columns (1) and (2) of Appendix Table C11 identifies a large and statistically significant increase in minority households' ownership of farm machinery along both the extensive and intensive margins. The intensive margin increase is equivalent to 50 percent of the treatment effect for farm bank loans. Additionally, column (3) shows that minority households in treated districts were also more likely to own transport for farm activities. Columns (4) and (6) point to higher values of farm transport owned, driven by a 57% increase in the value of tractors, although the point estimates are statistically significant only at the 15% level (p-value .116)

Alongside investing in farm machinery, farm loan recipients could have expanded their land holdings, or undertaken long-term investments in the quality of agricultural land. Alternatively, the use of farm machinery could have boosted crop yields, the sales of which could have been used to increase

land holdings or undertake long-term investments in land quality. Columns (1)-(3) of Appendix Table C12 suggests improvements in the quality of farm land owned by minorities in minority concentration areas. Minority households saw a 9 ppt. (p-value .103) increase in the likelihood of having irrigated land, and their irrigated farm area also increased by 0.12 hectares (p-value .110). The treatment effect on overall farm area in column (4) however is attenuated towards 0 and statistically non-significant. This is indicative of private investments in irrigation, as opposed to the acquisition of irrigated land. Columns (3) and (5) show large and statistically significant increases in both the value of irrigated farm land, and overall farm land. Comparing the treatment effects in columns (3) and (5), half of the increase in aggregate farm value emanated from the increase in the value of irrigated farm land owned by minorities. There is however no corresponding investment in the ownership of livestock, as seen from columns (6) and (7) of Appendix Table C12. Collectively, access to farm loans from banks increased minority households' ownership of farm machinery and irrigated farm lands. While overall acreage remained unchanged, there were improvements in the quality of agricultural land owned by minorities through private investments in irrigation, along with increased mechanization of farm activities.

C.4 Non-Farm Assets

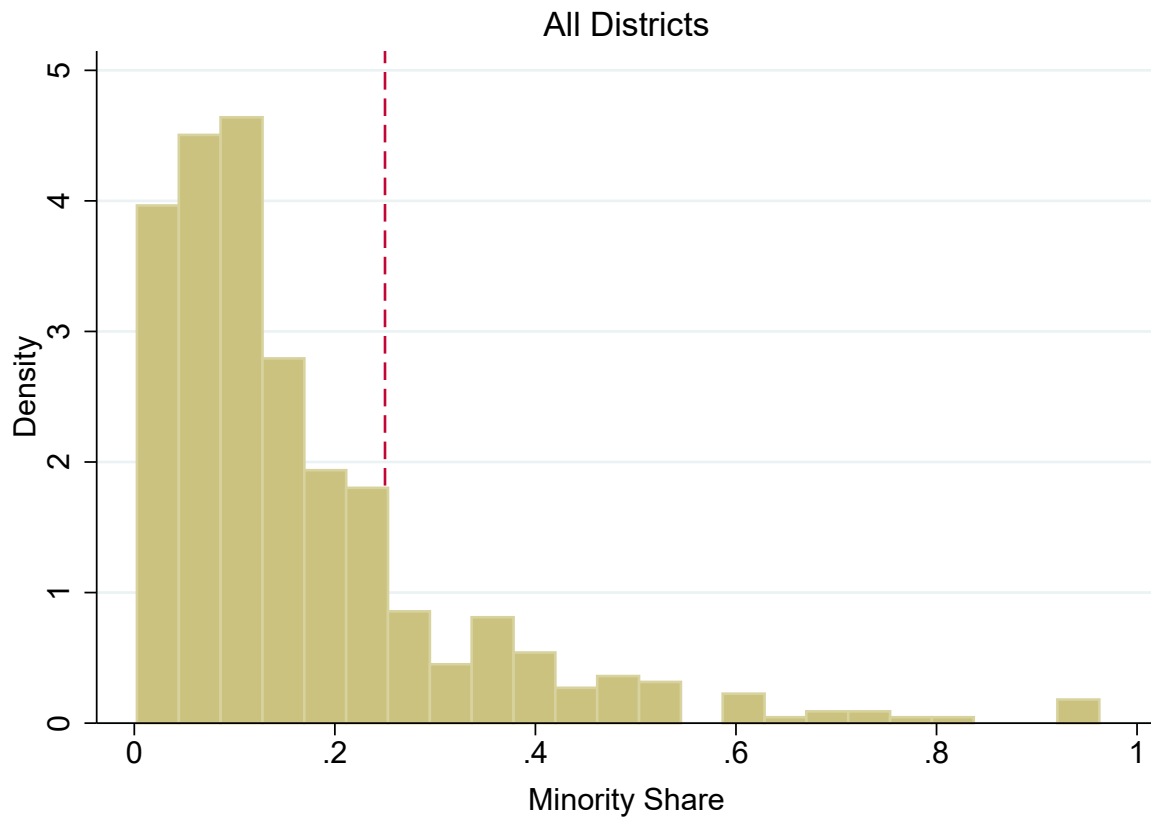
In addition to farm loans, Appendix Table C10 identified a positive treatment effect on bank credit for housing purposes. These could be for the construction of new houses, or improving existing structures. Consistent with the increase in housing credit, column (1) of Appendix Table C13 identifies a weakly significant (p-value .092) increase in the value of residential buildings owned by minority households in minority concentration areas. Akin to the findings on farm land, the increase in the value of buildings is unaccompanied by any increase in residential building area. This indicates that housing loans from banks is used to upgrade existing real estate, as opposed to purchasing additional properties. Not unsurprisingly, the positive treatment effects on farm value and residential buildings combine to yield a significant increase in the value of real estate owned by minority households in minority concentration areas [column (3)].

The treatment effect for housing values is an order of magnitude larger than the treatment effect for housing credit for banks. This leads to the question of whether the directed credit policy led to an overall appreciation of asset prices in minority concentration areas through general equilibrium effects.⁴ We test this by comparing the value of real estate owned by non-minority households across minority concentration and non-minority concentration areas. In the event of an overall increase in property prices, we would expect an appreciation of asset values for non-minority households also. Columns (1)-(4) of Appendix Table F4 shows overall expenditure loans and loans for housing purposes to be unchanged across the discontinuity threshold for non-minority households. However, as seen from column (2), there is a significant (p-value .082) increase in the value of residential buildings owned by non-minority households in minority concentration areas. The appreciation in real estate values occur, despite noisy evidence in column (1) of a 25 percent reduction in the area of buildings owned (p-value .133). Column (3) shows a weakly significant increase in the overall value of land and buildings owned by non-minority households in minority concentration areas. The treatment effect on overall real estate for non-minorities is almost identical in magnitude to the treatment effect for buildings, implying that the increase in the value of physical assets emanated solely from the increase in building values. Resultantly, while the directed credit policy had positive wealth effects for minority households, it also had an indirect spillover on the wealth of non-minority households through the appreciation of local asset prices.

⁴ Easy availability of bank credit could have incentivized households to move to minority concentration areas. Alterately, improvements in farm quality through investments in irrigation could have also made minority concentration areas more attractive.

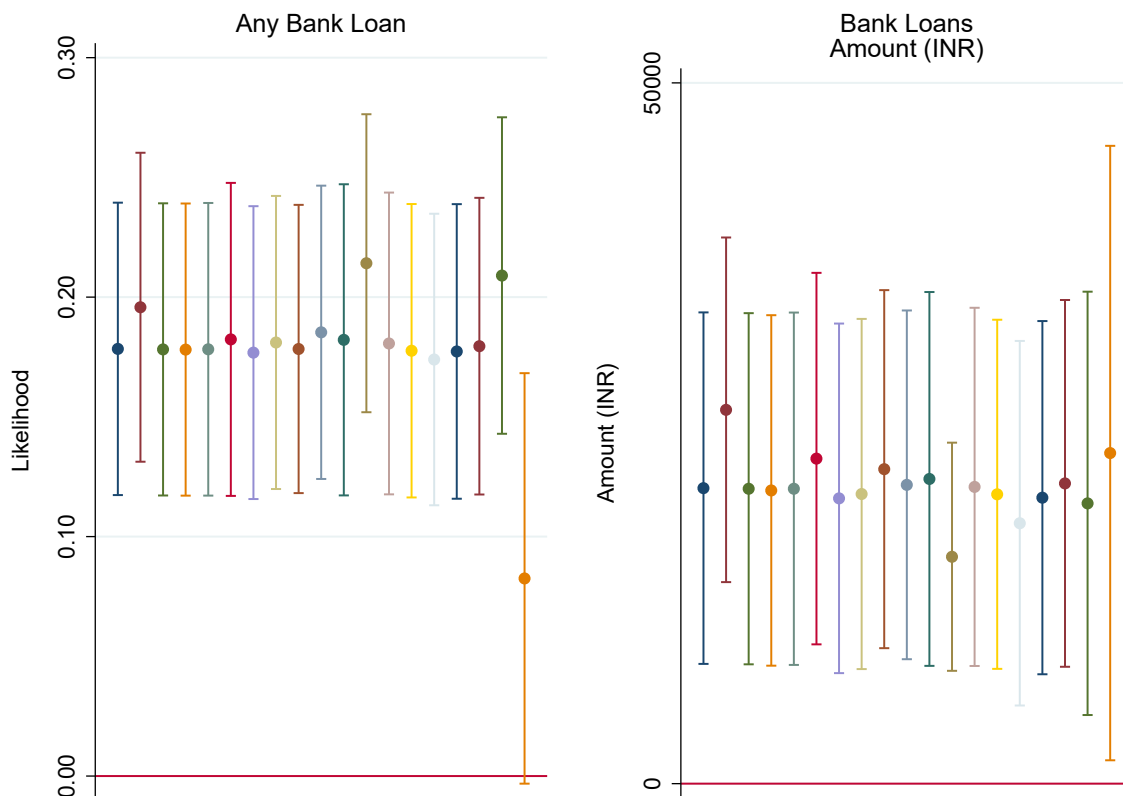
C.5 Figures

Figure C1: Distribution of Running Variable



Notes: This figure shows the distribution of the share of religious minorities across districts in India. Religious minorities include the religious denominations Muslims, Christians, Sikhs, Buddhists and Zoarastrians, classified as religious minorities in the Census 2001. The population shares are computed using the 2001 Census. The red vertical line shows the population threshold of 25 percent, using to classify districts as “minority concentration”.

Figure C2: Bank Credit for Minority Households: Robustness to Dropping Individual States



Notes: This figure shows the robustness of the baseline results to dropping individual states. The unit of observation is the household. The outcome of interest in the left-panel is a dummy equaling 1 if the household has any bank loan; in the right-panel, the volume of bank loans (initial loan value). The vertical lines denote 95% confidence intervals of the coefficient estimates. The coefficients are obtained from estimating our baseline specification, but dropping one state at a time. All specifications include state fixed effects, household covariates and a linear polynomial in the running variable. Standard errors are clustered by district. Estimates are weighted using a triangular kernel and household specific weights provided by the AIDIS.

C.6 Tables

Table C1.A: Summary Statistics: Household Characteristics

	N	Mean	SD
Rural	97154	0.673	0.469
Minority	97154	0.150	0.357
Muslim	97154	0.113	0.316
SC/ST	97154	0.287	0.452
OBC	97154	0.456	0.498
Any secondary educated individual	97154	0.614	0.487
Any higher educated individual	97154	0.234	0.423
Farm	97154	0.372	0.483
Non-farm self-Employed	97154	0.155	0.362
Non-farm casual Work	97154	0.183	0.387
Non-farm salaried Work	97154	0.197	0.398
Monthly Per Capital Household Expenditures	97154	2643.365	2333.916

Notes: This table shows the summary statistics for household demographic characteristics from the AIDIS, 2019

Table C1.B: Summary Statistics: Household Credit, Extensive Margin

	N	Mean	SD
Any loan	97154	0.417	0.493
Any bank loan	97154	0.231	0.422
Any commercial bank loan	97154	0.171	0.377
Any bank-SHG loan	97154	0.073	0.261
Any non-bank financial institution loan	97154	0.106	0.308
Any money-lender loan	97154	0.131	0.338
Any community network loan	97154	0.085	0.279
Any farm loan	97154	0.132	0.338
Any non-farm business loan	97154	0.037	0.189
Any expenditure loan	97154	0.168	0.374

Notes: This table shows the summary statistics for household credit along the extensive margin. Bank-SHG loans are loans issued by bank-linked SHGs. Non-bank financial institution loans include loans issued by co-operative banks and other non-banking financial corporations. Money-lender loans include loans obtained from input-suppliers. Community network loans include loans obtained from friends, relatives, employers and landlords. Expenditure loans include loans obtained for consumption, housing, health and education.

Table C1.C: Summary Statistics: Household Credit, Intensive Margin

	N	Mean	SD	P25	P50	P75
Total loan	60683	216998	442825.5	30501	73337	203339
Total bank loan	30003	235294	497198.3	40000	75787	201669
Commercial bank loan	24553	298823	562999.3	50835	101669	285750
Bank-SHG loan	6342	44964	33412.7	20334	40000	54570
Non-bank financial institution loan	12886	138326	286264.6	28000	50835	105501
Money-lender loan	16864	111975	159061.9	20334	50835	130968
Com loan	14125	70170	121709.9	10167	26501	70000
Farm loan	15534	147068	242104.8	32742	71169	159008
Non-Farm business loan	5555	222833	456930.3	39222	75000	203339
Health/education loan	8334	106350	230131.5	20334	50000	101669
Expenditure loan	22540	98220	172915.8	20334	45751	100835

Notes: This table shows the summary statistics for household credit along the intensive margin. The sample is limited to households with some outstanding loan in each category. Bank-SHG loans are loans issues by bank-linked SHGs. Non-bank financial institution loans include loans issued by co-operative banks and other non-banking financial corporations. Money-lender loans include loans obtained from input-suppliers. Community network loans include loans obtained from friends, relatives, employers and landlords. Expenditure loans include loans obtained for consumption, housing, health and education.

Table C1.D: Summary Statistics: Loan Characteristics

	N	Mean	SD
Any long-term loan	60683	0.835	0.371
Any long-term bank loan	30003	0.824	0.381
Any unsecured loan	60683	0.622	0.485
Any unsecured bank loan	30003	0.370	0.483
Any unsecured NBFI loan	60683	0.074	0.261
Any unsecured money-lender loan	16864	0.798	0.402
Any unsecured friend/relatives loan	14125	0.914	0.280
Interest rate	60675	0.140	0.126
Bank interest rate	30003	0.106	0.048
NBFI interest rate	12886	0.122	0.076
Money-lender interest rate	16864	0.303	0.175
Friends/relatives interest rate	14125	0.021	0.090
Loan delinquent	57286	0.347	0.476
Bank loan delinquent	28766	0.264	0.441
Informal loan delinquent	27154	0.413	0.492

Notes: This table shows the summary statistics for other loan characteristics. Bank-SHG loans are loans issues by bank-linked SHGs. Non-bank financial institution (NBFI) loans include loans issued by co-operative banks and other non-banking financial corporations. Money-lender loans include loans obtained from input-suppliers. Community network loans include loans obtained from friends, relatives, employers and landlords. Expenditure loans include loans obtained for consumption, housing, health and education. Interest rates are weighted using loan size. Summaries are computed only for households with outstanding loans in the referred categories.

Table C3: Robustness of Baseline Treatment Effects to Alternate Specification Choices

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Loan = 1)			Loan Amount (INR)		
Treat	.295*** (.048)	.177*** (.023)	.178*** (.063)	35987.118** (14191.902)	59420.081*** (5312.745)	21048.977** (8476.572)
Observations	1765	1586	1765	1765	1586	1765
Dep Var Mean	.115	.115	.115	20801.635	20801.635	20801.635
Polynomial	Quadratic	Linear	Linear	Quadratic	Linear	Linear
Dist. Controls	N	Y	N	N	Y	N
Dist. Cluster	Y	Y	N	Y	Y	N

Notes: The above table shows robustness of the baseline treatment effects to alternate specification and sample choices. The unit of observation is the household. The outcome of interest in columns (1)-(3) is a dummy equaling 1 if the household had any outstanding bank loan; in columns (4)-(6), the amount of bank loans received by the household. Columns (1) and (3) include a quadratic polynomial in the running variable. Columns (2) and (5) extend the covariate vector to include select district covariates. Columns (3) and (6) use robust standard errors as opposed to clustering by district. Except for columns (2) and (5), all specifications include a linear polynomial in the running variable, state fixed effects, and household covariates. All specifications are restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using household-specific weights, and a triangular kernel. Standard errors in parentheses, clustered by district, except for columns (3) and (6). Significant levels: *10%, **5%, and ***1%

Table C4: Robustness of Baseline Treatment Effects to Alternate Samples

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Loan = 1)			Loan Amount (INR)		
Treat	.180*** (.032)	.170*** (.028)	.298*** (.107)	21362.226*** (6176.088)	21339.415*** (7120.914)	39614.209** (15574.208)
Observations	1721	2480	3776	1721	2480	3776
Dep Var Mean	.115	.148	.166	20801.763	28452.875	24523.878
Excl. New Dist.	Y	N	N	Y	N	N
Incl. Oth. Minorities	N	Y	Y	N	Y	Y
RD Type	Sharp	Sharp	Fuzzy	Sharp	Sharp	Fuzzy

Notes: The above table shows robustness of the baseline treatment effects to alternate specification and sample choices. The unit of observation is the household. The outcome of interest in columns (1)-(3) is a dummy equaling 1 if the household had any outstanding bank loan; in columns (4)-(6), the amount of bank loans received by the household. Columns (1) and (4) exclude new districts formed from existing Census districts, whose treatment status are unspecified. Columns (2) and (5) expand the sample to include other religious minorities. Columns (3) and (6) use a fuzzy RD specification to include all districts classified as minority concentration. All specifications include a linear polynomial in the running variable, state fixed effects, and household covariates. All specifications restrict the sample to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using household-specific weights, and a triangular kernel. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table C5: Treatment Effect on Overall and Non-Bank Sources of Credit for Minority Households

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pr(Loan = 1)				Loan Amount (INR)			
	Any Source	Bank	Informal	Non-Bank Financial Institutions	Total Loans	Banks	Informal	Non-Bank Financial Institutions
Treat	.015 (.080)	.178*** (.031)	.040 (.066)	-.144*** (.044)	21564.692 (17613.971)	21048.977*** (6386.733)	10418.295 (8538.272)	-9891.569 (7429.366)
Observations	1765	1765	1765	1765	1765	1765	1765	1765
Dep Var Mean	.323	.115	.165	.091	44059.071	20801.635	14671.783	8566.542

Notes: The above table shows the treatment effect on household credit across alternate sources of credit. The unit of observation is the household. The outcome of interest in columns (1)-(4) is a dummy equaling 1 if the household had any outstanding loan from the sources mentioned; in columns (5)-(8), the amount of bank loans received by the household. *Informal* loans include loans from professional money lenders, input suppliers, and friends, relatives and landlord. Non-bank financial institutions include co-operative banks and non-bank financial corporations, including micro-finance institutions. All specifications include state fixed effects, a linear polynomial in the running variable, and controls for household size and rural location. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table C6: Treatment Effect on Sources of Informal Credit for Minority Households

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Pr(Any Loan = 1)		Pr(Bank and Informal Loan = 1)			Loan Amount (INR)	
	Professional Money Lenders	Community Networks	Any Informal Source	Professional Money Lenders	Community Networks	Professional Money Lenders	Community Networks
Treat	-.051 (.039)	.092 (.081)	.036* (.019)	-.001 (.013)	.037** (.015)	-238.169 (1217.230)	10656.464 (9051.379)
Observations	1765	1765	1765	1765	1765	1765	1765
Dep Var Mean	.064	.103	.030	.012	.018	4668.111	10003.671

Notes: The above table shows the treatment effect across sources of informal credit. The unit of observation is the household. The outcome of interest in columns (1)-(5) is a dummy equaling 1 if the household had any outstanding loan from the sources mentioned; in columns (6)-(7), the amount of bank loans received by the household. *Professional Money Lenders* include loans from professional money lenders and input suppliers. *Community Networks* include loans from friends, relatives, employers and landlord. The outcome of interest in column (3) is a dummy equaling 1 if the household has an outstanding loan from a bank and any informal source; in column (4), a dummy equaling 1 if the household has an outstanding loan from a bank and a professional money lender; in column (5), a dummy equaling 1 if the household has an outstanding loan from a bank and community networks. All specifications include state fixed effects, a linear polynomial in the running variable, household size and a rural indicator. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table C7: Treatment Effect on Loan Delinquency

	(1)	(2)	(3)	(4)
	Pr(Any Delinquent Loan = 1)			
	Banks	Bank SHG	Commercial Banks	Any Source
Treat	-.045 (.066)	-.250*** (.076)	.022 (.070)	.064 (.056)
Observations	465	95	376	991
Dep Var Mean	.292	.348	.281	.315

Notes: The above table shows the treatment effect on the likelihood of having a delinquent loan for minority households. The unit of observation is the household. The outcome of interest is a dummy equaling 1 if the household has a delinquent loan from the sources mentioned. A loan is classified as delinquent if there has been no repayment towards that loan for at least 6 months preceding that survey. *Bank-SHG* in column (2) refers to loans received through bank-linked self-help groups. *Commercial Banks* in column (3) refers to loans received directly from commercial banks. All specifications include state fixed effects, a linear polynomial in the running variable, and controls for household size and rural location. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table C8: Treatment Effect on Loan Duration: Minority Households

	(1)	(2)	(3)	(4)
	Pr(Loan Duration > 1 Yr = 1)			
	Banks	Informal	Non Bank Financial Institution	Any Source
Treat	.125*** (.026)	-.022 (.057)	-.118*** (.028)	-.070 (.067)
Observations	1765	1765	1765	1765
Dep Var Mean	.101	.141	.071	.275

Notes: The above table shows the treatment effect on the likelihood of having a long-term loan for minority households. Long-term loans are defined as a loan whose duration exceeds 1 year. The unit of observation is the household. The outcome of interest is a dummy equaling 1 if the household has a long-term loan from the sources mentioned. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table C9: Purpose of Bank Credit: Minority Households

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Loan = 1)			Loan Amount (INR)		
	Farm	Non Farm	Expenditure	Farm	Non Farm	Expenditure
Treat	.051*** (.019)	.018 (.024)	.092*** (.027)	6359.716** (2701.171)	-862.725 (2177.439)	15231.539** (6554.996)
Observations	1765	1765	1765	1765	1765	1765
Dep Var Mean	.033	.014	.051	3909.646	2064.700	11634.652

Notes: The above table shows the treatment effect for the purpose of bank borrowing for minority households. The unit of observation is the household. The outcome of interest in columns (1)-(3) is a dummy equaling 1 if the household had any outstanding bank loan for the purpose mentioned; in columns (4)-(6), the amount of bank loans received by the household for the purpose mentioned. *Expenditure* loans include household loans obtained for expenditure on consumption items, health and education, and housing. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table C10: Purpose of Bank Expenditure Loans: Minority Households

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Loan = 1)			Loan Amount (INR)		
	Health and Education	Housing	Consumption	Health and Education	Housing	Consumption
Treat	.018*** (.005)	.050** (.023)	.045*** (.016)	609.541** (254.883)	13271.860** (6604.250)	1350.138 (1685.966)
Observations	1765	1765	1765	1765	1765	1765
Dep Var Mean	.004	.020	.027	773.008	8902.867	1958.777

Notes: The above table shows the treatment effect on three types of expenditure loans for minority households. The unit of observation is the household. The outcome of interest in columns (1)-(3) is a dummy equaling 1 if the household had any outstanding bank loan for the purpose mentioned; in columns (4)-(6), the amount of bank loans received by the household for the purpose mentioned. *Consumption* loans include household loans obtained for household expenditures which does not include spending on health or education, or spending on housing materials, land, or the direct purchase of housing. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table C11: Treatment Effect on Minority Household Farm Assets

	(1)	(2)	(3)	(4)	(5)	(6)
	Farm Machinery			Farm Transport		
	Any Machine	Machine Value	Any Farm Transport	Farm Transport Value	Any Tractor	Tractor Value
Treat	.261*** (.100)	3007.985*** (1072.893)	.036*** (.014)	1570.863 (1143.729)	.001 (.003)	1686.520 (1067.701)
Observations	1765	1765	1265	1765	1765	1765
Dep Var Mean	.390	2772.477	.049	3322.419	.012	2961.867

Notes: The above table shows the treatment effect on household farm machinery and farm transport assets. The unit of observation is the household. The outcome of interest in column (1) is a binary equaling 1 if the household owned any farm machinery; in column (2), value of farm machinery (INR); in column (3), a binary equaling 1 if the household owns any farm transport; in column (4), the value of farm transport (INR); in column (5), a binary equaling 1 if the household owned a tractor; in column (6), the value of tractors (INR). All specifications include state fixed effects, a linear polynomial in the running variable and household covariates. All specifications are weighted using a triangular kernel and AIDIS assigned household weights. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table C12: Treatment Effect on Minority Household Farm Land and Livestock

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Farm Land					Livestock	
	Any Irrigated Land	Irrigated Farm Area	Irrigated Farm Value	Farm Area	Farm Value	Any Livestock	Livestock Value
Treat	.094 (.058)	.120 (.075)	144539.375***** (43900.653)	-.027 (.061)	308104.549***** (93520.890)	.115 (.126)	-751.228 (1696.726)
Observations	1765	1765	1765	1765	1765	1765	1765
Dep Var Mean	.157	.207	198955.22	.615	372481.147	.311	9218.514

Notes: The above table shows the treatment effect for farm and livestock outcomes for minority households. The unit of observation is the household. The outcome of interest in column (1) is a dummy equaling 1 if the household owns any irrigated land; in column (2), the area of irrigated farm land owned; in column (3), value of irrigated farm area; in column (4), total farm land owned; in column (5), value of farm land owned; in column (6), a binary equaling 1 if the household owns any livestock; in column (7), the value of livestock owned. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. All specifications are weighted using a triangular kernel and AIDIS assigned household weights. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table C13: Treatment Effect on Real Estate and Household Assets: Minority Households

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Real Estate			Machinery			Transport		
	Res.Bld. Area	Res. Bld. Value	Land and Buildings	Any Machine	Machine Value	Any Trans.	Trans. Value	Any Household Trans.	Household Trans. Value
Treat	-.088 (8.772)	238534.989* (141360.523)	945713.840*** (176309.735)	-.020 (.055)	-1478.255 (1069.128)	.018 (.083)	11914.879** (5651.738)	.064 (.058)	5782.258 (3787.665)
Observations	1765	1765	1765	1765	1765	1765	1765	1265	1765
Dep Var Mean	63.692	549001.824	1527689.182	.078	2197.101	.720	61334.582	.946	50310.033

Notes: The above table shows the treatment effect on minority households' ownership of real estate and non-farm productive assets. The unit of observation is the household. The outcome of interest in column (1) is the area of residential real estate; in column (2), the value of residential real estate; in column (3), the combined value of land and buildings; in column (4), a binary equal to 1 if the household owns any non-farm machinery; in column (5), the value of non-farm machinery; column (6), a binary equal to 1 if the household owns any transport equipment; in column (7), the value of transport equipment; in column (8), a binary equaling 1 if the household owns any transport equipment for household use; in column (9), the value of transport owned for household use. All specifications include state fixed effects, a linear polynomial in the running variable and household covariates. All specifications are weighted using a triangular kernel and AIDIS assigned household weights. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table C14: Treatment Effects on the Composition of Bank-Linked Self-Help Groups

Panel A: 2008-2019							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	SHGs Formed					SHG Size	
	All SHG	Total Muslims	Some Muslim	Only Muslim	All SHG	All SHG	Only Muslim
Treat	6.283 (23.646)	78.083* (41.148)	5.851 (5.089)	5.357* (2.818)	.352 (2.494)	-.311 (.306)	-.155 (.319)
Observations	3682	3682	3682	3682	3682	3682	1255
Dep Var Mean	74.694	87.793	12.734	4.230	8.368	11.511	11.026
Panel B: 2014-2019							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	SHGs Formed					SHG Size	
	All SHG	Total Muslims	Some Muslim	Only Muslim	All SHG	All SHG	Only Muslim
Treat	17.058 (33.242)	116.917** (55.439)	9.582 (6.805)	8.000** (3.610)	1.400 (3.475)	-.208 (.285)	-.015 (.232)
Observations	2151	2151	2151	2151	2151	2151	797
Dep Var Mean	105.092	122.681	18.196	5.464	12.552	11.390	11.005

Notes: This table shows the treatment effect on the social composition of bank-linked self-help groups (SHG). The unit of observation is the district-bank. Panel A considers bank-linked SHGs registered between 2008 and 2019; panel B considers bank-linked SHGs registered between 2014 and 2019. The outcome of interest in column (1) is the total number of bank-linked SHGs registered; in column (2), the total number of Muslim members in bank-linked SHGs; in column (3), the number of bank-linked SHGs with a Muslim member; in column (4), the number of bank-linked SHGs registered with only Muslim members; in column (5), the number of bank-linked SHGs registered with both Muslim and non-minority members; in column (6), average SHG size; in column (7), the average size of SHGs with only Muslim members. All specifications include state, bank and SHG formation year fixed effects, along with a linear polynomial in the running variable. All specifications are weighted using a triangular kernel. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table C14: Treatment Effects on the Composition of Bank-Linked Self-Help Group Formation: All Minorities

	(1)	(2)	(3)
	Total Minorities	Some Minority	Only Minorities
Treat	100.697 (67.827)	2.945 (9.740)	8.843** (4.058)
Observations	2151	2151	2151
Dep Var Mean	160.504	25.885	6.745

Notes: This table shows the treatment effect on the social composition of bank-linked self-help groups (SHG). The unit of observation is the district-bank. The outcome of interest in column (1) is the total number of religious minority members in bank-linked SHGs; in column (2), the number of bank-linked SHGs with a religious minority member; in column (3), the number of bank-linked SHGs registered with only religious minority members. All specifications include state, bank and SHG formation year fixed effects, along with a linear polynomial in the running variable. All specifications are weighted using a triangular kernel. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table C15: Minority Credit Access and District Population

	(1)	(2)	(3)	(4)	(5)
	Share of Households		Population		
	Muslim	Hindu	Muslim	Hindu	Total
Treat	-.043 (.078)	.132 (.127)	.010 (.175)	.353 (.382)	.310 (.551)
Observations	61	61	61	61	61
Dep Var Mean	.147	.700	.280	1.338	1.743

Notes: The above table shows the treatment effect on district population. The unit of observation is the district. The outcome of interest in the first two columns is the share of households in the community of interest; in columns (3)-(4), the district population (in millions) for the community of interest; in column (5), the total district population. The data is from the AIDIS, conducted in 2019. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel. Standard errors in parentheses. Significant levels: *10%, **5%, and ***1%

D Appendix: Labor Market Outcomes

Recent work by Breza and Kinan (2021) documented the adverse labor market impacts of credit contraction by micro-finance institutions in India, both through its direct impact on business operations, and indirect effects by lowering aggregate demand. Bruhn and Love (2014) showed financial deepening in Mexico to allow self-employed business owners to continue operating their business, and not have to seek wage employment. This section details the impact of increased credit access on labor market outcomes for minority households.

At the outset, the directed credit policy could have altered the occupational choices of minority workers through two possible channels. First, higher farm credit could have directly affected farm labor. Appendix C.3.1 points to the increased mechanization of farm work through investments in farm machinery and irrigation, but no accompanying increase in land holdings. If households used farm credit from banks to invest in labor saving technology, it could have freed labor for non-farm activities – the labor push channel, seen in Bustos et al. (2016). Alternately, the mechanization of agriculture could have improved farm yields, leading to higher local demand. This in turn could have generated higher local non-farm employment to meet the additional demand for goods and services – the labor pull channel, evidenced by Emerick (2018). Relatedly, expenditure loans from banks could also have increased local demand for goods and services, generating higher demand for non-farm labor.

We examine the labor market impacts of the directed credit policy using data from the Primary Labour Force Survey (PLFS). The PLFS is a nationally representative survey conducted by the NSS and provides data on labor force characteristics. It is the successor to the quinquennial employment-unemployment surveys conducted by the NSS between 1987 and 2011. We use weekly data from the PLFS which inquires individuals for their labor force participation for each of the 7 days preceding the survey. Conditional on participation, the PLFS also provides detailed information on the type of activity undertaken, as well as the hours of work accorded to that activity in each of the seven days. Locational identifiers at the level of district, along with information on individual age, educational attainment, caste and religion are also provided in the PLFS.

Summary statistics of weekly labor market activities are shown in Appendix Tables D1.A and D1.B. Religious minorities comprised 15% of working-aged individuals and overall weekly participation in the labor force participation was 55%. Conditional on participation in the labor force, individuals worked 46 hours per week, with 40% of workers engaged in farm activities. 30% of the workers were engaged in trade or service activities, with manufacturing and construction work accounting for approximately 10% of workers. 12% of weekly non-farm workers reported being self-employed. Amongst manufacturing workers, a third were self-employed, and three-fourths of self-employed manufacturing workers operated establishments located within their residence.

We extend our RD strategy for causal identification and estimate the following specification:

$$Y_{ids} = \alpha_s + \beta Treat_{ds} + f(Runvar_{ds}) + \gamma \mathbf{Z}_{ids} + \epsilon_{ids} \quad (D5)$$

The unit of observation is now the working-aged individual, defined as individuals aged between 18 and 65. $\mathbf{Z}$ includes individual controls for gender, and a quadratic in age, in addition to binary dummies for individual caste categories. We continue to include a linear polynomial in the running variable. A triangular kernel is used to weight the regressions, along with individual-specific weights provided by the PLFS. Standard errors are clustered by district for inference and the sample is restricted to districts located within a bandwidth of .058 around the discontinuity threshold. We identify the impact of the directed credit policy on labor market outcomes along both the extensive and intensive margins.

Columns (1) and (2) of Appendix Table D2 shows that the directed credit policy left unaffected overall labor force participation, or unemployment rate for minority workers. Column (5) rules out the labor push channel by identifying a positive, but statistically non-significant coefficient corresponding

to farm work. Instead, columns (3) and (4) points to a reallocation of labor from salaried work to self-employment: minority workers in minority concentration areas have a 11 ppt. higher likelihood of being self-employed during the week, and a 9 ppt. lower likelihood of being engaged in salaried work. Columns (7) and (8) offer evidence of sectoral reallocation in response to the directed credit policy: minority workers in treated districts were 6 ppt. more likely to be employed in the manufacturing sector, with a comparable decline in their likelihood of working in trade and services. Appendix Table D3 identifies very similar effects along the intensive margin. While total labor hours during the week was unaffected by the treatment (column (1)); column (2) shows 5 additional hours of self-employment for minority workers, accompanied by a decline in wage work of 6 hours (column (3)). Likewise, column (4) shows time spent in farm activities to be unimpacted by the directed credit policy, and column (6) reports 3 additional hours of manufacturing work. As seen in Appendix Tables D4 and D5, the increase in manufacturing employment is primarily in the form of self-employment, as opposed to wage work.

Appendix Tables D4-D5 examine this reallocation across occupation type and sectors by combining workers' occupation with their sector of employment. Across both tables, columns (3) and (6) show that the treatment led to minority workers reallocating their labor from wage work in trade and services, to manufacturing self-employment. The findings are similar to Bruhn and Love (2014), who showed that the sudden expansion of bank branches in Mexico led to higher survival of small business owners, and reduced their propensity to engage in wage work. This is however contrary to Gupta and Dehejia (2021), who found that financial deepening by state-owned banks in India facilitated a shift away from self-employment towards wage work. Disaggregating the manufacturing, trade and service sectors into its largest components, Appendix Table D6 shows that the positive treatment effect for minority workers in manufacturing employment was driven by higher self-employment in textiles, with labor being drawn away from wage work in the transport and hospitality sectors.

Overall, there is little evidence to suggest that the mechanization of farm work led to labor being pushed out from farm to non-farm activities, negating the labor push channel. Instead, the directed bank credit resulted in the reallocation of minority workers away from wage work in the transport and hospitality sectors, to self-employment in textile manufacturing.

D.1 Tables

Table D1.A: Summary Statistics: Working-Aged Individuals

	N	Mean	SD
Hindu	229488	0.843	0.363
Muslim	229488	0.116	0.320
Christian	229488	0.018	0.134
Sikh	229488	0.004	0.065
SC/ST	229488	0.278	0.448
OBC	229488	0.438	0.496
Rural	229488	0.692	0.462
Female	229488	0.501	0.501
Age	229488	35.797	12.060
Secondary or Higher Educated	229488	0.380	0.485
Higher Educated	229488	0.133	0.340
LFP	229488	0.556	0.497

Notes: This table shows the summary statistics for working-aged individuals from the Primary Labour Force Survey, 2017-18. Working-aged individuals are individuals aged between 18 and 60. LFP is labour force participation rate.

Table D1.B: Summary Statistics: Workforce Characteristics

	N	Mean	SD
Unemployed	126361	0.087	0.282
Farm	126361	0.378	0.485
Manufacturing	126361	0.114	0.318
Construction	126361	0.108	0.311
Trade	126361	0.095	0.293
Services	126361	0.207	0.405
Hours Worked	126361	46.267	19.816
Farm Hours	126361	17.366	23.608
Manufacturing Hours	126361	6.187	17.871
Trade Hours	126361	5.577	17.550
Construction Hours	126361	5.110	15.246
Service Hours	126361	11.416	23.029
Non-Farm Self-Employed	126361	0.122	0.328
Non-Farm Self-Employed, Home Establishment	126361	0.066	0.247
Manufacturing Self-Employed	126361	0.043	0.202
Manufacturing Self-Employed, Home Establishment	126361	0.030	0.172

Notes: This table shows the summary statistics from the Primary Labour Force Survey, 2017-18, for individuals participating in the labour force. The sample is restricted to individuals participating in the labour force during the week and aged between 18 and 60. Hours worked refer to hours worked in the 7 days preceding the survey.

Table D2: Treatment Effects on Weekly Labour Market Activities: Extensive Margin

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pr(Outcome = 1)							
	Labour Force Participation	Unemployed	Self Employed	Wage Work	Farm	Non Farm	Manufacturing	Trade and Services
Treat	.020 (.027)	.002 (.012)	.107*** (.027)	-.089*** (.026)	.026 (.042)	-.016 (.032)	.060*** (.013)	-.058** (.027)
Observations	3379	3379	3379	3379	3379	3379	3379	3379
Dep Var Mean	.497	.048	.232	.217	.093	.352	.103	.187

Notes: The above table shows the treatment effect on weekly labour market outcomes along the extensive margin for minority working-age individuals. The unit of observation is the working-age individual. Working-age individuals are individuals aged between 18 and 60 years. The outcome of interest in column (1) is a dummy equal to 1 if the individual participated in the labour force in any of the 7 days preceding the survey; in column (2), a dummy equal to 1 if the individual was unemployed in any of the 7 days preceding the survey; in columns (3)-(8), a dummy equal to 1 if the individual was engaged in the employment category/sector referred. All specifications include state and survey month fixed effects; a linear polynomial in the running variable; controls for demographic and education covariates at the individual level; household level controls. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table D3: Treatment Effects on Weekly Labour Market Activities: Intensive Margin

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Hours Worked in						
	Week	Self Employed	Wage Work	Farm	Non Farm	Manufacturing	Trade and Services
Treat	-.232 (1.260)	5.411*** (1.463)	-5.692*** (1.304)	.215 (1.889)	-.447 (1.979)	3.120*** (.923)	-2.492 (1.884)
Observations	3379	3379	3379	3379	3379	3379	3379
Dep Var Mean	23.854	12.317	11.501	4.343	19.511	5.565	10.747

Notes: The above table shows the treatment effect on weekly labour market outcomes along the intensive margin for minority working-age individuals. The unit of observation is the working-age individual. Working-age individuals are individuals aged between 18 and 60 years. The outcome of interest is hours worked per week in the sector/occupation mentioned. All specifications include state and survey month fixed effects; a linear polynomial in the running variable; controls for demographic and education covariates at the individual level; household level controls. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table D4: Treatment Effects on Occupation Type and Sector: Extensive Margin

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Outcome = 1)					
	Farm		Manufacturing		Trade and Services	
	Self Employed	Wage Work	Self Employed	Wage Work	Self Employed	Wage Work
Treat	.040 (.032)	-.014 (.016)	.046*** (.007)	.014 (.009)	.011 (.017)	-.069*** (.016)
Observations	3379	3379	3379	3379	3379	3379
Dep Var Mean	.067	.027	.053	.051	.099	.088

Notes: The above table shows the treatment effect on sectoral employment for minority working-age individuals, disaggregated by type of employment in each sector. The unit of observation is the working-age individual. Working-age individuals are individuals aged between 18 and 60 years. The outcome of interest is a binary equal to 1 if the individual is employed in the sector mentioned. The outcome of interest in the odd-numbered columns refer to self-employment; in the even-numbered columns, employment in wage work. All specifications include state and survey month fixed effects; a linear polynomial in the running variable; controls for demographic and education covariates at the individual level; household level controls. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table D5: Treatment Effects on Occupation Type and Sector: Intensive Margin

	(1)	(2)	(3)	(4)	(5)	(6)
	Hours Worked in					
	Farm		Manufacturing		Trade and Services	
	Self Employed	Wage Work	Self Employed	Wage Work	Self Employed	Wage Work
Treat	1.030 (1.416)	-.815 (.691)	2.433*** (.444)	.687 (.606)	1.478 (1.231)	-3.971*** (1.159)
Observations	3379	3379	3379	3379	3379	3379
Dep Var Mean	3.074	1.269	2.728	2.837	5.819	4.928

Notes: The above table shows the treatment effect on sectoral employment for minority working-age individuals, disaggregated by type of employment in each sector. The unit of observation is the working-age individual. Working-age individuals are individuals aged between 18 and 60 years. The outcome of interest is the hours of work undertaken in the week preceding the survey in the sector mentioned. The outcome of interest in the odd-numbered columns refer to self-employment; in the even-numbered columns, employment in wage work. All specifications include state and survey month fixed effects; a linear polynomial in the running variable; controls for demographic and education covariates at the individual level; household level controls. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table D6: Disaggregating Non-Farm Employment for Minority Workers, Extensive Margin

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pr(Outcome = 1)							
	Manufacturing			Trade and Services				
	Food	Textiles	Other	Trade	Services	Transport	Hospitality	Others
Treat	.002 (.002)	.061*** (.009)	-.003 (.006)	.017 (.025)	-.075*** (.017)	-.038** (.015)	-.064*** (.015)	.027* (.017)
Observations	3379	3379	3379	3379	3379	3379	3379	3379
Dep Var Mean	.017	.045	.041	.082	.105	.039	.011	.054

Notes: The above table shows the treatment effect on weekly labour market outcomes for major sub-sectors in manufacturing, trade and services for minority working-age individuals. The unit of observation is the working-age individual. Working-age individuals are individuals aged between 18 and 60 years. All outcomes of interest are a binary, which equal 1 if the individual is employed in the sector mentioned. Columns (1)-(3) refer to employment in the manufacturing sector; columns (4)-(8) refer to employment in the trade and service sectors. All specifications include state and survey month fixed effects; a linear polynomial in the running variable; controls for demographic and education covariates at the individual level; household level controls. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table D7: Treatment Effect on Non-Farm and Manufacturing Employment: by Concentration of Religious Groups in Industry

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pr(Employment in Sector = 1)							
	High Diversity		High Muslim		High Hindu		High Hindu Self-Employed	
	Non Farm	Mfg.	Non Farm	Mfg.	Non Farm	Mfg.	Non Farm	Mfg.
Treat	.025 (.038)	.061*** (.012)	.004 (.036)	.063*** (.013)	.020 (.016)	-.003* (.002)	-.040 (.032)	.049*** (.009)
Observations	3379	3379	3379	3379	3379	3379	3379	3379
Dep Var Mean	.314	.091	.317	.095	.029	.009	.307	.091

Notes: The above table shows the treatment effect on weekly labour market outcomes along the extensive margin for minority working-age individuals. The unit of observation is the working-age individual. Working-age individuals are individuals aged between 18 and 60 years. The outcome of interest in column (1) is a dummy equal to 1 if the individual participated in the labour force in any of the 7 days preceding the survey; in column (2), a dummy equal to 1 if the individual was unemployed in any of the 7 days preceding the survey; in columns (3)-(6), a dummy equal to 1 if the individual was engaged in the sector referred. All specifications include state and survey month fixed effects; a linear polynomial in the running variable; controls for demographic and education covariates at the individual level; household level controls. The sample is restricted to a bandwidth of .06 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

E Appendix: Outcome Specific MSERD Bandwidths

We replicate in this section some of our key results using outcome-specific data driven MSERD bandwidths recommended by Calonico et al. (2020).

Table E1: Treatment Effect Across Credit Sources for Minority Households: MSERD Bandwidth

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Loan = 1)			Loan Amount (INR)		
	Any Source	Informal	Non-Bank Financial Institutions	Total Loans	Informal	Non-Bank Financial Institutions
Treat	.246*** (.086)	-.021 (.057)	-.139*** (.043)	59844.760*** (16729.043)	21767.700** (9712.502)	-1.00e+04 (7373.080)
Observations	1354	2245	1695	1378	1378	1765
Dep Var Mean	.363	.161	.086	41968.809	10211.738	8566.542
Bandwidth	.042	.072	.057	.045	.046	.058

Notes: The above table shows the treatment effect on non-bank sources of credit for minority households. The unit of observation is the household. The outcome of interest in columns (1)-(3) is a dummy equaling 1 if the household had any outstanding loan from the sources mentioned; in columns (4)-(6), the amount of bank loans received by the household. *Informal* loans include loans from professional money lenders, input suppliers, and friends, relatives and landlord. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. All outcomes are estimated using data-driven MSERD optimal bandwidths, as prescribed by Calonico et al. (2020). All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table E2: Purpose of Bank Credit: Minority Households, MSERD Bandwidth

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Loan = 1)			Loan Amount (INR)		
	Farm	Non Farm	Expenditure	Farm	Non Farm	Expenditure
Treat	.058** (.023)	.038 (.025)	.075*** (.027)	6326.148** (2972.968)	-1092.835 (1918.556)	7459.322 (6868.065)
Observations	1378	1465	2057	1378	2606	2449
Dep Var Mean	.025	.021	.050	3372.215	2764.343	12081.604
Bandwidth	.045	.050	.070	.045	.083	.076

Notes: The above table shows the treatment effect on purpose of bank credit access for minority households. The unit of observation is the household. The outcome of interest in columns (1)-(3) is a dummy equaling 1 if the household had any outstanding bank loan for the purpose mentioned; in columns (4)-(6), the amount of bank loans received by the household for the purpose mentioned. *Expenditure* loans include household loans obtained for expenditure on consumption items, health and education, and housing. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. All outcomes are estimated using data-driven MSERD optimal bandwidths, as prescribed by Calonico et al. (2020). All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table E3: Mechanism: Nature of Bank Lending in Minority Concentration Districts, MSERD Bandwidths

	(1)	(2)	(3)	(4)
	Pr(Loan = 1)		Loan Amount (INR)	
	Commercial Bank	Bank SHG	Commercial Bank	Bank SHG
Treat	.140*** (.027)	.110*** (.020)	16242.768*** (5934.496)	3944.881*** (1333.656)
Observations	1378	1378	1835	1505
Dep Var Mean	.106	.033	20046.831	1105.784
Bandwidth	.046	.047	.059	.052

Notes: The above table shows the treatment effect by type of bank loan for minority households. The unit of observation is the household. The outcome of interest in columns (1)-(2) is a dummy equaling 1 if the household has any outstanding loan from the source mentioned; in columns (3)-(4), the amount of outstanding loan from the source mentioned. *Commercial Bank* refers to loans obtained directly from commercial banks; *Bank SHG* refers to loans received through bank-linked SHGs. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. All outcomes are estimated using data-driven MSERD optimal bandwidths, as prescribed by Calonico et al. (2020). All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table E4: Mechanisms: Treatment Effects on Collateral Requirements and Cost of Credit, MSERD Bandwidth

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Unsecured Loan = 1)			Interest Rates		
	All Bank	Commercial Banks	Bank SHG	All Bank	Commercial Banks	Bank SHG
Treat	.118*** (.039)	.048** (.020)	.087*** (.032)	-.005 (.015)	-.017* (.009)	.026** (.012)
Observations	1930	1835	1695	782	454	98
Dep Var Mean	.044	.035	.045	.118	.106	.111
Bandwidth	.064	.060	.058	.089	.068	.058

Notes: The above table shows the treatment effect on bank loan securitization and interest rates for minority households. The unit of observation is the household. The outcome of interest columns (1)-(3) is a dummy equaling 1 if the household has any unsecured bank loan; in columns (4)-(6), the interest rate faced by the household for bank loans. The outcome of interest in columns (1) and (4) include all bank loans; in columns (2) and (5), loans obtained directly from commercial banks; in columns (3) and (6), loans obtained from bank-linked SHGs. Average household interest rates are weighted by loan volume. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. All outcomes are estimated using data-driven MSERD optimal bandwidths, as prescribed by Calonico et al. (2020). All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table E5: Treatment Effect on Loan Delinquency: Minority Households

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Any Delinquent Loan = 1)					
	Banks	Informal	Money Lenders	Community Networks	Non-Bank Financial Institutions	Any Source
Treat	-.019 (.054)	.282*** (.099)	.019 (.084)	.238* (.123)	.173 (.201)	.049 (.064)
Observations	531	360	131	228	151	792
Dep Var Mean	.286	.508	.451	.591	.045	.313
Bandwidth	.063	.051	.047	.053	.038	.047

Notes: The above table shows the treatment effect on the likelihood of having a delinquent loan for minority households. The unit of observation is the household. The outcome of interest is a dummy equaling 1 if the household has a delinquent loan from the sources mentioned. A loan is classified as delinquent if there has been no repayment towards that loan for at least 6 months preceding that survey. All specifications include state fixed effects, a linear polynomial in the running variable, and controls for household size and rural location. The sample is restricted to a bandwidth of .061 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table E6: Treatment Effects on Weekly Labour Market Activities: Extensive Margin, MSERD Optimal Bandwidth

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pr(Outcome = 1)							
	Labour Force Participation	Unemployed	Self Employed	Wage Work	Farm	Non Farm	Manufacturing	Trade and Services
Treat	.020 (.027)	-.019 (.012)	.131*** (.026)	-.086*** (.025)	.012 (.038)	-.019 (.027)	.034** (.014)	-.034 (.029)
Observations	2747	2506	2747	3596	3889	4452	2652	2652
Dep Var Mean	.510	.065	.249	.207	.137	.319	.040	.194
Bandwidth	.054	.048	.053	.059	.065	.070	.051	.051

Notes: The above table shows the treatment effect on weekly labour market outcomes along the extensive margin for minority working-age individuals. The unit of observation is the working-age individual. Working-age individuals are individuals aged between 18 and 60 years. The outcome of interest in column (1) is a dummy equal to 1 if the individual participated in the labour force in any of the 7 days preceding the survey; in column (2), a dummy equal to 1 if the individual was unemployed in any of the 7 days preceding the survey; in columns (3)-(6), a dummy equal to 1 if the individual was engaged in the sector referred. All specifications include state and survey month fixed effects; a linear polynomial in the running variable; controls for demographic and education covariates at the individual level; household level controls. The sample is restricted to a bandwidth of .06 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table E7: Treatment Effects on Weekly Labour Market Activities: Intensive Margin, MSERD Optimal Bandwidth

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Hours Worked in						
	Week	Self Employed	Wage Work	Farm	Non Farm	Manufacturing	Trade and Services
Treat	-.267 (1.206)	6.855*** (1.428)	-4.900*** (1.294)	-.541 (1.567)	-.044 (1.808)	2.587*** (.954)	-.102 (2.049)
Observations	3610	2747	3889	4649	3889	2747	2540
Dep Var Mean	23.776	12.173	10.898	5.760	17.548	2.765	10.862
Bandwidth	.060	.052	.063	.073	.063	.054	.049

Notes: The above table shows the treatment effect on weekly labour market outcomes along the intensive margin for minority working-age individuals. The unit of observation is the working-age individual. Working-age individuals are individuals aged between 18 and 60 years. The outcome of interest is hours worked per week in the sector mentioned. All specifications include state and survey month fixed effects; a linear polynomial in the running variable; controls for demographic and education covariates at the individual level; household level controls. The sample is restricted to a bandwidth of .06 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

F Treatment Effects for Non-Minorities

F.1 Tables

Table F1: Credit Access for Non-Minority Households

Panel A:								
All								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pr(Loan = 1)				Loan Amount (INR)			
	Bank	Informal	Non-Bank Financial Institutions	Any Source	Bank	Informal	Non-Bank Financial Institutions	All Sources
Treat	.052 (.051)	-.055 (.039)	.082*** (.020)	.050 (.041)	-7788.013 (36089.014)	-13520.081*** (5083.625)	8800.623*** (3079.208)	-13520.081 (34490.198)
Observations	9763	9763	9763	9763	9763	9763	9763	9763
Dep Var Mean	.200	.134	.068	.327	58710.405	10699.927	8548.257	78977.693
Panel B:								
SC/ST								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pr(Loan = 1)				Loan Amount (INR)			
	Bank	Informal	Non-Bank Financial Institutions	Any Source	Bank	Informal	Non-Bank Financial Institutions	All Sources
Treat	.083 (.077)	-.085** (.043)	-.010 (.025)	-.013 (.061)	-7934.688 (11383.284)	-8025.348 (5641.843)	2663.482 (2149.385)	-13237.598 (12038.035)
Observations	2544	2544	2544	2544	2544	2544	2544	2544
Dep Var Mean	.183	.164	.081	.348	25655.867	8714.584	4271.318	38807.468

Notes: The above table shows the treatment effect credit access for non-minority households. The unit of observation is the household. Panel A shows the results for all non-minorities; Panel B shows the results for non-minorities hailing from the *Dalit* or *Adivasi* (SC/ST) communities. The outcome of interest in columns (1)-(4) is a dummy equaling 1 if the household has an outstanding loan from the sources mentioned; in columns (5)-(8), the value of outstanding loans. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table F2: Mechanism: Nature of Bank Lending in Minority Concentration Districts

	(1)	(2)	(3)	(4)
	Pr(Loan = 1)		Loan Amount (INR)	
	Commercial Bank	Bank SHG	Commercial Bank	Bank SHG
Treat	.043 (.052)	.017 (.022)	-8138.938 (35833.548)	350.924 (552.129)
Observations	9763	9763	9763	9763
Dep Var Mean	.164	.040	57072.574	1637.831

Notes: The above table shows the treatment effect by type of bank loan for minority households. The unit of observation is the household. The outcome of interest in columns (1)-(2) is a dummy equaling 1 if the household has any outstanding loan from the source mentioned; in columns (3)-(4), the amount of outstanding loans from the source mentioned. *Commercial Bank* refers to loans obtained directly from commercial banks; *Bank SHG* refers to loans received through bank-linked SHGs. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table F3: Mechanisms: Treatment Effects on Collateral Requirements and Cost of Credit, Non-Minority Households

	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Unsecured Loan = 1)			Interest Rates		
	All Bank	Commercial Banks	Bank SHG	All Bank	Commercial Banks	Bank SHG
Treat	.021 (.037)	.010 (.024)	.018 (.023)	-.003 (.014)	-.000 (.011)	-.012 (.036)
Observations	9763	9763	9763	3158	2849	363
Dep Var Mean	.064	.036	.064	.110	.102	.110

Notes: The above table shows the treatment effect on bank loan securitization and interest rates for non-minority households. The unit of observation is the household. The outcome of interest in columns (1)-(3) is a dummy equaling 1 if the household has any unsecured bank loan; in columns (4)-(6), the interest rate faced by the household for bank loans. The outcome of interest in columns (1) and (4) include all bank loans; in columns (2) and (5), loans obtained directly from commercial banks; in columns (3) and (6), loans obtained from bank-linked SHGs. Average household interest rates are weighted by loan volume. All specifications include state fixed effects, a linear polynomial in the running variable, and household covariates. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. All specifications are weighted using a triangular kernel and household-specific weights. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table F4: Treatment Effect on Real Estate and Household Assets: Non-Minority Households

	(1)	(2)	(3)
	Res.Bld. Area	Res. Bld. Value	Land and Buildings
Treat	-16.900 (11.240)	344313.928* (197798.773)	344301.179* (203742.969)
Observations	9763	9763	9763
Dep Var Mean	64.106	555767.77	1559916.2

Notes: The above table shows the treatment's impact on real estate owned by non-minority households. The unit of observation is the household. The outcome of interest in column (1) is area of residential buildings owned; in column (2), the value of residential buildings owned; in column (3), the total value of land and residential buildings owned. All specifications include state fixed effects and household covariates. All specifications are weighted using a triangular kernel and AIDIS assigned household weights. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

G Appendix: Treatment Effects for Electoral Outcomes

This section shows additional results for electoral outcomes.

Table G1: Summary Statistics: Electoral Outcomes

	N	Mean	SD
Any Muslim Candidate	8381	0.467	0.499
Muslim Candidates	8381	1.133	1.900
Any Muslim Candidate, Non-Majoritarian Party	8381	0.131	0.337
Muslim Vote Share	8381	0.067	0.162
Muslim Vote Share, Non-Majoritarian	8381	0.030	0.103
Muslim Win	8381	0.058	0.233
Muslim Win, Non-Majoritarian	8381	0.035	0.185
BJP Vote Share	8381	0.251	0.199
BJP Win	8381	0.324	0.468
INC Vote Share	8381	0.218	0.188
INC Win	8381	0.210	0.407
Turnout	8381	0.685	0.115
Candidates	8381	12.158	5.410

Notes: This table shows the summary statistics for electoral outcomes. The unit of observation is the electoral constituency, for elections conducted between 2008 and 2019. .

Table G2: Directed Credit Policy and Electoral Outcomes: Electoral Wins and Voter Participation

	(1)	(2)	(3)	(4)	(5)
	Pr(Electoral Victory = 1)			Citizen Participation	
	Muslim Candidate	Muslim Candidate Major Party	BJP Candidate	Voter Turnout	Registered Voters
Treat	.090*** (.034)	.076*** (.028)	-.052 (.111)	.034 (.035)	4115.641 (8272.015)
Observations	1064	1064	1064	1064	1064
Dep Var Mean	.057	.018	.420	.639	188938.53

Notes: This table identifies the impact of the directed credit policy on electoral outcomes for politicians from minority communities and those representing the right-wing Bharatiya Janata Party (BJP). The unit of observation is the electoral constituency. The outcome of interest in column (1) is a binary variable equaling 1 if a Muslim candidate won the election; in column (2), a binary variable equaling 1 if a Muslim candidate contesting from a major political party won the election; in column (3), a dummy equaling 1 if the BJP won the election; in column (4), overall voter turnout; in column (5), the total number of registered voters. All specifications include state and electoral year fixed effects, in addition to a linear polynomial in the running variable. All specifications restrict the sample to a bandwidth of .058 around the discontinuity threshold, and are weighted using a triangular kernel. Standard errors are in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table G3: Treatment Effects on Electoral Outcomes: Initial vs Later Years

Panel A: 2014-2019						
	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Candidate = 1)		Vote Shares			
	Any Muslim	Any Muslim Major Party	Muslim Candidates	Muslim Candidates Major Party	Muslim Candidates Major Party	BJP
Treat	.162** (.064)	.085 (.058)	.047** (.022)	.038** (.018)	.142** (.070)	-.061 (.053)
Observations	574	574	574	574	88	574
Dep Var Mean	.586	.116	.078	.026	.224	.338
Panel B: 2008-2013						
	(1)	(2)	(3)	(4)	(5)	(6)
	Pr(Candidate = 1)		Vote Shares			
	Any Muslim	Any Muslim Major Party	Muslim Candidates	Muslim Candidates Major Party	Muslim Candidates Major Party	BJP
Treat	.198*** (.056)	.071 (.057)	.037 (.024)	.053** (.021)	.143*** (.045)	-.030 (.051)
Observations	490	490	490	490	102	490
Dep Var Mean	.574	.185	.090	.030	.163	.231

Notes: This table disaggregates the treatment effects on electoral outcomes over two time-periods. Panel A restricts the sample to elections conducted between 2008 and 2013; Panel B restricts the sample to elections conducted between 2014 and 2019. The unit of observation is the electoral constituency. The outcome of interest in column (1) is a dummy equaling 1 if there was a Muslim candidate contesting the election; in column (2), a dummy equaling 1 if there was a Muslim candidate contesting the election on behalf of a major party; in column (3), the total vote share received by all Muslim candidates; in column (4), vote share received by Muslim candidates contesting on a major party ticket; in column (5), vote share received by Muslim candidates contesting on a major party ticket, conditional on a Muslim candidate contesting on a major party ticket; in column (6), the vote share received by the BJP candidate. Vote share is the number of votes received by a candidate, divided by the total votes cast in the constituency. All specifications include state and electoral year fixed effects, in addition to a linear polynomial in the running variable. All specifications restrict the sample to a bandwidth of .058 around the discontinuity threshold, and are weighted using a triangular kernel. All specifications include state and election year fixed effects, along with a linear polynomial in the running variable. All specifications are weighted using a triangular kernel. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table G4: Directed Credit Policy and Electoral Participation of Muslim Politicians: Heterogeneity by Share of Muslim Voters

Panel A: High Muslim							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Pr(Muslim Candidate = 1)		Muslim Candidate Vote Shares				
	Any Party	Major Party	All Parties	Major Parties	Muslim Candidate Win	BJP Vote Share	BJP Win
Treat	.125*** (.044)	.337*** (.115)	.078*** (.023)	.154*** (.023)	.276*** (.043)	-.043 (.043)	-.177* (.092)
Observations	346	346	346	346	346	346	346
Dep Var Mean	.787	.320	.161	.050	.107	.276	.435
Panel B: Low Muslim							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Pr(Muslim Candidate = 1)		Muslim Candidate Vote Shares				
	Any Party	Major Party	All Parties	Major Parties	Muslim Candidate Win	BJP Vote Share	BJP Win
Treat	.349*** (.091)	.144*** (.031)	.068*** (.018)	.038*** (.014)	.047** (.021)	-.047 (.055)	.039 (.119)
Observations	569	569	569	569	569	569	569
Dep Var Mean	.388	.046	.039	.010	.021	.237	.350

Notes: This table shows the treatment's impact on electoral outcomes, conditional on the share of minorities in the constituency. The unit of observation is the electoral constituency. Panel A restricts the sample to constituencies with a relatively high (above median) share of registered Muslim voters; Panel B restricts the sample to constituencies with a relatively low share of registered Muslim voters. The outcome of interest in columns (1)-(2) is a binary equaling 1 if a minority Muslim candidate contested elections; in columns (3)-(4), the vote shares received by minority Muslim candidates; in column (5), a binary equaling 1 if the Muslim candidate won the election; in columns (6)-(7), electoral support for the right-wing BJP. *Major Party* refers to major non-majoritarian parties. All specifications include state and election year fixed effects, in addition to a linear polynomial in the running variable. All specifications are weighted using the number of registered voters in the constituency and a triangular kernel. The sample is restricted to districts with a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table G5: Treatment Effects on Politician Assets

	(1)	(2)	(3)	(4)	(5)	(6)
	Assets		Liabilities		Net Assets	
	All Candidates	Major Party	All Candidates	Major Party	All Candidates	Major Party
Treat	4.356 (5.483)	16.664** (7.070)	1.315 (1.279)	6.711** (3.352)	3.956 (5.022)	20.979*** (5.807)
Observations	1159	185	1159	185	1159	185
Dep Var Mean	9.392	20.754	2.247	4.828	8.381	18.068

Notes: This table compares the assets and liabilities of minority Muslim politicians across minority and non-minority concentration districts. The unit of observation is candidates contesting elections to the state legislative assembly. *Net Assets* equal assets less liabilities. Assets and liabilities are measured in millions of rupees (INR). The sample is restricted to Muslim politicians contesting between 2008 and 2019. The even numbered columns consider all Muslim politicians; the odd numbered columns restrict the sample to Muslim politicians contesting on behalf of major non-right wing parties. All specifications include state and electoral year fixed effects, in addition to a linear polynomial in the running variable. All specifications restrict the sample to a bandwidth of .058 around the discontinuity threshold, and are weighted using a triangular kernel and the number of electors in the constituency. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table G6: Treatment Effects on Politician Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Age		Pr(College = 1)		Criminal Cases		Pr(Conviction = 1)	
	All Candidates	Major Party	All Candidates	Major Party	All Candidates	Major Party	All Candidates	Major Party
Treat	1.633* (.914)	5.174** (2.431)	.100*** (.035)	.333*** (.107)	-.047 (.231)	.164 (.293)	-.007 (.035)	-.042 (.045)
Observations	1148	185	849	178	1159	185	1157	183
Dep Var Mean	44.586	47.592	.186	.317	.611	.503	.095	.170

Notes: This table compares observable characteristics of minority Muslim politicians across minority and non-minority concentration districts. The unit of observation is candidates contesting elections to the state legislative assembly. The sample is restricted to Muslim politicians contesting between 2008 and 2019. The even numbered columns consider all Muslim politicians; the odd numbered columns restrict the sample to Muslim politicians contesting on behalf of major non-right wing parties. The outcome of interest in columns (1)-(2) is the politician's age; in columns (3)-(4), a binary equal to 1 if the politician has a college degree; in columns (5)-(6), the number of criminal cases registered; in columns (7)-(8), a binary equal to 1 if the politician has been convicted in a criminal case. All specifications include state and electoral year fixed effects, in addition to a linear polynomial in the running variable. All specifications restrict the sample to a bandwidth of .058 around the discontinuity threshold, and are weighted using a triangular kernel and the number of electors in the constituency. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

H Appendix: Treatment Effects for Religious Violence

This section shows additional results for religious violence.

Table H1: Summary Statistics: Religious Conflicts, Administrative Data

	N	Mean	SD
Religious Riots	2573	1.640	9.613
Religious Riots Per Capita	2573	0.662	4.562
Non-Religious Riots	2573	128.342	168.690
Non-Religious Riots Per Capita	2573	47.780	52.014
Murder	2573	58.754	41.675
Murder Per Capita	2573	23.787	14.679

Notes: This table shows the summary statistics for district-level crimes obtained from the administrative National Crime Records Bureau (NCRB). The unit of observation is the district and the data covers the years 2014, 2015, 2017-2019. Per capita crimes divides the total incidence of crimes by the district population (in millions).

Table H2: Summary Statistics: Religious Conflicts, ACLED Database

	N	Mean	SD
Hindu Muslim Riots	1824	0.085	0.351
Hindu Muslim Riots Per Capita	1824	0.031	0.143
Other Riots	1824	4.901	8.264
Other Riots Per Capita	1824	1.939	5.049
Hindu-Instigated Riots	1824	0.020	0.175
Muslim-Instigated Riots	1824	0.002	0.045

Notes: This table shows the summary statistics for district-level crimes obtained from the ACLED database. The unit of observation is the district and the data covers the years 2016-2019. Per capita crimes divides the total incidence of crimes by the district population (in millions).

Table H3: Treatment Effect on Religious Violence: ACLED Database

	(1) Hindu Muslim Riots	(2) Riots Per Capita	(3) Hindu Instigated Riots	(4) Riots Per Capita	(5) Other Riots	(6) Riots
Treat	.070** (.029)	.171*** (.050)	.019*** (.007)	.112*** (.033)	-2.102* (1.138)	-.465 (2.726)
Observations	222	222	222	222	222	222
Dep Var Mean	.036	.109	.023	.081	2.621	5.085

Notes: The above table shows the treatment effect on religious violence from the ACLED database. The unit of observation is the district. Columns (1)-(2) consider Hindu-Muslim religious riots; columns (3)-(4) consider Hindu-Muslim riots which are instigated by Hindu actors; columns (5)-(6) consider other non-religious incidence of rioting. The outcome of interest in columns (1) and (6) is a binary equaling 1 if there was any incidence of rioting; columns (2), (4) and (7) scale the outcome of interest by district population in 2001 (in millions); columns (2), (3), (4) and (6) use the raw number of riot incidents. All specifications include state fixed effects and a linear polynomial in the running variable. All specifications are weighted using a triangular kernel and district population. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%

Table H4: Treatment Effects on Mechanisms of Religious Violence: ACLED Database

	(1)	(2)	(3)	(4)	(5)
	Hindu Muslim Riots Per Capita			Hindu Muslim Riots	
	1 Yr Post-Elections	1 Yr Pre-Elections	Other Years	High Muslim	Low Muslim
Treat	.047* (.025)	-.012 (.009)	.011 (.011)	.147* (.079)	-.006 (.024)
Observations	218	218	218	554	836
Dep Var Mean	.005	.005	.023	.254	.082

Notes: The above table shows the treatment effect on religious violence from the ACLED database. The unit of observation in columns (1)-(3) is the district; in columns (4)-(5), the electoral constituency. The outcome of interest in columns (1)-(3) is Hindu-Muslim riots per capita; in columns (4)-(5) incidents of Hindu-Muslim riots. Column (1) considers Hindu-Muslim riots occurring in the twelve months following elections to the state legislative assembly; column (2) considers Hindu-Muslim riots occurring in the twelve months prior to elections to the state legislative assembly; column (3) considers Hindu-Muslim riots in all other periods between elections; column (4) considers Hindu-Muslim riots in electoral constituencies with a relatively high share of Muslim voters; column (5) considers Hindu-Muslim riots in electoral constituencies with a relatively low share of Muslim voters. All specifications include state fixed effects and a linear polynomial in the running variable. Specifications in columns (1)-(3) are weighted using a triangular kernel and district population; specifications in columns (4)-(5) are weighted using a triangular kernel and the number of registered voters. The sample is restricted to a bandwidth of .058 around the discontinuity threshold. Standard errors in parentheses, clustered by district. Significant levels: *10%, **5%, and ***1%